VETERINARY SERVICES FOR HOUSEHOLD PETS

Appendix B: The impact of corporate acquisitions on treatment costs

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The Competition and Markets Authority has excluded from this published version of the final report information which the inquiry group considers should be excluded having regard to the three considerations set out in section 244 of the Enterprise Act 2002 (specified information: considerations relevant to disclosure). The omissions are indicated by [✄]. Some numbers have been replaced by a range. These are shown in square brackets. Non-sensitive wording is also indicated in square brackets.

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1. Introduction and summary

1.1 A major development in the veterinary sector over the last 10 years has been the rapid and significant growth of a few large veterinary groups (**LVGs**).[1] In 2013, around 10% of veterinary practices belonged to LVGs, but that share is now almost 60%, and many of the large groups have expressed an intention to continue expanding their business through the acquisition of independently owned practices.[2] As a result, in our Issues Statement we committed to investigate whether LVGs might have incentives to act in ways which reduce choice and weaken competition.[3] An important question in this respect is whether these acquisitions have had an impact on the costs incurred by customers to treat their pets. This appendix summarises the work we have undertaken to estimate this impact empirically, using data on insurance claims.

Data and descriptive statistics

1.2 To perform this analysis, we requested the records of insurance claims processed by two [✂] insurers: [✂]. For each insurance claim submitted to the insurers, the data we obtained reports the total amount claimed by the policy holder, some information about the pet and condition treated, and the name and location of the practice where the pet was treated. The practices issuing the claims are categorised as First Opinion Practices (**FOPs**) or Referral Centres (with a small number of practices belonging to other categories).[4]

1.3 The data covers the 2003-2024 period for Insurer 1, and the 2015-2024 period for Insurer 2.[5] Our analysis focuses on the most recent 10 years since these have better coverage (in terms of number of practices and number of observations). For the most recent year (July 2023 – June 2024), the total value of claims from FOPs and Referral Centres included in the data was £[✂] for Insurer 1 and £[✂] for Insurer 2. The value of claims issued by LVG-owned practices was £[✂] for Insurer 1 and £[✂] for Insurer 2, or £[✂] in total, which represents approximately [✂]% of the total turnover reported by LVGs for their most recent reporting periods.[6]

1.4 The dataset obtained from Insurer 1 also provides more granular information on the different items invoiced under each claim, and the price of each item. Each

[1] There are six large groups currently operating in the UK market: CVS, IVC, Linnaeus, Medivet, Pets at Home and VetPartners. In common with the industry, we refer to these as Large Veterinary Groups, or LVGs, and we refer to other suppliers as 'independents'.
[2] CMA, Veterinary services for household pets in the UK: Decision to make a market investigation reference, 23 May 2024, paragraph 4.7.
[3] CMA, Veterinary services for household pets in the UK:- Issues Statement, 9 July 2024.
[4] Footnote 33 in this paper provides more details on the categorisation of practices in this data.
[5] [✂].
[6] The CMA estimates the total turnover of the six LVGs in their last reporting year to be approximately £2,939m, based on LVG responses to RFI [✂].

item is categorised by Insurer 1 into [✂] categories of clinical procedures and treatments [✂]. In turn, these categories are aggregated into eight broader groups: [✂]. We excluded all treatments categorised as 'other' from the analysis, as this group includes a mix of unrelated treatments that do not fit into a clear or consistent classification. In this appendix, for simplicity, we refer to all the clinical procedures and treatments that can be administered to diagnose or treat pets as **treatments**, and we refer to the average invoiced item amount within a given treatment category in the Insurer 1 data as a treatment's **average price**.

1.5 The amount claimed by a policy holder before any excess or deductions (the **claim value**) reflects the treatments that were used and the prices of these treatments. It is common for pet owners or their vets to submit multiple claims for a given pet and condition, and different vets might have different policies in terms of aggregating invoices into claims (and these policies might vary over time). To obtain a proxy for the total cost of a course of treatment that is more consistent across suppliers and across time, we created a summary variable which, for every given combination of a pet and a condition appearing in the data, sums the values of all claims submitted within 12 months of the first claim. We refer to this variable as the **first-year treatment cost** incurred to treat a given pet for a given condition. We have used these two different metrics – claim values and first-year treatment costs – throughout our analysis.

1.6 The Insurer 1 dataset covers only a selected sample of claims, [✂] and there are indications that the process that generated the sample has varied over time and across practices. For these reasons, we have used the Insurer 1 data primarily to analyse average prices (which are less likely to be affected by this issue), while our analysis of claim values and first-year treatment costs relies primarily on the Insurer 2 data.

1.7 In the Insurer 2 data, between 2016 and 2023, average first-year treatment costs grew by 53% for FOPs (46% overall for FOPs and Referral Centres combined).[7] In the Insurer 1 data, over the same period, average prices grew by 63% for FOPs (62% overall for FOPs and Referral Centres combined).[8] By way of comparison, between January 2016 and December 2023, the ONS Consumer Price Index (**CPI**) for veterinary and other pet services grew by 54%,[9] the ONS CPI for services grew by 32%,[10] and the ONS estimate of average weekly earnings grew

[7] This estimate is based on a balanced sample, comprised of all observations related to conditions and practices present consistently in the data between 2016 and 2023. Where multiple observations are available for a given combination of a practice, a condition, and a year, these are averaged prior to computing the overall average.
[8] This estimate is based on unweighted yearly averages of prices for all observations related to practice-treatment combinations present consistently in the data between 2016 and 2023. This is set out in paragraphs 2.30 to 2.30 in more detail.
[9] Source: ONS, CPI INDEX 09.3.5.0 Veterinary and other services for pets. Our understanding is that this series consists of prices for annual vaccinations and kennelling services.
[10] Source: ONS, CPI Index: services.

by 37%.[11] This indicates that the cost of treating pets has grown faster than the price of other goods and services in the economy, and faster than income.

1.8 These increases in first-year treatment costs and average prices were generally larger for practices that were acquired by LVGs during the period than for practices that remained independent. Our analysis of the Insurer 2 data indicates that, for the most recent year (2023/24),[12] first-year treatment costs were on average 12% higher at LVG-owned FOPs than at independents, after controlling for observable differences in pet characteristics (age, breed, and condition treated) and practice location. Each of the six LVGs has higher first-year treatment costs compared to the average of independents, albeit the magnitude of that difference varies, ranging from [5-10%] to [10-20%]. Our analysis of variations in claim values yields similar results directionally, with the exception of [✂] whose claim values do not appear higher than independents.

1.9 Considering variations in average prices, our analysis of the Insurer 1 data indicates that, in 2023-24, LVG-owned FOPs charged prices on average [✂]% higher than those of independents, after controlling for observable differences in the types of treatment administered and practice location. Each of the six LVGs has higher average prices compared to the average of independents, albeit the magnitude of that difference varies, ranging from [5%] to [25%].

1.10 These observed differences in first-year treatment costs and average prices between LVG-owned and independent practices might reflect the effect of unobserved differences in the operating environment of these practices, as well as the effect of differences in their ownership models. For example, if the practices acquired by LVGs tend to operate in areas with higher costs, or with more affluent customers, then the observed difference in treatment costs might reflect differences in the economic environments of these practices rather than differences in their business models. To isolate the causal effect of a change in ownership model, we estimated an econometric model of claim values and average treatment prices.

Econometric analysis

1.11 The particular econometric method we used to isolate the causal effect of a change in ownership is known as difference-in-differences (**DiD**). Instead of simply comparing claim values between LVG-owned and independent practices, this method compares changes in claim values for LVG-owned practices around the time of their acquisition with changes in claim values for independent practices around the same period. If corporate acquisitions are not systematically correlated

[11] Source: ONS, Average weekly earnings in Great Britain: November 2024.
[12] Specifically, for the period between January 2023 and July 2024.

with changes in the operating environment of veterinary practices, then this method can isolate the causal effect of these acquisitions.

1.12 Pets at Home (Vets4Pets) did not acquire any practices in the 2015-2024 period, and therefore our analysis of the effect of corporate acquisitions does not apply to Pets at Home. For clarity, in the context of this analysis, we refer to the group of five LVGs excluding Pets at Home as LVG5s.

1.13 We implemented the DiD method using both the Insurer 2 data (using claim values as the outcome variable of interest), and the Insurer 1 data (using average treatment prices as the outcome variable of interest), and the results were broadly consistent:

(a) When using the Insurer 2 data, this analysis shows that an acquisition causes claim values to increase by 5% within four years.[13] We also applied the DiD methodology to individual LVGs with a view to understanding whether the effect of an acquisition varied across the set of LVG5s. This analysis necessarily relied on smaller samples and produced estimates that were less precise, especially for those LVGs which acquired fewer practices. Notwithstanding these limitations, this exercise indicates that an acquisition increases treatment costs for [at least three LVGs] but not for Linnaeus. [✄].[14]

(b) When using the Insurer 1 data, this analysis shows that an acquisition causes average treatment prices to increase by 9.2% within four years.[15] We again applied the DiD methodology to individual LVGs, subject to the same limitations of smaller samples and less precise estimates. The results indicate that an acquisition increases average prices for [at least three LVGs] [✄], but not for Linnaeus. [✄].[16] In addition, we conducted the same analysis for each broad treatment category and found that the effect was 10.4% for drugs, 9.9% for consultations, and 4.4% for diagnostics (the three largest categories in the data).

Discussion and limitations

1.14 The most important limitation of our analysis is that we do not observe any indicator of quality in the data obtained from insurers. In principle, the increase in

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[13] The estimated impact of an acquisition on claim values is 1.2% one year after the acquisition, 2.9% two years after the acquisition, 3.5% three years after of the acquisition, and 5.0% four years after the acquisition. The estimated impact averaged over all periods of observation (including periods beyond four years) was 3.2%.
[14] [✄].
[15] The estimated impact of an acquisition on average treatment prices is 3.3% one year after the acquisition, 5.2% two years after the acquisition, 6.7% three years after of the acquisition, and 9.2% four years after the acquisition. The estimated impact averaged over all periods of observation (including periods beyond 4 years) was 7.4%.
[16] [✄].

claim values observed for practices acquired by LVGs could be accompanied by an increase in the quality of the service provided, either in terms of the clinical effectiveness of the treatments administered, or in terms of the broader customer experience. This limitation applies to our analysis of average prices as well as our analysis of claim values, insofar as the treatment categories available in the Insurer 1 data can cover different variants of a given treatment, with different levels of sophistication or effectiveness.

1.15 All LVGs have submitted that they seek to improve quality at the practices they acquire, in terms of clinical capability or service quality, and that they seek to position themselves as offering higher-quality care and outstanding customer service compared to independents. The LVGs have provided numerous examples of capital investments and changes in management practices designed to meet these goals, including investments in staffing and training, changes in equipment, and access to central resources and staff.

1.16 We do not dispute that LVGs often invest resources in the practices they acquire. However, there is documentary evidence from three of the LVG5s suggesting that the price increases implemented upon acquisitions reflect their assessment of customers' willingness to pay and comparisons with competitor pricing, rather than their investments into quality (paragraph A.18). The CMA provisionally considers that the evidence submitted to-date by LVGs on investments in quality (to the extent such investments have been made) is not sufficient to fully explain the scale of the price increases and the price differentials with independent practices that we observe. In fact, we have evidence that customers often perceive the quality of independent practices as higher. Specifically, in our pet owner survey, pet owners reported significantly higher satisfaction with independent practices than with LVGs for the quality of information and advice, the quality of service, the outcome of their visit, and the cost of service. Taken together, this evidence suggests that quality improvements at acquired practices are unlikely to explain the results of our analysis. [17]

1.17 Our analysis relies on data on insured customers, so the results might not be fully generalisable to uninsured customers. However, we believe that the price effect we have identified is directly relevant for uninsured customers, as insurance companies and many veterinarians have told us that practices do not charge different prices for treatments depending on whether the customer is insured or uninsured.[18] It is possible that treatment intensity is higher for insured customers, particularly for complex conditions, because they do not bear the full cost of treatment and may therefore be more inclined to accept recommendations for more complex or expensive procedures. As a result, a sample based on insured

[17] This is discussed in our Overview of our working papers

[18] This is explicitly prohibited by section 9.34 of the supporting guidance to the RCVS Code of Professional Conduct for Veterinary Surgeons (available here: Practice information, fees and animal insurance).

customers could over-represent complex and costly treatments compared with one based solely on uninsured customers. That said, much of our sample consists of treatments commonly purchased by both insured and uninsured customers and typically standard for treating a given condition. For example, in the Insurer 1 data, the largest treatment category is 'drugs'. For a given mix of conditions, we would expect insured and uninsured pets to receive broadly similar prescriptions.[19] Notwithstanding this, given that 3.7 million pet owners in the UK currently purchase pet insurance,[20] we also consider our findings as significant in their own right.[21]

Summary

1.18 In summary, this analysis suggests that the acquisition of independent practices by at least some LVGs leads to substantial increases in the treatment costs incurred by customers. Specifically, our analysis suggests that an acquisition causes a 5% increase in claim values within four years (based on the Insurer 2 data) and a 9% increase in average prices over the same period (based on the Insurer 1 data). For context, between 2016 and 2023, first-year treatment costs at all FOPs (acquired and not acquired) appear to have increased by approximately 53%, and average prices appear to have increased by approximately 63%. The analysis also suggests that there are some economically significant differences between individual LVGs. Table 1 below summarises our current findings at this stage.

Table 1: Current findings

	Average treatment prices *(Insurer 1)*	*First-year treatment costs* *(Insurer 2)*
Overall increase across all providers (2016-2023, FOPs)	63%	53%
Difference between LVG-owned practices and independents (2023/24, FOPs)	16.6%	12.0%
Causal effect of an acquisition (four years after the acquisition, FOPs)	9%	5%

1.19 The remainder of this appendix is structured as follows: section 2 provides more details on the data we used for the analysis and some descriptive statistics; section 3 provides more details on the econometric approach we used to identify the effect of LVG acquisitions; and the annexes summarise the submissions we

[19] An exception could occur if practices systematically prescribed higher dosages or more expensive brands to insured pets, although the scope to recommend more complex or costly drugs is likely to be much smaller than for other, more sophisticated treatments.
[20] Association of British Insurers (ABI), Paw-sitively protected - pet insurers processed a record £2.4 million a day in 2021, 24 May 2022.
[21] Another aspect to consider is that, while uninsured customers may be less likely to accept complex treatments, they still have the option to do so, meaning our findings for these treatments are relevant for them as well.

have received from LVGs on a preliminary version of this analysis, and our assessment of these submissions.

2. Data and descriptive statistics

2.1 The analysis summarised in this appendix relies on the records of insurance claims collected by two large insurers: Insurer 1 and Insurer 2. In 2023, Insurer 1 and Insurer 2 accounted for [✂]% and [✂]% of the UK pet insurance market, respectively (in terms of shares of gross written premium).[22] This section describes the structure and content of this data, and provides some stylised facts on claim values and prices in the UK market.

Data on insurance claims

2.2 For Insurer 1, the data obtained by the CMA covers the period from July 2003 to July 2024, and the total number of claims is [✂]. This data covers a subset of the claims submitted to Insurer 1, [✂].[23] [24] For Insurer 2, the data obtained by the CMA covers the period from January 2015 to June 2024, and the total number of claims was [✂].[25]

2.3 For the year 2023/24 (July 2023 – June 2024), the total value of claims from FOPs and Referral Centres included in the data was £[✂] for Insurer 1 and £[✂] for Insurer 2.[26] The value of claims associated with LVG-owned practices was £[✂] for Insurer 1 and £[✂] for Insurer 2, or £[✂] in total. This represents approximately [✂]% of the total turnover reported by LVGs for their most recent reporting periods.[27]

2.4 For each claim, both datasets report:

- some characteristics of the pet treated (species, breed and age);
- the condition treated, which is coded by the claim assessor based on each insurer's own categorisation;
- the 'claimed amount', which represents the exact amount claimed by the policy holder before any decision is made by the insurer;
- the dates when the treatment started and ended;

[22] Based on Mintel data provided by Many Pets by email, 17 September 2024.
[23] [✂].
[24] Phone call with [✂].
[25] The total number of claims was calculated before any data cleaning.
[26] The total value of claims was calculated after data cleaning, which involved removing claims with negative values and observations with incorrectly formatted dates. For this calculation, treatments classified as "other" [✂] were retained; these are excluded from subsequent analyses, as discussed in paragraph 3.5. [✂].
[27] The CMA estimates the total turnover of the six LVGs in their last reporting year to be approximately £2,939m, based on LVG responses to RFI [✂].

- whether another claim has been submitted previously for the same animal and condition (claims that meet this definition are hereafter referred to as ongoing); and

- the name and postcode of the site where the treatment was administered.

2.5 The Insurer 1 dataset has an additional level of granularity in that, for each claim, it also reports the different items invoiced along with the price of each item. Each item is categorised by the Insurer 1 claim assessor into one of over 400 categories of clinical procedures and treatments (for example ‘Diagnostics – rhinoscopy’, ‘Surgery – hemilaminectomy’, ‘Drugs – ‘Metacam’). In turn, these categories are aggregated into eight broader groups: ‘consultation’, ‘diagnostics’, ‘drugs’, ‘hospitalisation’, ‘supplement’, ‘surgery’, ‘therapy’, and ‘other’.[28] We exclude all treatments categorised as ‘other’ from the analysis, as this group includes a mix of unrelated treatments that do not fit into a clear or consistent classification.[29] In the remainder of this appendix, for simplicity, we refer to all the clinical procedures and treatments that can be administered to diagnose and treat pets as treatments, and the categories used by Insurer 1 to label individual items as treatment categories.

2.6 Observations in a given treatment category can display significant variation, both in terms of the type of items included (as indicated by the free text descriptions provided by the vets) and the prices of these items. This variation can stem from variations in the particular types of treatment administered,[30] variations in invoicing practices,[31] or variation in quantities (especially for drugs). We discuss the implications of this heterogeneity in Annex A. In our main specifications, we use the average invoiced amount of all items labelled under a given treatment category within a given year as the representative price for that category. Accordingly, in the remainder of this appendix, we refer to this measure as the **average price** of that treatment category.[32]

2.7 The amount claimed by a policy holder before any excess or deduction (hereafter referred to as the **claim value**) reflects the set of treatments administered, and the prices of these treatments. It is common for pet owners or their vets to submit multiple claims for the same pet and condition. For some of our analysis, we have

[28] In addition to the eight broader groups, Insurer 1 includes an additional category, ‘unknown’, which comprises treatments that are not assigned to any specific classification. We have excluded these observations from our analysis.
[29] [✂].
[30] [✂].
[31] [✂].
[32] Our aim is to assess overall price differences across the full basket of goods and services offered by practices, without placing disproportionate weight on frequently provided treatments. If common treatments are of relatively low value, changes in their prices may have limited impact on overall customer welfare. In addition, because these treatments are typically more visible to customers, practices may face stronger competitive pressure, reducing their incentive to increase prices. For these reasons, by including all observations within each treatment category, we may understate the true extent of price differences between LVG-owned and independent practices. We discuss these issues further in the Annex and provide sensitivity tests using alternative approaches.

aggregated claim values by pet and condition, with a view to approximating the total cost of a course of treatment (paragraph 2.20).

2.8 The Insurer 2 dataset includes a variable identifying the category of the site where the treatment was administered. For our analysis, we include only practices classified as FOPs or Referral Centres. All other types (which include sites such as catteries, kennels, and pharmacies) are excluded.[33, 34] We use the same categorisation of practices when analysing the Insurer 1 dataset. When a pet is referred from a FOP to a Referral Centre, the FOP and the Referral Centre will typically issue separate claims for the treatments and procedures they have administered.[35]

2.9 We restricted our analysis to the period from 2015 to 2024 because the level of coverage in both datasets was much lower prior to 2015 (in terms of the number and value of claims processed).

2.10 The data contains some claims with very high values (for thousands of pounds). We understand that such claim values are plausible, and therefore, with one exception,[36] we have not excluded them from the data. We discuss the sensitivity of our analysis to high- and low-value claims in paragraph A.35 and following of Annex A.

Data on ownership

2.11 We separately asked each of the six LVGs to provide the list of all the practices they currently own and, for each practice, the date at which they acquired it (or opened it). This information was then merged with the Insurer 2 and Insurer 1 datasets. Any practices present in the insurance data but not identified as owned by an LVG – either for the entire sample period or in specific years prior to acquisition dates – were coded as independent. Pets at Home did not acquire any practices during our sample period, and therefore our analysis of the effect of corporate acquisitions is not relevant to it.

2.12 Table 2.1 below shows the distribution of our sample by practice owner and practice type for the period 2023-2024. Approximately 98% of practices in both Insurer 2's and Insurer 1's data are FOPs. Our main analysis focuses on FOPs, but we also report results that include referral centres for comparison. Generally,

[33] [✂].
[34] [✂].
[35] As explained in paragraph 2.7, for some aspects of our analysis, we have aggregated claims by pet and condition to approximate the total cost of treatment, and the resulting total may include claims made by different practice types (FOPs and referral centre).
[36] [✂].

the number of claims per practice is higher for LVG-owned practices than for independent practices.

Table 2.1: Sample Distribution by Practice Type

	Insurer 2		Insurer 1	
	FOP	**Referral**	**FOP**	**Referral**
Number of claims				
All practices	[✂]	[✂]	[✂]	[✂]
Independents	[✂]	[✂]	[✂]	[✂]
LVGs	[✂]	[✂]	[✂]	[✂]
Unique practices	[✂]	[✂]	[✂]	[✂]
All practices	[✂]	[✂]	[✂]	[✂]
Independents	[✂]	[✂]	[✂]	[✂]
LVGs	[✂]	[✂]	[✂]	[✂]
Claims per practice	[✂]	[✂]	[✂]	[✂]
All practices	[✂]	[✂]	[✂]	[✂]
Independents	[✂]	[✂]	[✂]	[✂]
LVGs	[✂]	[✂]	[✂]	[✂]

Source: CMA analysis using both Insurer 2 and Insurer 1 data
Note: The sum of unique practices for independents and LVGs does not always match the total number of practices. This discrepancy arises because, during the 2023-2024 period, some independent practices were acquired by LVGs, resulting in the same practice being included in both groups.

2.13 Our methodology uses changes in ownership to identify the effect of different ownership models on outcomes. Figure 2.1 shows that, over the 2015-2024 period, 1,847 practices were acquired by the LVG5s, with the bulk of these acquisitions happening around the middle of this period. IVC was the most active, securing ownership of [✂] practices, followed by VetPartners with [✂], Medivet with [✂], Linnaeus with [✂], and CVS with [✂].

Figure 2.1:

[✂]

Characteristics of the Insurer 1 data

2.14 As explained in paragraph 2.2, the data provided by Insurer 1 is in fact a subset of the claims processed by the insurer: [✂].[37] [38] This feature of the Insurer 1 data has two important implications for the coverage of the data and our analysis.

2.15 First, the coverage of the Insurer 1 data in terms of practices is lower than that of the Insurer 2 data. In 2023-24, the number of practices present in the datasets

[37] [✂].
[38] [✂].

was 3,112 for Insurer 1 and 4,979 for Insurer 2. When the sample is restricted to practices owned by LVGs, for which the CMA has up-to-date information obtained from the LVGs themselves, the number of practices present in the datasets was 2,085 for Insurer 1 and 2,566 for Insurer 2, representing 75% and 92% of known sites, respectively.[39]

2.16 Second, even for those practices present in the [subset of data] [✂] is unlikely to be a random sample of claims. [✂].

2.17 To understand whether the selection process changed with acquisitions, we tested whether the number of claims observed in the Insurer 1 dataset changed differently for acquired practices compared to those that remained independent throughout the period. This analysis uses the same 'difference-in-differences' design as employed in our econometric analysis of claim values (paragraphs 3.1 and following). The dependent variable in the model is the number of claims submitted by a practice in a given year (log-transformed to allow the coefficients to be interpreted as approximate percentage changes). The sample includes all practice types, including FOPs and Referral Centres. The results (reported in Figure 2.2) show that corporate acquisitions are associated with significant changes in the number of claims reported for Insurer 1. More specifically, the number of claims reported by acquired practices in the year of the acquisition decreases by 12% compared to non-acquired practices, with the effect statistically significant at the 1% level. In subsequent years, the number of claims processed by acquired practices increases faster than at independent practices, albeit the difference is not statistically significant. This suggests that practices that are acquired change their reporting practices with respect to customers insured by Insurer 1.

Figure 2.2:

[✂]

2.18 We consider that this limitation of the Insurer 1 data could potentially affect the reliability of comparisons of claim values (between suppliers or over time). However, we think that this limitation is less likely to affect the reliability of comparisons of average prices (between suppliers or over time). [✂]. Average prices are unlikely to vary between the selected sample of claims available to the CMA, and the whole sample of claims processed by Insurer 1. For these reasons, we have used the Insurer 1 data primarily to analyse average prices, while our analysis of claim values relies primarily on the Insurer 2 data (albeit we do report some results from analysis of the Insurer 1 data for completeness).

[39] [✂].

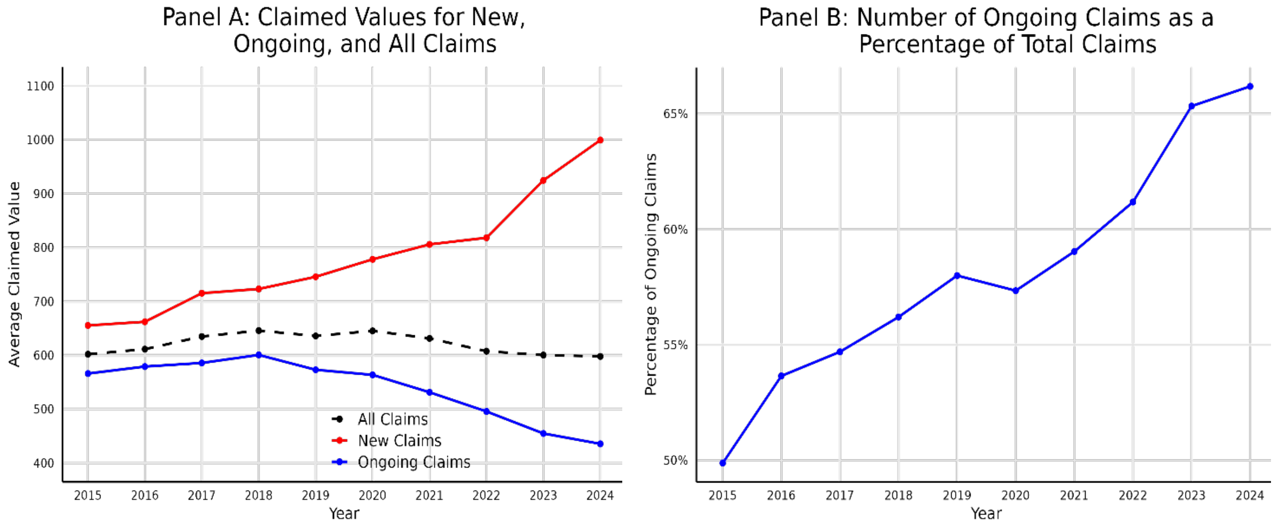
Changes in claim values over time

2.19 We sought to use the Insurer 2 data to estimate changes in claim values over time. We made a series of adjustments to the sample and the level of aggregation of the data to ensure that our estimates of these changes were not confounded by changes in the composition of the sample in terms of condition, practice type (FOP vs Referral Centre), or claim type (new vs ongoing).

2.20 First, we summed all claims related to a given pet-condition combination over the first 12 months following the issuance of the first claim for that combination.[40] This adjustment is necessary because it is common for pet owners and their vets to submit multiple claims for a given condition, and there are indications that the relative importance of the initial and ongoing claims changed over time. By way of illustration, Figure 2.3 below shows the average value of new and ongoing claims (Panel A) and the ratio between the number of ongoing and total claims (Panel B) over the 2015-2024 period. This shows that the share of ongoing claims has increased, but the value of individual ongoing claims has decreased. This indicates that the total cost of a course of treatment tends to be increasingly ‘parcelled out’ in multiple claims submitted over time (whether this is for clinical or accounting reasons). Against that backdrop, computing a simple average of claim values over time would understate the true increase in treatment costs faced by pet owners. We interpret the sum of all claims submitted over the first 12 months for a pet-condition combination as a proxy for the overall cost of a course of treatment, and we refer to this metric as the **first-year treatment cost** charged by a supplier for a given pet and condition.[41]

[40] Specifically, for each pet-condition combination appearing in the data, we identify the first claim submitted and add to its value the values of all subsequent claims submitted for the same pet and the same condition within a 12-month period. The total cost is assigned to the year in which the last claim occurs. For example, if a pet was treated for a condition from January 2023 to January 2024, we sum all claimed amounts over this period and assign the total to January 2024. Claims submitted beyond the 12-month window are excluded from the sample.
[41] In principle, it would be possible to compute the total cost of a course of treatment for a pet-condition combination by including claims beyond the 12-month window. However, because some pet owners occasionally submit claims over several years, this would involve losing observations for later years in the sample.

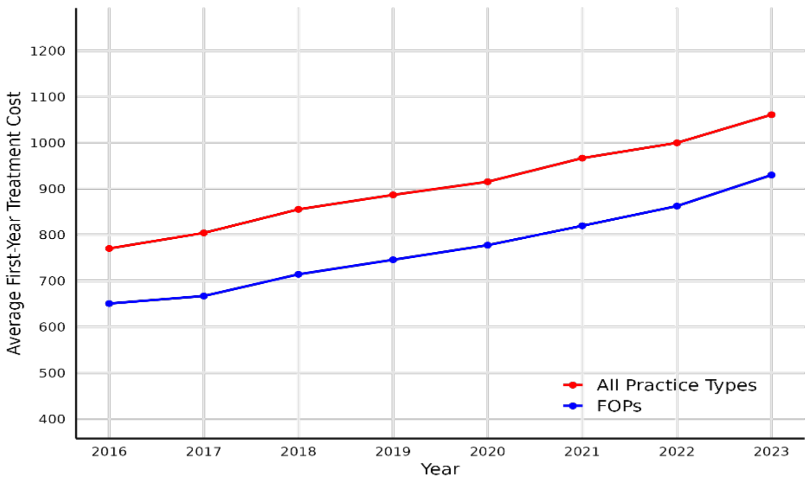
Figure 2.3: Relative importance of new and ongoing claims over the 2015-2024 period (Insurer 2 data)



Source: CMA analysis using Insurer 2 data.

2.21 Figure 2.4 below shows average first-year treatment costs over time, for all practices (FOPs and referral centres) and for FOPs only.[42] This figure suggests that average first-year treatment costs have increased materially over the past decade. Specifically, over the period 2016 – 2023, average first-year treatment costs appear to have increased by 38% for all practices, and by 43% for FOPs only.

Figure 2.4: Average first-year treatment costs for all practices and FOPs (£) (Insurer 2 data)



Source: CMA analysis using Insurer 2 data

2.22 Second, as an additional test, we only kept observations related to practice-condition combinations that were consistently present in the data from 2016 to

[42] The first-year treatment cost calculated for a particular pet-condition combination may incorporate claims submitted by both FOPs and referral centres. We categorise a pet-condition combination as associated with a FOP or a referral centre based on the type of the practice which submitted the last claim in the sequence.

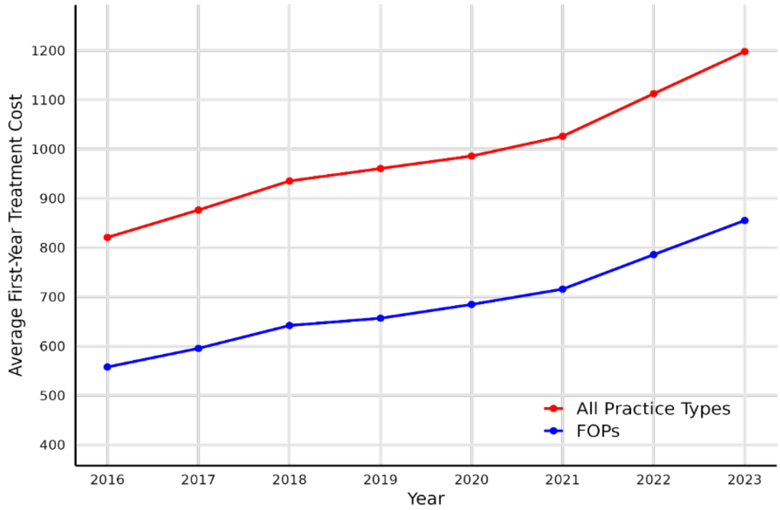
2023 (in this way, we used a 'balanced sample').[43] For instance, if a practice consistently issued claims for lameness every year from 2016 to 2023, observations related to the treatment of lameness at that practice are included in the sample. In contrast, if a practice only starts issuing claims for lameness in 2017 (either because this practice only enters the dataset in 2017, or because it only starts issuing claims for that particular condition in that year), then all observations related to the treatment of lameness at that practice are excluded from the sample. This adjustment is necessary to avoid capturing 'compositional effects' in our averages of first-year treatment costs over time (for example if more expensive practices are added to the sample, or if practices start submitting claims for more complex conditions). It implies that the resulting estimates of inflation in first-year treatment costs is likely to put more weight on observations related to the most common conditions treated at the largest practices.

2.23 Additionally, we aggregated the data by calculating the average first-year treatment costs for each combination of practice, condition, and year. This ensures that each condition is given equal weight when plotting average first-year treatment costs, regardless of how frequently it appears in the data. As a result, we obtained a balanced sample consisting of one observation for each practice-condition combination in each year. Specifically, after making these adjustments, we were left with 50,072 practice-condition-year combinations spread over 1,358 practices.

2.24 Figure 2.5 below shows average first-year treatment costs over time for the balanced sample, both for all practices and for FOPs only. The trend is consistent with what is shown in Figure 2.4, which uses an unbalanced sample, although the estimated percentage increases are higher in the balanced sample. Overall, the figure suggests that average first-year treatment costs have increased materially over the past decade. Specifically, between 2016 and 2023, average first-year treatment costs increased by 46% for all practices, and by 53% for FOPs only.[44]

[43] We did not use 2015 and 2024 as we did not have data for the whole of these two years.
[44] We consider this second approach, which uses a balanced sample and aggregates first-year treatment costs by practice-condition-year, to be more robust and informative. Despite relying on a smaller sample of practices and conditions, it eliminates compositional effects and ensures that each condition contributes equally to the analysis. This allows for a more reliable assessment of trends in treatment costs over time.

Figure 2.5: Average first-year treatment costs for all practices and FOPs (£) (balanced sample, Insurer 2 data)



Source: CMA analysis using Insurer 2 data

2.25 We also considered whether the increase in first-year treatment costs varied between LVG-owned and independent practices. To this end, we divided the sample of practices into two groups: practices that remained independent throughout the 2016-2023 period, and practices that were acquired by a LVG at any point after 2016 (Pets at Home practices are entirely excluded from the sample for the purpose of this analysis).[45] The latter group therefore includes practices that were initially independent but were later acquired by a LVG, with the share of LVG-owned practices in that group increasing progressively over time, from 0% in 2016 to 100% in 2023. Figure 2.6 shows average first-year treatment costs for these two groups using the unbalanced sample, while Figure 2.7 presents the same comparison based on the balanced sample.

2.26 Figure 2.6 suggests that, between 2016 and 2023, first-year treatment costs increased by 34% for practices that remained independent throughout the period, and by 43% for practices that were acquired. In addition, the difference in average first-year treatment costs between acquired practices and those that remained independent widened over time—from £47 in 2016 to £135 in 2023.

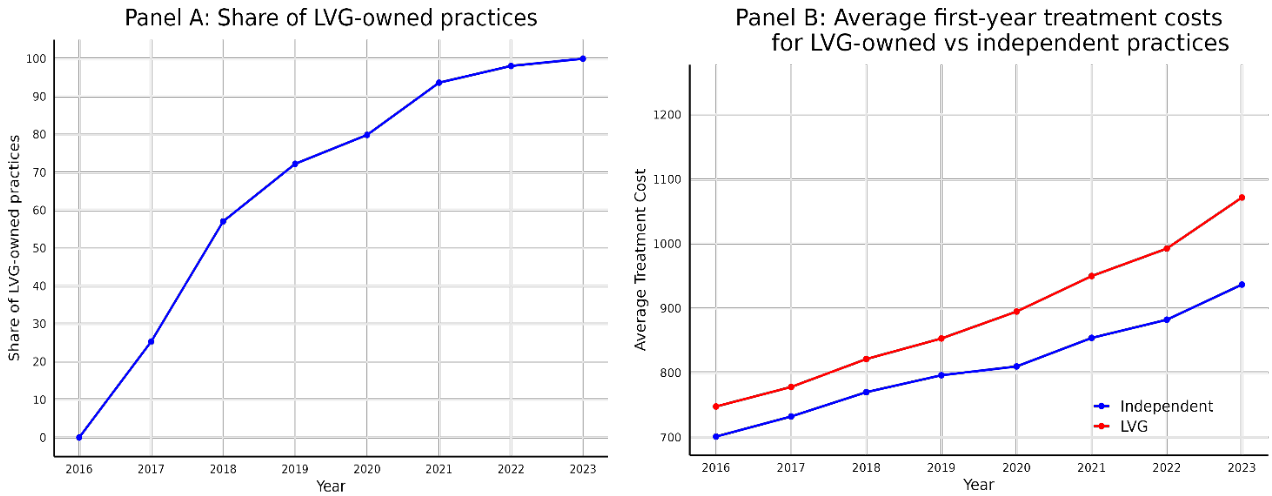
2.27 Figure 2.7 presents the same comparison using the balanced sample of practices and conditions. This analysis shows a similar picture: between 2016 and 2023, first-year treatment costs increased by 43% for practices that remained independent and by 50% for those that were acquired. The difference in average first-year treatment costs between acquired and independent practices widened from £56 in 2016 to £136 in 2023.

[45] We exclude practices acquired in 2024. As with the previous figures, the sample is balanced, including only practice-condition combinations consistently present from 2016 to 2023.

2.28 Overall, both figures point to the same trend—first-year treatment costs have risen more sharply for practices that were acquired by LVGs, and the gap between acquired and independent practices has widened over time.

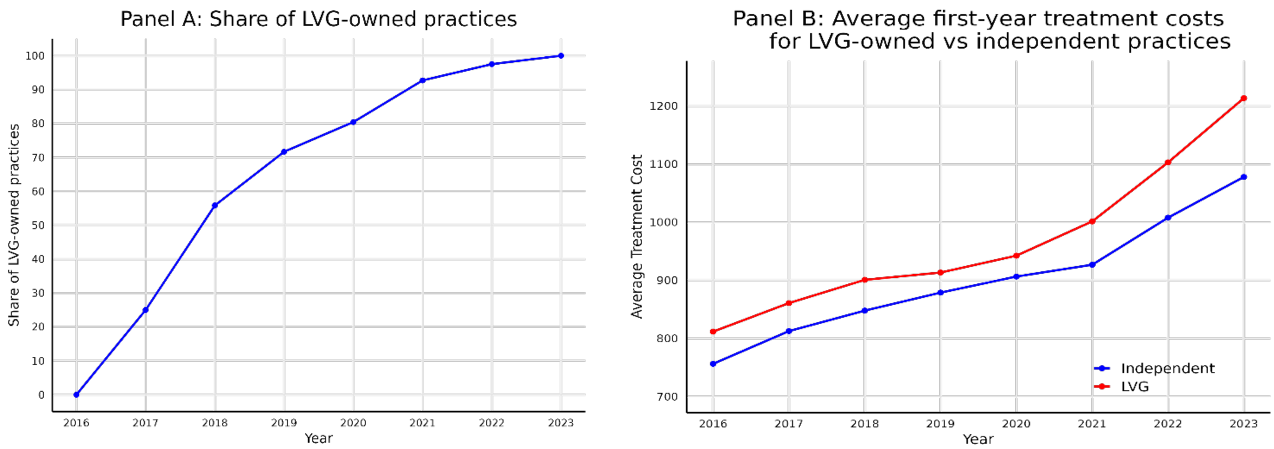
2.29 In addition, the fact that first-year treatment costs initially follow similar trends for the two groups, and then start diverging when a significant number of practices in the second group are acquired is consistent with the proposition that changes in ownership have a causal effect on first-year treatment costs. In other words, the widening of the difference in first-year treatment cost that is observed at the end of the period appears to have been caused by the fact that some practices in the second group were acquired. We use these arguments more formally in the econometric analysis presented in section 3.

Figure 2.6: Average first-year treatment costs for LVG-owned and independent practices (£) (Insurer 2 data)



Source: CMA analysis using Insurer 2 data

Figure 2.7: Average first-year treatment costs for LVG-owned and independent practices (£) (balanced sample, Insurer 2 data)



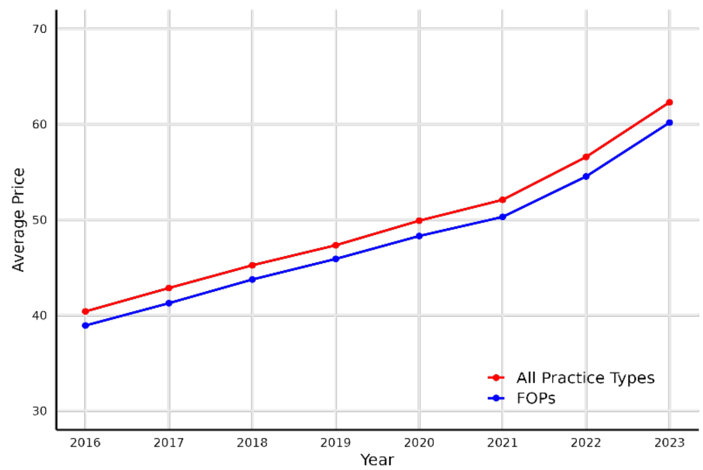
Source: CMA analysis using Insurer 2 data

Changes in average treatment prices over time

2.30 We sought to perform a similar analysis of average treatment prices. As for our analysis of first-year treatment costs, we first used all observations (ie an unbalanced sample), and then restricted the analysis to practice-treatment combinations that were consistently present in the data from 2016 to 2023, resulting in a balanced sample with one observation for each practice-treatment combination in each year. After making this adjustment, we were left with 268,176 practice-treatment-year combinations spread over 1,242 practices.[46]

2.31 Figure 2.8 and Figure 2.9 below show our analysis of average prices for the unbalanced and balanced samples, respectively. Figure 2.8 indicates a consistent upward trend in average prices over the last decade. Specifically, for all practices, the average price rises from £40.4 to £62.3, an increase of 54%. When focusing on FOPs alone, we observe a comparable percentage increase (55%). This increase is of a similar magnitude as that reported by the ONS CPI for veterinary and other pet services, which was 54% between January 2016 and December 2023.[47]

Figure 2.8: Average prices for all practice types and FOPs (£) (Insurer 1 data)



Source: CMA analysis using Insurer 1 data.

2.32 Figure 2.9 presents the same comparison based on the balanced sample of practices and treatments. The results show a similar pattern: between 2016 and 2023, average prices increased by 62% for all practices, and by 63% for FOPs alone.

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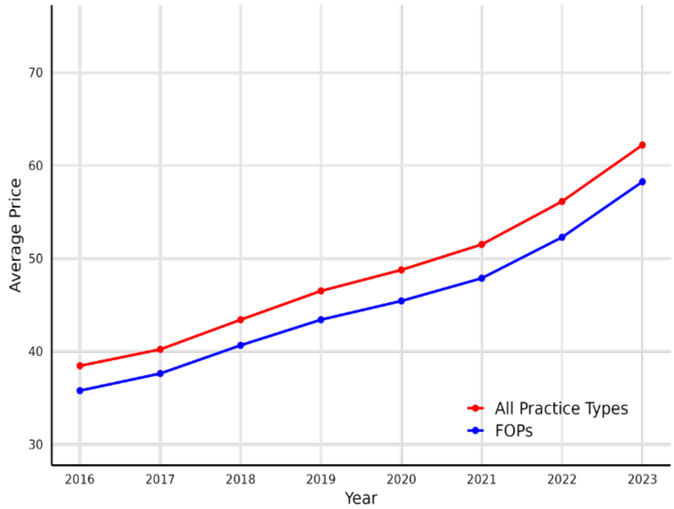
[46] As explained in paragraph 1.6, we aggregated the data by calculating average prices for each combination of a practice-treatment-year. This is necessary because the data often include multiple observations for the same treatment for a given practice in a year (for example, a practice may prescribe the same antibiotic to multiple pets each year), typically with minimal variation in prices. Taking the simple average of all observations would give undue weight to treatments that are used frequently.

[47] Office for National Statistics, CPI INDEX 09.3.5.0, Veterinary and other services for pets.

Figure 2.9: Average prices for all practice types and FOPs (£) (balanced sample, Insurer 1 data)



Source: CMA analysis using Insurer 1 data.

2.33 Table 2.2 presents our analysis of changes in average prices for the different categories used by Insurer 1 to categorise individual treatments. This suggests significant increases in average prices across all treatment categories from 2016 to 2023.

Table 2.2: Price changes by treatment category, all practice types (Insurer 1 data)

	Average Price (£)					
	2016		**2023**		**Percentage Change**	
	Unbalanced sample	Balanced sample	Unbalanced sample	Balanced sample	Unbalanced sample	Balanced sample
Drugs	24.7	23.9	35.0	37.3	42%	56%
Consultations	43.5	33.6	65.0	53.2	49%	58%
Diagnostics	88.2	76.7	126.0	124.0	43%	62%
Surgeries	198.0	139.0	356.0	263.0	80%	89%
Hospitalisation[48]	43.5	39.5	69.3	62.0	59%	57%
Therapy	112.0	94.1	155.0	142.0	38%	51%
Supplement	55.2	48.7	68.7	57.0	24%	17%

Source: CMA analysis using Insurer 1 data.

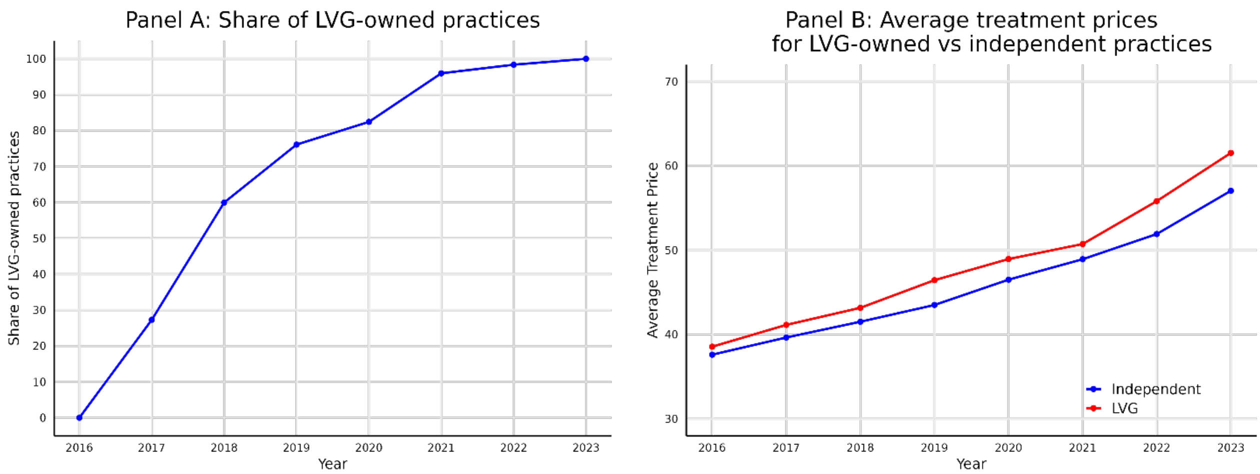
2.34 Figure 2.10 and Figure 2.11 below present average prices for LVG-owned and independent practices separately. The two groups of LVG-owned and independent practices are defined using the approach set out in paragraph 2.25. Pets at Home practices are also excluded. Figure 2.10 shows the results based on the

[48] The ‘Hospitalisation’ category in the Insurer 1 data includes four main subcategories: Hospitalisation, Inpatient Care, Intensive Care, and Nurse Care. These categories cover various costs, such as charges for weekend stays, inpatient services, intensive care days, and nursing care during hospitalisation.

unbalanced sample, while Figure 2.11 presents the same comparison using the balanced sample of practices and treatments.

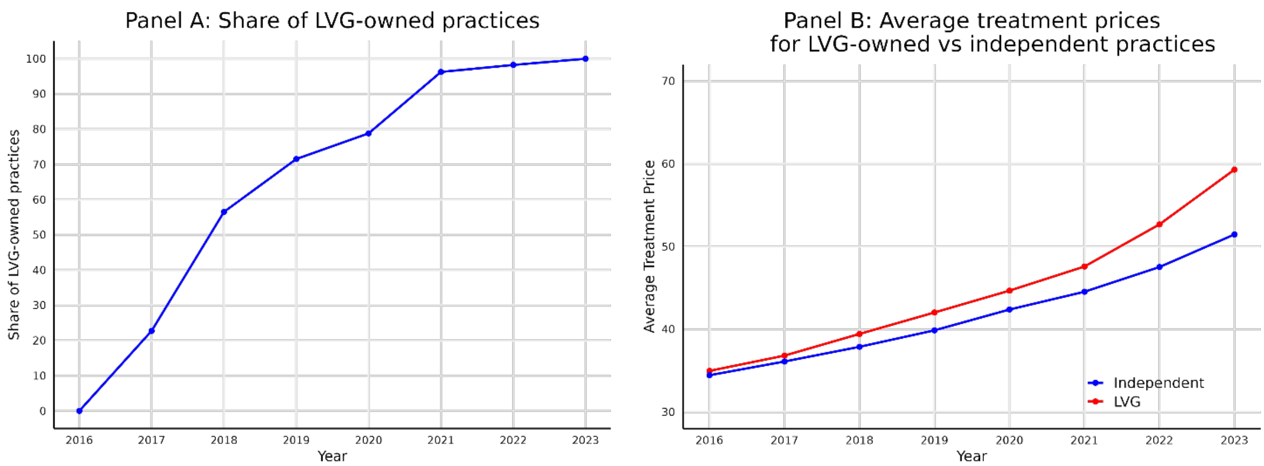
2.35 Both figures show a pattern similar to that observed for first-year treatment costs: average prices were broadly similar for the two groups in earlier years but began to diverge as the number of acquired practices increased. In the unbalanced sample, between 2016 and 2023, average prices increased by 52% for practices that remained independent and by 59% for those acquired by LVGs. In addition, the difference in average prices between acquired and independent practices widened from £1 in 2016 to £4.4 in 2023. When using the balanced sample, the increases for the two groups were 50% and 69%, respectively. In this case, the difference in average prices between acquired and independent practices widened from £0.6 in 2016 to £7.8 in 2023.

Figure 2.10: Differences in average treatment prices between LVGs and independents over time (£) (Insurer 1 data)



Source: CMA analysis using Insurer 1 data.

Figure 2.11: Differences in average prices between LVGs and independents over time (£) (balanced sample, Insurer 1 data)



Source: CMA analysis using Insurer 1 data.

Differences in claim values between LVGs and independents for 2023/24

2.36 Table 2.3 shows the average claim values and average prices for the whole sample, and for independent and LVG-owned practices separately, for the period 2023–2024. These results suggest that claim values and average prices are significantly higher for LVGs compared to independents. Considering the claims recorded in the Insurer 2 data in 2023/24 for all practice types (including FOPs and Referral Centres), average claim values were 17% higher at LVG-owned practices compared to independents, and average first-year treatment costs were 13% higher at LVG-owned practices compared to independents. Considering the claims recorded in the Insurer 1 data in 2023/24 for all practice types (including FOPs and Referral Centres), average prices in our analysis were 9.4% higher for LVGs than for independents.[49]

Table 2.3: Descriptive Statistics for Key Variables (Insurer 2 and Insurer 1 data, Jan 2023 to July 2024)

	All Observations		**Independents**		**LVGs**	
	Obs.	Mean	Obs.	Mean	Obs.	Mean
Insurer 2						
Claimed Value (£)	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]
First-year Treatment Cost (£)	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]
Pet Age (years)	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]
Share of ongoing claims	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]
Share of claims in rural areas	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]
Insurer 1	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]
Average Price (£)	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]
Share of treatments in rural areas	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]

Source: CMA analysis using both Insurer 2 and Insurer 1 data

2.37 Table 2.4 below provides the same descriptive statistics split by LVG. This shows that one LVG's practices have lower claim values and average prices than independents (reported in Table 2.3). Another LVG has higher claim values but lower average prices. All other LVGs have higher claim values and average prices compared to independents.

[49] In our analysis, we also control for pet breed, though breed-specific statistics are excluded here due to the large number of breeds. Additionally, the cross-sectional regression controls for practice type, as categorised in Table 2.1.

Table 2.4: Descriptive Statistics for Key Variables (Individual LVGs, Insurer 2 and Insurer 1 data)

	[✂]		[✂]		[✂]		[✂]		[✂]		[✂]	
	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]
Insurer 2	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]
Claimed Value (£)	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]
First-year Treatment Cost (£)	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]
Pet Age (years)	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]
Share of ongoing claims	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]
Share of claims in rural areas	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]
Insurer 1	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]
Average Price (£)	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]
Share of treatments in rural areas	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]

Source: CMA analysis using both Insurer 2 and Insurer 1 data

2.38 These differences in claim values and average prices may plausibly reflect differences in the mix of conditions or animals treated by these different groups, or differences in the type of practices owned. To control for these factors, we ran a series of simple regressions using data for 2023-24.[50] We used two different dependent variables: claim values and first-year treatment costs (both variables are log-transformed, such that the regression coefficients can be interpreted as approximate percentage changes). The independent variables are: pet age; pet condition; pet breed; practice type (FOP or not);[51] whether the customer has submitted a previous claim for the same condition;[52] and a variable indicating whether the practice issuing the claim is owned by an LVG or not.[53]

2.39 The results are presented in Table 2.5.[54] The coefficients in the table can be interpreted as approximate percentage differences in the dependent variable between LVG-owned and independent practices. For example, the coefficient in the top-left cell (0.062) means that, under this specification, claim values at LVG-owned practices appear to be approximately 6.2% higher than at independent

[50] Specifically, we use data from January 2023 to July 2024.
[51] This control variable is only used when the sample is not restricted to FOPs.
[52] This control variable is only used when the dependent variable is the value of an individual claim.
[53] We model claimed values as follows: $\log(claimed\_value_{i,j,t}) = \alpha + \beta_1 LVG_{i,t} + \beta_2 X_{i,j,t} + \varepsilon_{i,j,t}$, where: $claimed\_value_{i,j,t}$ is the claimed value for claim j, which is processed at practice i in year t; $LVG_{i,t}$ is an indicator variable equal to one if practice i is owned by a LVG in year t, and zero otherwise; $X_{j,t}$ is a vector of covariates including: the condition treated, the breed and age of the pet, a binary variable indicating whether the customer has submitted a previous claim for the same condition, an indicator variable denoting the practice type (FOP or not), a binary variable indicating whether the practices is located in rural or urban areas, and a series of indicator variables for the local authority. We use the same model specification when the dependent variable is the first-year treatment cost. However, in this version of the model, we exclude the variable "ongoing claim." Additionally, when we restrict the sample to FOPs, we exclude the variable “practice type”. When reporting results for individual LVGs, we replace $LVG_{i,t}$ with six separate dummy variables. Each dummy variable equals one if practice i is owned by a specific LVG in year t, and zero otherwise.
[54] For ease of exposition, we do not report coefficient estimates for the control variables, but we check that these are of the expected sign.

practices. The figures in brackets under each coefficient are the standard errors (a measure of the precision of the estimate), and the stars denote statistical significance (put simply, stars next to a coefficient indicate that we can be confident that the true value of the coefficient is not zero). Overall, these results appear to confirm that LVGs have, on average, higher claim values and first-year treatment costs than independents, even when controlling for observable determinants of these variables. The differences range between 6.2% and 12.0%, depending on the dependent variable and sample used.

Table 2.5: Cross-sectional regression for claim values and first-year treatment costs (all LVGs)

	Log (Claimed Value)		**Log (First-Year Treatment Cost)**	
	All Types	FOPs	All Types	FOPs
LVG	0.062***	0.065***	0.110***	0.120***
	(0.012)	(0.011)	(0.010)	(0.010)
Controlling for:				
Pet Characteristics (Age, Breed, Condition)	Yes	Yes	Yes	Yes
Practice Type	Yes	No	Yes	No
Ongoing Claim	Yes	Yes	No	No
Rural/urban classification	Yes	Yes	Yes	Yes
Local authority	Yes	Yes	Yes	Yes
Observations	521,407	484,129	215,988	199,403
Adjusted R^2	0.374	0.363	0.425	0.396

Source: CMA analysis using both Insurer 2 data
*Note: Standard errors are clustered at the practice level and are reported in parentheses. ***, **, and * correspond to statistical significance at the 1%, 5%, and 10% levels, respectively.*

2.40 To explore whether this difference varies across individual LVGs, we ran an additional regression model, replacing the general LVG indicator variable with individual indicators for each LVG. The results, reported in Table 2.6 below, show that the magnitude of that difference varies across LVG. For [one LVG] the coefficient is positive for first-year treatment costs, but not for claim values. For other LVGs, the coefficients are positive and statistically significant, and range roughly between 5% and 10% for claim values, and between 6% and 16% for first-year treatment costs.

Table 2.6: Cross-sectional regression for claim values and first-year treatment costs (individual LVGs)

	Log (Claim Value)		Log (First-Year Treatment Cost)	
	All Types	FOPs	All Types	FOPs
[✂]	[✂]	[✂]	[✂]	[✂]
	[✂]	[✂]	[✂]	[✂]
[✂]	[✂]	[✂]	[✂]	[✂]
	[✂]	[✂]	[✂]	[✂]
[✂]	[✂]	[✂]	[✂]	[✂]
	[✂]	[✂]	[✂]	[✂]
[✂]	[✂]	[✂]	[✂]	[✂]
	[✂]	[✂]	[✂]	[✂]
[✂]	[✂]	[✂]	[✂]	[✂]
	[✂]	[✂]	[✂]	[✂]
[✂]	[✂]	[✂]	[✂]	[✂]
	[✂]	[✂]	[✂]	[✂]
Controlling for:				
Pet Characteristics (Age, Breed, Condition)	Yes	Yes	Yes	Yes
Practice Type	Yes	No	Yes	No
Ongoing Claim	Yes	Yes	No	No
Rural/urban classification	Yes	Yes	Yes	Yes
Local authority	Yes	Yes	Yes	Yes
Observations	521,407	484,129	215,988	199,403
Adjusted R²	0.374	0.364	0.426	0.396

Source: CMA analysis using both Insurer 2 data
*Note: Standard errors are clustered at the practice level and are reported in parentheses. ***, **, and * correspond to statistical significance at the 1%, 5%, and 10% levels, respectively.*

Differences in average prices between LVGs and independents for 2023/24

2.41 We then tested whether average prices (rather than first-year treatment costs) were higher at LVGs compared to independent practices. To that end, we ran a similar set of regressions using the Insurer 1 data and the average price of a treatment as the dependent variable. For this reason, we do not control for pet characteristics in the regression. The independent variables included are a series of indicator variables for the type of treatment administered, an indicator variable denoting the practice type (FOP or not), a binary variable indicating whether the practice is located in a rural or urban area, and a series of indicator variables for the local authority.[55]

2.42 The results, shown in Table 2.7 below, indicate that LVGs generally have higher average prices than independents, with an average effect of 16.5% across all practice types and 16.6% for FOPs. Again, all LVGs appear to exhibit higher average prices, with the difference for FOPs ranging from [5]% to [25]%.

Table 2.7: Cross-sectional regression for average price (Insurer 1 data)

	Log (Average Price)		**Log (Average Price)**	
	All Types	FOPs	All Types	FOPs
LVG	0.165***	0.166***		
	(0.008)	(0.008)		
[✂]			[✂]	[✂]
			[✂]	[✂]
[✂]			[✂]	[✂]
			[✂]	[✂]
[✂]			[✂]	[✂]
			[✂]	[✂]
[✂]			[✂]	[✂]
			[✂]	[✂]
[✂]			[✂]	[✂]
			[✂]	[✂]
[✂]			[✂]	[✂]
			[✂]	[✂]
Controlling for:				
Pet Characteristics (Age, Breed, Condition)	No	No	No	No
Practice Type	Yes	No	Yes	No
Ongoing Claim	No	No	No	No
Rural/urban classification	Yes	Yes	Yes	Yes

[55] We model the price of a treatment as follows: $\log(average\_price_{i,j,t,o}) = \alpha + \beta_1 LVG_{i,t,o} + \beta_2 X_{i,j,t,o} + \varepsilon_{i,j,t,o}$, where: $average\_price_{i,j,t,o}$ is the price for treatment type j, averaged by practice i and year t, and ownership status *o* (independent or LVG). $LVG_{i,t,o}$ is an indicator variable equal to one if practice i is owned by a LVG in year t, and zero otherwise. $X_{i,j,t,o}$ is a vector of covariates including the specific treatment type administered, an indicator variable denoting the practice type (FOP or not), a binary variable indicating whether the practice is located in rural or urban areas, and a series of indicator variables for the local authority. When reporting results for individual LVGs, we replace $LVG_{i,t,o}$ with six separate dummy variables. Each dummy variable equals one if practice i is owned by a specific LVG in year t, and zero otherwise.

Local authority	Yes	Yes	Yes	Yes
Treatment type	Yes	Yes	Yes	Yes
Observations	350,064	338,308	350,064	338,308
Adjusted R^2	0.680	0.677	0.681	0.678

Source: CMA analysis using Insurer 1 data.
*Note: Standard errors are clustered at the practice level and are reported in parentheses. ***, **, and * correspond to statistical significance at the 1%, 5%, and 10% levels, respectively.*

2.43 We note that the estimated difference in average prices between LVG-owned and independent practices (16.6% for FOPs - set out in paragraph 2.42) is larger than the estimated difference in first-year treatment costs (12.0% for FOPs - set out in paragraph 2.39). We consider that it is difficult to compare these two coefficients because they are measured on two different datasets, and therefore the difference between these two coefficients may conflate some economically meaningful factors (eg a systematic difference in treatment intensity between LVG-owned and independent practices) with less relevant factors stemming from differences in the coverage of the samples or the incentives facing the customers of the two insurers. For example, if one of the insurers tends to sell policies with more generous coverage (with lower excess or narrower exclusions), then the customers of that insurer might be more inclined to accept recommendations of more intensive treatment from some vets.

2.44 Overall, our descriptive statistics appear to show that prices and treatment costs have significantly increased over the past decade. LVGs consistently show higher claimed amounts and prices than independents. However, in and of itself this set of results would not establish that the LVG ownership model is the cause of higher treatment costs. In the next section, we outline our approach to identify the causal effect of LVG ownership on claimed amounts and prices.

3. The causal effect of corporate acquisitions

3.1 This section summarises the econometric analysis we undertook to estimate the causal effect of corporate acquisitions on claim values and prices. It starts by discussing our identification strategy, that is the general approach we took to isolating the effect of changes in ownership models from that of confounding factors. It then provides more details on the specific econometric models we took to the data and the main results. Finally, it explores the mechanisms underpinning these results and discusses sensitivities and robustness checks.

3.2 Pets at Home did not acquire any practice in the period of observation, so this strand of our analysis does not apply to them. For clarity, in the context of this section, we refer to the group of five LVGs excluding Pets at Home as LVG5s.

Identification strategy

3.3 As set out in paragraphs 2.38 to 2.42 above, simple regressions suggest that claim values, first-year treatment costs and average prices tend to be higher at LVG-owned practices, and this difference remains when we control for observable characteristics of pets and practices. In and of itself, this set of results would not prove conclusively that the LVG ownership model *causes* these variables to be higher. It is conceivable that some unobserved factors that are positively correlated with LVG ownership also cause higher claim values at LVG-owned practices. Suppose, for example, that LVGs tend to own practices located in areas characterised by higher demand (or higher costs): then the observed difference in claim values between LVG-owned and independent practices may reflect differences in the operating environments of these practices, rather than differences in their ownership models.

3.4 To identify the effect of LVG ownership separately from that of other factors, we use an econometric approach known as ‘difference-in-differences’ (DiD), which exploits the staggered acquisition of independent practices by corporate groups over the period considered. Instead of comparing the *levels* of claim values for LVG-owned and independent practices at a given point in time, this approach compares *changes* in claim values at practices acquired by LVG5s with *changes* in claim values at independent practices around the time of each acquisition.

3.5 The key identifying assumption is that, in the absence of the acquisitions, claim values for the acquired and independent practices would have followed ‘parallel trends’, ie they would have evolved in the same way. This assumption allows us to attribute any deviations from expected trends around the acquisition to the acquisition itself. In other words, the group of practices that remain independent serves to establish a counterfactual, showing what would likely have happened to practices that were acquired if they had remained independent. We can partially validate this assumption by examining whether parallel trends existed in the pre-acquisition periods.[56]

3.6 The staggered nature of acquisitions offers an important advantage. A key challenge in DiD studies with a single event is the risk of confounding due to economic shocks that are concurrent with the change of interest. In our case, the acquisitions are staggered over a decade, from 2015 to 2024, occurring under different economic conditions.

3.7 With that in mind, we conduct several tests to assess the validity of these arguments and our identification strategy. Specifically, we first evaluate whether the parallel trends assumption holds before acquisitions. This can help determine whether changes in claim values and prices are driven by the acquisition itself or reflect pre-existing trends in the acquired practices. We also examine whether claim values and prices for acquired practices respond relatively sharply around acquisition dates.

3.8 Our identification strategy also assumes that the practices that remain independent (our control group) do not change their pricing policies in response to the acquisition of other practices in the sample. We recognise that this is a strong assumption in a context where many independent practices compete with LVG5-owned practices. If acquired practices increase prices, the independent practices that compete with these acquired practices are likely to also increase their prices, and our analysis is likely to understate the effect of acquisitions. However, if acquired practices increase both price and quality, in theory rival practices may respond by increasing or decreasing prices, depending on the characteristics of their cost base and their customer clienteles. On balance, because the documentary evidence provided by some of the LVGs, indicates that the initial price increases they applied at acquired practices are motivated by their assessment of customer willingness to pay and comparisons with competitors rather than quality improvements (paragraph A.18), we consider that such spillover effects are more likely to imply a downward bias in our estimate of the effect of acquisitions.

Model Specifications

3.9 In the terminology of panel data analysis, the data provided by the insurers is a 'repeated cross-section', which reports the value of different claims processed at different practices in different years, with some covariates providing further information on potential drivers of treatment costs. We estimate slightly different

[56] While we can examine whether acquired and non-acquired practices followed parallel trends before the acquisition, this only suggests – but does not confirm – that the assumption holds. In our context, we use a relatively long pre-acquisition period (four years) to test this. Although a longer timeframe increases the likelihood of detecting non-parallel trends, it also provides stronger reassurance that, if parallel trends hold, the assumption is likely valid.

models when using the Insurer 2 data and the Insurer 1 data, to reflect differences in the data structure and the outcome variable of interest.

3.10 When using the Insurer 2 data, this analysis focuses on claim values rather than first-year treatment costs as the dependent variable. This is because our identification strategy relies on the precise timing of acquisitions, and first-year treatment costs can aggregate claim values from two different years. The inclusion of a control variable indicating whether an observation relates to the first claim submitted for a specific condition or an ongoing claim mitigates the impact of changes in the relative value of first and ongoing claims documented in paragraph 2.20 (for completeness, we also present results using first-year treatment costs as the dependent variable in Annex B). In this setting, we model the value of a claim as follows:

$$\log\left(claim\_value_{j,i,t}\right) = \alpha_i + \alpha_t + \beta_1 LVG_{i,t} + \beta_2 X_j + \varepsilon_{j,i,t}$$

where:

$claim\_value_{j,i,t}$ is the claim value for claim j, which is processed at practice i in year t;

α_i is the fixed effect associated with practice i;

α_t is the fixed effect associated with year t;

$LVG_{i,t}$ is an indicator variable equal to one if practice i is owned by a large veterinary group (excluding Pets at Home) in year t, and zero otherwise;

X_j is a vector of attributes of claim j including: the condition treated, the breed and age of the pet, and a binary variable indicating whether the customer has submitted a previous claim for the same condition.

3.11 When using the Insurer 1 data, the dependent variable is the average price of a specific treatment type, averaged by practice-year. Covariates from previous models are excluded, and the analysis now controls solely for the specific treatment type administered.[57] We therefore model the average price of a treatment as follows:

$$\log\left(average\_price_{j,i,t}\right) = \alpha_i + \alpha_t + \beta_1 LVG_{i,t} + \beta_2 X_j + \varepsilon_{j,i,t}$$

where:

[57] Since we aggregate to one observation per practice-treatment-year, some covariates (eg pet age and breed) cannot be included in the model.

$average\_price_{j,i,t}$ is the price for treatment type j, averaged by practice i and year t;

α_i is the fixed effect associated with practice i;

α_t is the fixed effect associated with year t;

$LVG_{i,t}$ is an indicator variable equal to one if practice i is owned by a LVG in year t, and zero otherwise;

X_j is the specific treatment type administered (eg *Cerenia* for nausea, *chemotherapy,* etc).

3.12 In both models, the fixed effects for practices and years ensure that the parameter of interest (β_1) is identified from differences between changes in claim values at acquired practices and changes in claim values at independent practices, as required by our identification strategy. The covariates X_j are additional controls that can increase the precision of parameter estimates and strengthen the validity of the parallel trends assumption. In both cases, the dependent variable is log-transformed, implying that the estimated coefficient β_1 can be interpreted as the approximate percentage change in claim values caused by the acquisition (for small values of that coefficient). Standard errors are clustered at the practice level.

3.13 In our data, different practices are acquired in different years – in the terminology of causal inference research, the 'treatment' is 'staggered'. Recent research has shown that, when that is the case, standard 'two-way fixed effects' regressions do not provide valid estimates of the treatment effect (eg Callaway and Sant'Anna, 2021; Sun and Abraham, 2021). This issue, known as the 'bad comparisons' problem, arises when earlier-treated units are used as controls for later-treated units, potentially introducing bias in the presence of treatment effect heterogeneity. Therefore, in each of our main specifications, we adopt the alternative estimator developed by Sun and Abraham (2021), which ensures that acquired practices are not compared to those acquired earlier by corporate groups. Specifically, as outlined in our identification strategy, we use never-acquired practices as a clean control group (we discuss the suitability of this control group further in paragraph A.47 to A.51 in Annex A). Additionally, we exclude practices acquired before 2016 to ensure that all practices have a sufficient pre-acquisition period within our sample, which begins in 2015.

3.14 The descriptive analysis summarised in paragraph 2.40 above suggests that treatment costs vary across LVGs. For this reason, we have also estimated the models set out above 'by LVG'. For example, to estimate the causal effect of an acquisition by [✂], we have dropped observations related to claims processed by other LVG5s, and we have estimated the model set out in paragraph 3.9 above with the variable $LVG_{i,t}$ set equal to one where practice i is owned by [✂] in year

t, and zero otherwise. While this approach provides more granular insights into the causal effects of different owners, it reduces the number of observations used in the analysis, and therefore the precision of the estimates (especially for LVGs which have acquired smaller numbers of practices).

3.15 As discussed in paragraph 3.5 above, in our setting it is both important and possible to test whether the parallel trend assumption holds in the periods following each acquisition. To this end, as it is common practice, we also re-estimate the model set out in paragraph 3.9 by replacing the $LVG_{i,t}$ indicator with eight separate indicators, each representing a different year from t – 4 to t + 4 relative to the acquisition year (t = 0). For non-acquired practices, these indicators are always equal to zero. Year t – 1 is omitted, serving as a benchmark with a coefficient value of zero and no corresponding confidence interval. This model is known as a 'leads and lags' model since it estimates the effect of the treatment for each period preceding the treatment (the 'leads') and each period following the treatment (the 'lags').

Results – claim values

3.16 Our estimates of the effect of a corporate acquisition on claim values for FOPs are presented in Table 3.1. The results indicate that, when an independent practice is acquired by a LVG, claim values increase by approximately 3.2% more than for practices that are not acquired (on average across all acquired practices and all years following the acquisitions).[58] This result is statistically significant – that is, it is likely to reflect a genuine economic effect rather than noise in the data. The results for individual LVG5s suggest that the effect of an acquisition is positive and statistically significant for [at least three] LVGs bar Linnaeus.

Table 3.1: Effect of a corporate acquisition on claim values (FOPs only, all and individual LVG5s, Insurer 2 data)

[✂].

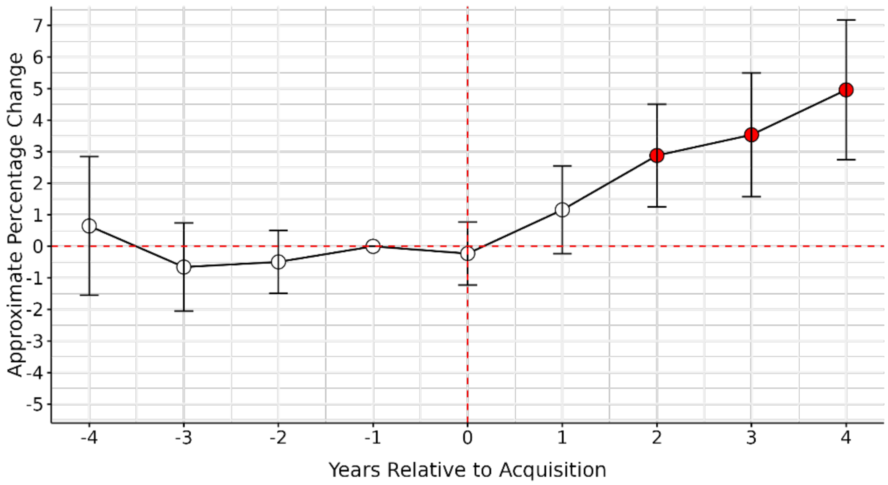
3.17 Figure 3.1 below shows the result of the 'leads and lags' specification described in paragraph 3.14, which tests the parallel trends assumption and decomposes the effect of an acquisition by year. Each dot on this chart shows the estimated difference in claim values between acquired and independent practices, relative to the difference in the year preceding each acquisition (which is normalised to zero).

[58] This figure should not be interpreted as an annual rate of change, nor does it show how the acquisition effect on claim values evolves over time. In our baseline model, described in paragraph 3.10, we consider just two periods — before and after the acquisition — and calculate an average acquisition effect using all years in each period. This approach does not reveal *when* the effect occurs or *how* it changes over time. The acquisition could cause claim values to rise immediately, increase only after a certain number of years, or grow or shrink gradually. Such patterns are not captured in this analysis. For example, if the acquisition effect grows over time, the baseline model's average would understate the actual impact observed in later years. Analysing changes on an annual basis requires a different approach, which we implement using the methodology described in paragraph 3.15 and discuss the results in paragraphs 3.17–3.18.

For example, the red dot for year 2 indicates that two years after an acquisition, the difference in claim values between acquired and independent practices has grown by approximately 3% compared to two years earlier. The vertical bars crossing the dots represent the 95% confidence interval for these estimates. One way of interpreting these confidence intervals is that when the bar does not cross the horizontal axis, one can be confident that the effect is not null and that the estimate is not merely due to noise in the data.

3.18 The chart indicates that, before each acquisition (ie in the years indexed from -4 to -1), the claim values of acquired and control practices moved in the same way. This suggests that the 'parallel trend' assumption is likely to hold throughout the observation period, ie changes in claim values at independent practices are likely to be good estimates of the changes in claim values that would have occurred at acquired practices if these had remained independent. Relative to the claim values of independent practices, the claim value of acquired practices starts to increase one year after an acquisition. After four years, the gap between the two groups has increased by 5% on average.[59]

Figure 3.1: Leads and lags results (effect on claim values, FOPs only, [at least three] LVG5s, Insurer 2 data)



Source: CMA analysis using Insurer 2 data.
Note: The figure plots the coefficient estimates together with their 95% confidence intervals. The dots are red if coefficients are significantly different from zero. Standard errors are clustered at the practice level.

3.19 [✂].

[59] This estimate of the effect in year 4 is higher than the estimate of the effect produced by our baseline model (3.2%). This is because the baseline model averages the estimated effect over all years of observation, including the early years following an acquisition when the effect is smaller.

Figure 3.2:

[✂]

Results – average prices

3.20 The results of our analysis of average prices are set out in Table 3.2 (for the average effects over post-acquisition years) and Figure 3.3 (for the annual effects over individual years).

3.21 Figure 3.2 shows that, when an independent practice is acquired by [✂] LVG, average prices increase by approximately 7.4% more than for practices that are not acquired (on average across all acquired practices and all years following the acquisitions).[60] Figure 3.3 suggests that, before each acquisition, the average prices of acquired and control practices moved in the same way. This indicates that the 'parallel trend' assumption is likely to hold throughout the observation period, that is, changes in average prices at independent practices are likely to be good estimates of the changes in average prices that would have occurred at acquired practices if these had remained independent. Relative to the average prices of independent practices, the average prices of acquired practices start to increase in the year when the practice is acquired. After four years, the gap between the two groups has increased by 9.2% on average. [✂].[61]

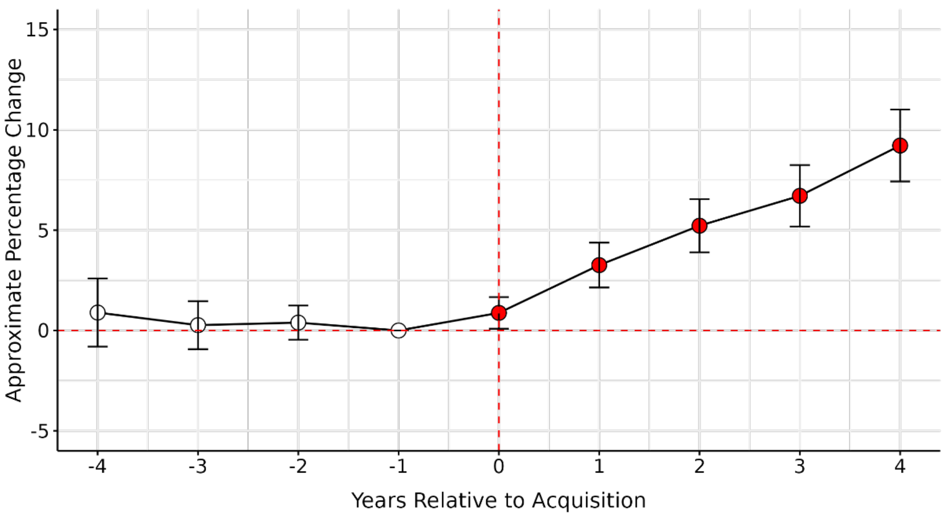
[60] The interpretation of this result follows the same logic as explained in footnote 58: it represents an average effect across all years before and after the acquisition and should not be interpreted as an annual rate of change. We analyse changes on an annual basis in Figures 3.3 and 3.4.
[61] [✂].

Table 3.2: Effect of a corporate acquisition on average prices (FOPs only, all and individual LVG5s, Insurer 1 data)

	All	[✂]	[✂]	[✂]	[✂]	[✂]
LVG	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]
Treatment-type FEs	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]
Practice FEs	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]
Year FEs	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]
Observations	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]
Adjusted R^2	[✂]	[✂]	[✂]	[✂]	[✂]	[✂]

Source: CMA analysis using Insurer 1 data.
*Note: Standard errors are clustered at the practice level and are reported in parentheses. ***, **, and * correspond to statistical significance at the 1%, 5%, and 10% levels, respectively.*

Figure 3.3: Leads and lags results (effect on average prices, FOPs only, [at least three] LVG5s, Insurer 1 data)



Source: CMA analysis using Insurer 1 data.
Note: The figure plots the coefficient estimates together with their 95% confidence intervals. The dots are red if coefficients are significantly different from zero. Standard errors are clustered at the practice level.

3.22 Figure 3.4 below shows the DiD estimates for individual LVGs. Consistently with what is shown before for claimed values, these results suggest that the effect of an acquisition is positive and statistically significant for [✂] [at least three] LVGs bar Linnaeus. [✂].

Figure 3.4: Leads and lags results (effect on average prices, FOPs only, individual LVGs, Insurer 1 data)

[✂].

3.23 Table 3.3 shows the DiD estimates for the individual treatment categories used by Insurer 1. These results suggest that the effect of an acquisition varies across different treatment categories. It is the highest for drugs (8.2%), followed by surgeries (7.6%), consultations (7.2%) and diagnostics (5.5%) while not significant for hospitalisations and therapy, and negative and significant for supplements (-12.5%).[62] It is important to note that hospitalisations, therapy, and supplements have a substantially lower number of observations, which limits the robustness of the estimates for these categories

Table 3.3: Effect of a corporate acquisition on average prices (individual treatment categories, FOPs only, all LVG5s, Insurer 1 data)

	Drugs	Diagnostics	Consultations	Surgeries	Hospitalisations	Therapy	Supplement
LVG	0.082*** (0.008)	0.055*** (0.016)	0.072*** (0.014)	0.076*** (0.028)	-0.037 (0.029)	0.080 (0.050)	-0.125*** (0.038)
Treatment-type FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Practice FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	796,120	106,231	38,283	40,577	15,231	9,360	12,936
Adjusted R^2	0.642	0.615	0.680	0.813	0.495	0.618	0.630

Source: CMA analysis using Insurer 1 data.
*Note: Standard errors are clustered at the practice level and are reported in parentheses. ***, **, and * correspond to statistical significance at the 1%, 5%, and 10% levels, respectively.*

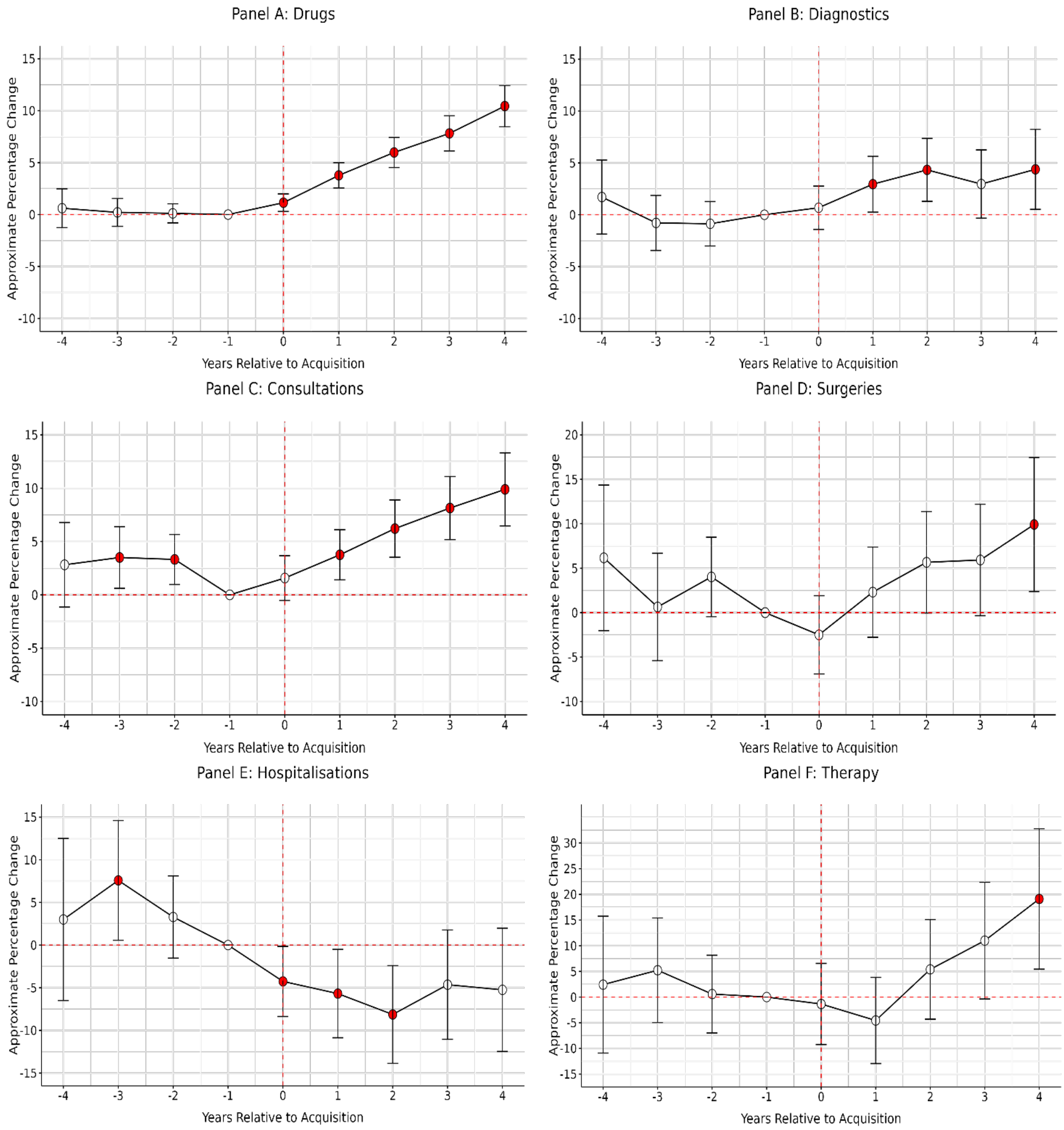
3.24 While we do not have any data on the costs incurred by suppliers to provide these different treatments, it seems unlikely that the effect of corporate acquisitions on average prices estimated in Table 3.3 could be explained by systematic increases in unit costs. If anything, it seems plausible that LVGs might achieve lower costs in some areas (for example drugs) due to economies of scale and higher bargaining power.

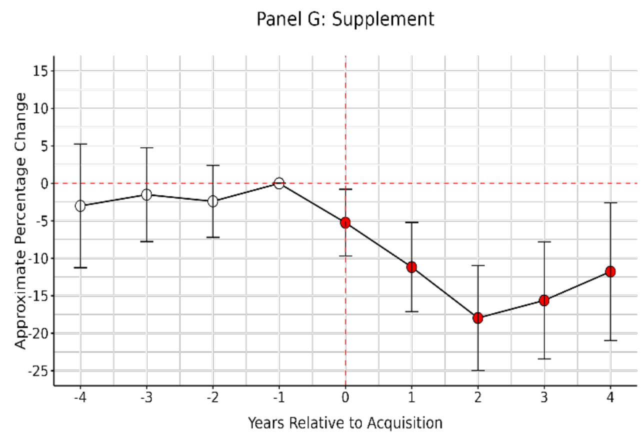
3.25 Figure 3.5 below presents the individual 'leads and lags' coefficients for each treatment category. Overall, this figure aligns closely with the results shown previously in Table 3.3. It suggests that the effect of an acquisition on average prices is positive and statistically significant for drugs, diagnostics, and consultations. After four years, the gap between acquired and independent practices has increased by 10.4% for drugs, 9.9% for consultations, and 4.4% for diagnostics. For drugs and diagnostics, the lead coefficients are not statistically different from zero, suggesting that the parallel trends assumption is likely valid for these categories. In contrast, for consultations, the lead coefficients at periods -3 and -2 are positive and statistically significant, suggesting that these results may be less reliable, as the parallel trends assumption may not hold. There is no strong

[62] The interpretation of these results follows the same logic as explained in footnote 58: they represent an average effect across all years before and after the acquisition and should not be interpreted as annual rates of change. We analyse changes on an annual basis for individual treatment categories in Figure 3.5.

evidence of an acquisition effect for surgeries and therapies, although the coefficient at period +4 is positive and statistically significant. For supplements, the acquisition effect seems to be negative.

Figure 3.5: Leads and lags results (effect on average prices, individual treatment categories, FOPs only, all LVG5s, Insurer 1 data)





Source: CMA analysis using Insurer 1 data.
Note: The figure plots the coefficient estimates together with their 95% confidence intervals. The dots are red if coefficients are significantly different from zero. Standard errors are clustered at the practice level.

Robustness checks

3.26 We have conducted a wide range of robustness checks in response to points raised by the LVGs, with the results reported in Annex A and Annex C. Specifically, we demonstrate that our results remain robust when we (i) remove control variables from the regressions; (ii) estimate the acquisition effect using the Callaway and Sant'Anna estimator; (iii) use an alternative control group—specifically, not-yet-acquired practices rather than never-acquired ones; (iv) exclude the COVID-19 pandemic period (COVID-19); and (v) apply the confidence sets generated by the Rambachan and Roth (2023) method to assess the validity of the parallel trends assumption. Additionally, we show that the estimated effects are even stronger when we weight observations by value, thereby giving greater importance to treatments that are likely to have a bigger impact on customer welfare. We also run the analysis for prices focusing on a single drug—Metacam—while controlling for key drug characteristics that may affect its price. The results hold in this specification as well, providing further reassurance that our main findings are not purely driven by compositional effects within treatment categories.

3.27 Further robustness checks are presented in Annex B. We show that expanding the sample to include referral centres produces DiD results of a similar order of magnitude to those estimated for FOPs only. We also confirm that our results for claim values hold when using a dataset with an explicit 'panel structure' rather than a 'repeated cross-section'. Specifically, we conduct a DiD analysis on panel data where the dependent variable is the average claim value at: (i) the practice-year level; and (ii) the practice–condition–year level. In both cases, the results remain consistent with those reported previously. We also re-estimate the model used for claim values, using first-year treatment costs as the dependent variable (the rationale and methodology underpinning this metric is set out in paragraph 2.20), and find that the results are again broadly consistent with our baseline specification.

3.28 We further show that our DiD results for prices hold when using all treatment observations, rather than average prices per treatment category, and controlling for covariates, although the estimated effect is smaller at 5.5%.[63] In addition, we show that when we restrict the sample to treatments where the start and end dates are identical— meaning they are more likely to include only a single invoice, as the treatment typically occurs within a single day or visit—the results for Insurer 2 and Insurer 1 are more consistent in terms of magnitude. Specifically, the estimated acquisition effect on claim values increases from 3.2% to 7%, which is much closer to the estimate obtained for average prices (7.4%). Finally, we re-estimate our claim value model using new data recently obtained from Insurer 1, which includes all their claim submissions [✂] and find that the DiD results remain broadly consistent with our previous findings.

[63] This reduction in magnitude is expected, as using all observations gives more weight to more frequently occurring treatments. If these treatments are of low value, changes in their prices may not have a particularly large bearing on customer welfare. Moreover, because these treatments are purchased more frequently, customers may be better informed about their prices, and veterinary practices may have less incentive to increase prices in these cases.

Annex A: Assessment of LVG submissions

A.1 We shared a preliminary version of this appendix (referred to hereafter as the working paper) with the LVGs' advisers (in a confidentiality ring) on 13 December 2024, and a redacted version with the LVGs themselves on 15 January 2024. We have received responses from the six LVGs and/or their advisers, which we considered carefully and, where appropriate, used to revise our methodology and analysis presented in the main body of the appendix.[64] This annex to the appendix summarises our assessment of these responses.

A.2 Four LVGs have submitted that our analysis is technically flawed, such that it overstates the effect of corporate acquisitions, and is difficult to interpret, primarily because it does not control for quality.

A.3 Linnaeus submitted a brief response, in which it emphasised that it does not have a policy to increase prices following acquisitions, and therefore the CMA's finding that treatment costs have not increased following its acquisitions is not surprising.[65] Linnaeus also stated that it made significant investments in quality in its practices, and that it is not possible to properly assess the impact of acquisitions by Linneaus, and most likely by other LVGs, without due consideration of quality benefits arising from ownership by larger groups. We do not discuss Linnaeus' response further in this annex.

A.4 Pets at Home (**PaH**) emphasised that it has not acquired any practices over the period of observation, and therefore our analysis of the impact of corporate acquisitions is not relevant to it.[66] [✂].

A.5 This annex focuses on the comments related to the accuracy of our analysis and the direct interpretation of the results, rather than to any issues with wider interpretation or how the results fit into our inquiry. We start by discussing common themes raised by several LVGs, before moving to more detailed points raised by individual LVGs.

A.6 In this annex, we also conduct several tests to address points raised by the LVGs and their advisers. In some cases, the results shown here differ from those in the main body of the appendix, as the current version incorporates multiple adjustments suggested by the advisers that we considered appropriate, and the final results therefore reflect the combined effect of these changes. In addition, when revising the draft working paper, we updated certain terminology and the

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64 Specifically, we made the following adjustments: [✂].
65 Linnaeus response to the draft working paper on the impact of corporate acquisitions on treatment costs, Linnaeus also stated that it made significant investments in quality in its practices, and that it is not possible to properly assess the impact of acquisitions by Linneaus, and most likely by other LVGs, without due consideration of quality benefits arising from ownership by larger groups.
66 PaH response to the draft working paper on the impact of corporate acquisitions on treatment costs.

numbering of paragraphs and tables. For example, we replaced the term 'unit prices' with 'average price', and some tables cited in the LVGs' responses may now have different numbers. As a result, some points in the LVGs' responses may no longer be relevant if they refer to terminology, results, or paragraph references that have since been updated in response to their comments.

Common themes

No control for quality

A.7 All LVGs stated that they seek to improve quality at the practices they acquire (in terms of clinical capability or service quality), and that they seek to position themselves as offering higher-quality care and outstanding customer service. Many respondents stated that, since the CMA's analysis did not incorporate any controls for quality, it was impossible to conclude on whether corporate acquisitions worked to the benefit or detriment of customers and their pets overall. For example CVS stated that, to the extent that increases in prices or claim values represent improvements in the quality and quantity of treatment available, and that customers have chosen that more modern and effective set of treatments, then these increases would represent a consumer benefit rather than a cost.[67]

A.8 LVGs put forward numerous examples of investments and changes in management practices designed to increase quality. These examples generally fell into the following three categories: staffing and training, equipment, and access to central resources and staff. Below we summarise some of the submissions made under these three headings.

Staffing and training

A.9 CVS submitted that most CVS acquisitions will be followed by the recruitment of additional staff – usually vets and nurses – with recruitment starting immediately post-acquisition and an expected 3-6 month lead time until these staff are in place. CVS will generally offer better training and support to veterinary professionals and the wider staff team than independent FOPs are able to offer their staff.[68]

A.10 VetPartners stated that, compared to independents, it provides more favourable employment terms and conditions, and provides better working conditions for veterinary professionals to ensure their retention and the long-term stability of practices. VetPartners also invests in clinical training to improve decision-making and encourage greater veterinary discretion.[69] This implies that VetPartners FOPs

[67] CVS response to the draft working paper on the impact of corporate acquisitions on treatment costs.
[68] CVS response to draft working paper.
[69] Vet Partners response to the draft working paper on the impact of corporate acquisitions on treatment costs.

may be offering a larger number of complex and more expensive treatments in house.

A.11 IVC told us that it has invested heavily in clinical staff pay and benefits in recent years, which also benefits staff at IVC's newly acquired practices. IVC has also implemented an extensive and rigorous training and development programme to ensure that clinical excellence is maintained, including a graduate programme to train the next generation of veterinary professionals. [✂].[70]

Equipment and systems

A.12 CVS submitted that, where an acquired practice is not bound into long-term onerous equipment leases, within three months of acquisition CVS will seek to replace the entire suite of lab equipment with its own diagnostic lab machines ("MiLab") and to ensure that the new equipment includes at least one item that the practice did not previously have on site. Having a uniform set of equipment allows CVS to operate more efficiently but also ensures that the site has access to modern and effective laboratory equipment that has been through CVS's thorough clinical governance process. CVS will also often seek to add new diagnostic equipment (it estimates in at least half of acquisitions) – usually within the first year of ownership. This could include for example dental x-ray, better ultrasound equipment, improved endoscopic equipment, CT scanners – all of which are expensive capital investments less commonly found at independent practices.[71]

A.13 IVC said that it often invests in the facilities and equipment of its newly acquired practices. [✂] – demonstrating the additional investment made in newly acquired practices. IVC also introduces a range of features designed to improve the digital client experience (for example digital appointment reminders, improved email communication, and new practice websites).[72]

A.14 One LVG stated that, compared to independents, it has more professional management systems in place and greater financial resources to make quality-enhancing investments in practices.[73] The LVG states that a number of quality improvements can be implemented in a relatively short period of time, for example those required to ensure compliance with the Practice Standards Scheme, or the purchase of equipment that could be introduced in short order.[74]

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[70] [✂].
[71] CVS response to draft working paper.
[72] IVC narrative response to draft working paper.
[73] [✂] response to draft working paper.
[74] [✂] response to draft working paper.

A.15 Another LVG submitted that [✂] have made investments in training and equipment which enables more advanced treatments to be offered in a FOP setting (up to CT imaging).[75]

Access to central resources and staff

A.16 CVS submitted that independent veterinary practices joining the CVS group will also benefit from centralised resources that help them to correctly diagnose and effectively treat pets – particularly in the more unusual/difficult cases that an individual veterinary surgeon will see less often, and where peer support can add particularly significant value. This includes access to training (such as CVS's "Knowledge Hub") as well as to an internal system of expert advice (via "Vet Oracle") and the support of Regional Clinical Directors.[76]

A.17 IVC said that, immediately on joining IVC, colleagues at newly acquired practices are provided with access to a wide range of clinical support and resources, including access to the wider IVC clinical network, clinical boards and a quality improvement team that can provide vets with tailored advice and support to inform their clinical decision making. IVC also provides practical support and guidance to practices on how to deliver exceptional client service, including guidance with advice, tips and practice checklists across all aspects of client service.[77]

A.18 **CMA assessment.** We do not dispute that LVGs often invest resources in the practices they acquire. To understand the relative impact of these investments on pricing decisions, we reviewed the internal documents at our disposal where LVGs discussed pricing adjustments post acquisitions. We had such documents from three of the LVGs, and they can be summarised as follows:

(a) [✂].[78]

(b) [✂].[79]

(c) [✂].[80] [81] [82]

A.19 Overall, these documents indicate that the price increases implemented following these acquisitions were motivated by the new owners' assessment of customers' willingness-to-pay and comparisons with market benchmarks, rather than by quality improvements. In addition, the LVGs have not provided evidence that their investments in quality — to the extent such investments have been made —

[75] [✂] response to the draft working paper on the impact of corporate acquisitions on treatment costs [✂].
[76] CVS response to draft working paper
[77] IVC narrative response to draft working paper
[78] [✂].
[79] [✂].
[80] [✂].
[81] [✂].
[82] [✂].

partially or fully explain the scale of the price increases and the price differentials with independent practices that we observe.

A.20 Moreover, what matters for the interpretation of our DiD results is not whether LVGs have increased quality at acquired practices compared to pre-acquisition levels, but whether they have increased quality at acquired practices at a faster rate compared to non-acquired practices (which serve as the control group). Some independents have told us that over this period they have also introduced a number of changes to improve their quality, for example by lengthening the duration of consultations, offering better conditions to staff, or using external specialist vets to offer more advanced treatments at their premises. In fact, we have evidence that customers often perceive the quality of independent practices as higher than LVGs. Specifically, in our pet owner survey, pet owners reported significantly higher satisfaction with independent practices than with LVGs for the quality of information and advice, the quality of service, the outcome of their visit, and the cost of service.[83]

A.21 Taken together, this evidence suggests that quality improvements at acquired practices are unlikely to explain the results of our analysis.

The comparison of unit prices is not made on a like-for-like basis

A.22 Three responses have pointed out that the treatment categories we use in our analysis of unit prices are not homogeneous. That is, observations in a given category can display significant variation, both in terms of the type of items included and the unit prices of these items. This creates the risk that our analysis is misinterpreting changes in treatment mix as changes in unit prices. Two of these responses point out that this issue also applies to the drugs categories, within which there is variation due to differences in quantities or modes of injection.

(a) IVC submitted that, because the CMA's analysis of unit prices is based on average treatment prices within treatment groups, the CMA's analysis potentially involves making comparisons of average treatment prices that are not on a like-for-like basis.[84] IVC pointed out that this issue also applies to drugs, a category where in principle it should be possible to ensure that comparisons are made on a like-for-like basis. This is particularly the case for the drugs category labelled [✂] which represents [20-30%] of all claims for drugs.[85]

(b) CVS submitted that the data on which the 'unit prices' analysis is based is not sufficiently granular to allow like-for-like comparisons to be made over time or

[83] Overview of our working papers.
[84] IVC response to the draft working paper on the impact of corporate acquisitions on treatment costs.
[85] IVC response to draft working paper

across FOPs. This is because there is significant heterogeneity in item types and prices within most treatment categories. This results in a serious risk that 'mix issues' drive much of the working paper's findings (ie that LVG customers are, through choice, shifting towards a more expensive mix of treatments, relative to the typical independent FOP customer).[86] CVS points out that this issue also applies to drugs categories, due to significant heterogeneity associated with differences in dosage, pack sizes, methods of delivery, and animal. Moreover [10-20%] of the underlying data in this category is in the general category [✂] which includes a wide range of medications.[87]

(c) [✂] submitted that the Insurer 1 insurance data lacks the necessary granularity to reflect the actual variety of treatments provided. While the Insurer 1 insurance data has greater ability than the Insurer 2 data to control for treatment, the underlying data is too aggregated to allow effective controls, and will assign different treatments, with different underlying costs, as if they were the same.[88]

A.23 **CMA assessment.** The categorisation of treatments that we used in our analysis was provided by Insurer 1 [✂]. It is correct that the observations in many treatment categories display significant heterogeneity, both in terms of the type of items included [✂] and in terms of the unit prices of these items. On that basis, it is correct that the results of our analysis are best interpreted as showing the impact of acquisitions on average prices within treatment categories, rather than the effect on unit prices for more tightly defined treatments. However, this does not invalidate our analysis or conclusions, for the following reasons.

A.24 First, any within-category heterogeneity will only affect our estimates of the acquisition effect to the extent that the mix of treatments applied within categories systematically changes after acquisitions. If that is not the case, then within-category heterogeneity does not affect the expected value of the coefficients (in econometric terms, it increases the variance of the estimator but not its expected value).

A.25 Second, even if acquisitions do have an effect on within-category treatment mixes and this does affect our estimates, we consider that, under most circumstances, the resulting estimates will remain economically meaningful. For example, if, within a given category, acquired practices start substituting away from cheaper items and towards more expensive items, then this is an economically meaningful effect that we would like to capture as part of our analysis. It may be the case that the change in treatment mix is associated with a change in treatment quality, but as

[86] CVS response to draft working paper.
[87] CVS response to draft working paper.
[88] [✂] response to the draft working paper on the impact of corporate acquisitions on treatment costs.

noted above, the evidence we have on quality suggests that improvements in quality at acquired practices are unlikely to explain the results of our analysis.

A.26 Third, we have seen some evidence suggesting that LVGs are less prone to waive charges for small items than independents.[89] If this is correct, then this might bias our assessment of the acquisition effect downward. For example, if, when invoicing for a consultation, most independents charge only for the consultation but acquired practices start charging for small ancillary items (for example for consumables or the disposal of clinical waste) which are still classified by Insurer 1 as consultation items, then the average fee for items categorised as consultations for acquired practices will go down, even if the main fee for the consultation goes up.

A.27 Fourth, while it is correct that even the drugs categories used by Insurer 1 exhibit some variation in the types and values of items listed, much of this variation appears to be associated with differences in the quantities prescribed and the modes of administration. Respondents have not put forward a mechanism by which acquisitions would be associated with systematic increases in the quantities of drugs prescribed, or with systematic moves towards costlier modes of administration, which would bias our estimates of price increases upwards. It is reasonable to assume that these two dimensions of prescribing decisions (dosage and mode of administration) are tightly driven by clinical considerations and offer less scope than other categories to increase quality of treatment.

A.28 Notwithstanding the above considerations, we agree that this issue is made more acute by the presence of generic categories labelled as [✂] which can encompass broad ranges of different treatments. We have therefore excluded these categories from the sample prior to estimation and revised the analysis presented in the main body of this appendix accordingly.[90] By excluding treatments classified as [✂] we have removed 8,514,930 observations from the dataset. By implementing this change, the estimated acquisition effect on unit prices decreases from 8.0% to 7.8%.

A.29 In response to concerns that our average price comparisons may not be made on a like-for-like basis due to heterogeneity within treatment categories (as outlined in paragraph A.22), we ran a sensitivity test focusing on a single drug —Metacam — and controlled for key drug characteristics that affect its price, in order to further isolate any potential bias introduced by within-category variation. We selected Metacam, a commonly prescribed anti-inflammatory used for pain and inflammation, because there are relatively few variables that affect its price — primarily bottle size, quantity prescribed, and administration method — and these are described in a relatively clear and consistent way by vets in the free-text fields

[89] CMA working paper on business models, provision of veterinary advice and consumer choice, 2 February 2025, paragraphs 2.41 and 2.42.
[90] [✂].

when submitting claims. This made it easier to extract and control for the relevant information in the analysis. Metacam accounts for [✂] of total drug claims in 2023 and is the [✂] significant drug by total claimed value in the dataset.

A.30 We estimated a DiD model for Metacam prices, incorporating controls for key drug and animal characteristics that could influence its price. These included bottle size, quantity prescribed, animal type (dog, cat, or other), and administration method (oral vs. injection). This information was extracted from the [✂] claims. Observations for Metacam tablets were excluded due to limited data and differing drug characteristics, and we also excluded records with missing quantity information.[91] The model additionally controls for the type of condition treated.

A.31 Using this more granular approach, which is more likely to support a like-for-like comparison, we continued to find a similar acquisition effect on prices, suggesting that our broader findings are unlikely to be solely the result of changes in drug mixes or prescriptions. As shown in Table A.1 below, the estimated acquisition effect for Metacam remains positive and statistically significant after controlling for these key price determinants. Notably, the estimated acquisition effect in a model without any control variables is only 1.3 percentage points lower, decreasing from 10.7% to 9.4%. We acknowledge, however, that this result is based on a single drug and may not necessarily be generalisable to other treatments or categories. Nevertheless, it provides additional reassurance that our main findings are not purely driven by compositional effects within treatment categories.

Table A.1: Effect of acquisitions on Metacam prices, controlling for drug characteristics (FOPs only, all LVGs, Insurer 1 data)

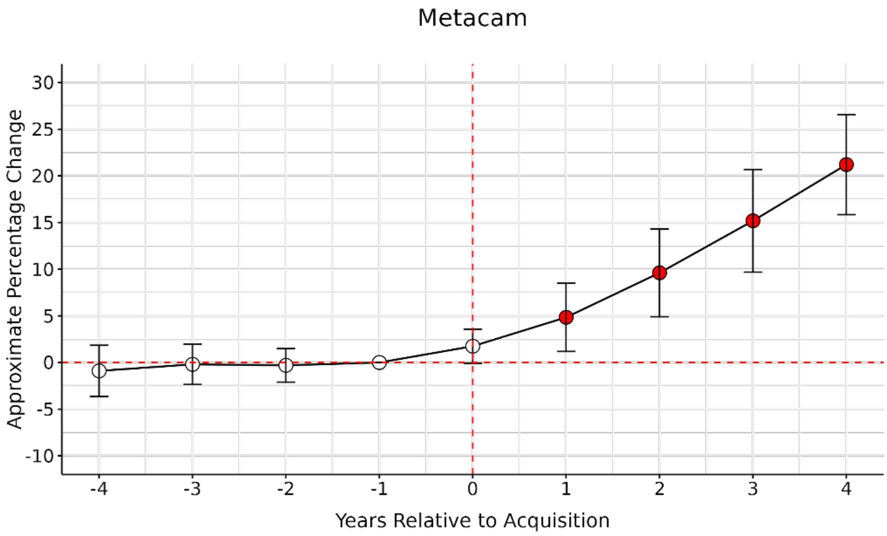
	Log(Price)	
	(1)	(2)
LVG	0.094***	0.107***
	(0.023)	(0.019)
Bottle size (ml)		0.010***
		(0.000)
Quantity		0.018**
		(0.009)
Cat		0.255***
		(0.043)
Dog		0.172***
		(0.040)

[91] Results remain similar when including observations with missing quantity information and estimating the DiD model without controlling for quantity. We also obtain consistent results when replacing missing quantities with 1, on the assumption that the information is missing because only one unit was prescribed.

Oral administration		0.175***
		(0.044)
Condition-type FEs	No	Yes
Practice FEs	Yes	Yes
Year FEs	Yes	Yes
Observations	167,844	167,726
Adjusted R^2	0.191	0.714

Source: CMA analysis using Insurer 1 data.
*Note: Standard errors are clustered at the practice level and are reported in parentheses. ***, **, and * correspond to statistical significance at the 1%, 5%, and 10% levels, respectively.*

Figure A.1: Impact of acquisitions on Metacam prices, controlling for drug characteristics (FOPs only, all LVGs, Insurer 1 data)



Source: CMA analysis using Insurer 1 data.
Note: The figure plots the coefficient estimates together with their 95% confidence intervals. The dots are red if coefficients are significantly different from zero. Standard errors are clustered at the practice level.

The comparison of claim values is not made on a like-for-like basis

A.32 Three LVGs pointed out that our analysis of claim values could not control for the treatment applied, and that the estimated increase in claim values at acquired practices could reflect a move towards more advanced treatments.

(a) IVC submitted that, because the Insurer 2 data does not have information on the treatments applied to treat a condition, and because a given condition can be treated using multiple different treatments, there is a risk that the CMA's analysis is misinterpreting changes in treatment mix as changes in prices.[92]

[92] IVC response to draft working paper.

(b) Vet Partners pointed out that the Insurer 2 data does not make it possible to differentiate between the different treatments that pets receive for a given condition. If LVG-owned practices are more likely to offer advanced treatments, then their average claim values will be higher than those of independents. Vet Partners estimated that, in 2023, LVG-owned practices exhibit a higher propensity to submit claims for advanced treatments than independent practices (for a list of 10 advanced treatments provided by VetPartners).[93]

(c) [✂] stated that the Insurer 2 insurance data only identifies the condition that is being treated and provides no information on the treatment provided in relation to this condition. This prevents the CMA from adequately controlling for treatment mix. This is important because the same condition may have different treatments applied because of the needs of the pet, and those treatments may differ substantially in their cost to provide.[94]

A.33 **CMA assessment.** We do not consider that the lack of controls for treatments is a limitation of this analysis. The purpose of the analysis of claim values and first-year treatment costs is to estimate the effect of an acquisition on the total cost of a course of treatment, rather than on the average prices of individual treatment categories (which is the object of a separate piece of analysis using the Insurer 1 data). An acquisition may affect the cost of a course of treatment by changing the mix of treatments applied, the unit prices of individual treatments, or both. Controlling for the mix of treatments would essentially 'shut down' the first of these channels, which would not be meaningful. We recognise that this analysis does not control for possible changes in the quality of care provided by LVGs, but as noted in paragraphs A.18 to A.19 above, we still consider that the findings are relevant for our investigation.

A.34 As a secondary point, we consider that, if the more advanced treatments examined by one LVG are invoiced relatively infrequently, then evaluating their availability at different practices by considering their occurrence in the Insurer 1 data (for a single year) might possibly overstate the difference between LVG-owned and independent practices. [✂]. To be clear, we are not disputing the finding that LVG-owned practices are more likely to offer a broader range of treatments, but we think that the difference may be overstated by this analysis of the insurer data.

[93] VetPartners supplementary response to the draft working paper on the impact of corporate acquisitions on treatment costs, [✂]
[94] [✂] response to draft working paper.

Trimming outliers

A.35 Three LVGs have submitted that the analysis should exclude observations in the top and bottom parts of the distributions of claim values and treatment costs.

(a) Vet Partners submitted that there are some unusually high claim values in the sample, and that it is likely that some of these observations result from data errors. These values are highly unusual and unlikely to be comparable with the rest of the data. Vet Partners proposes to exclude observations in the top and bottom 5% of the distribution in each year and finds that this reduces the estimate of the acquisition effect from 3.8% to 2.3% (in conjunction with the exclusion of covariates from the model),[95] and reduces the estimate of the acquisition effect on unit prices from 8.0% to 5.6%.[96]

(b) IVC also proposed to trim the top and bottom 5% of claim values within each condition-year combination. This is meant to facilitate like-for-like comparisons, and to lessen the impact of extreme claim values on the estimates, insofar as they are likely to relate to atypical treatments or simply reflect errors in the data. This sensitivity reduces the estimated effect of acquisitions on claim values from 3.8% to 1.9%.[97] IVC suggested the same adjustment with respect to our analysis of unit prices, in which case the impact is to reduce the estimate of the acquisition effect from 8.0% to 5.3%.[98]

(c) [✂] submitted that the presence of very high claim values in the Insurer 2 data for the three commonest conditions indicates that the dataset includes extreme outliers. If these outliers are not properly identified and removed, they distort the results of any subsequent statistical analyses conducted with the data.[99]

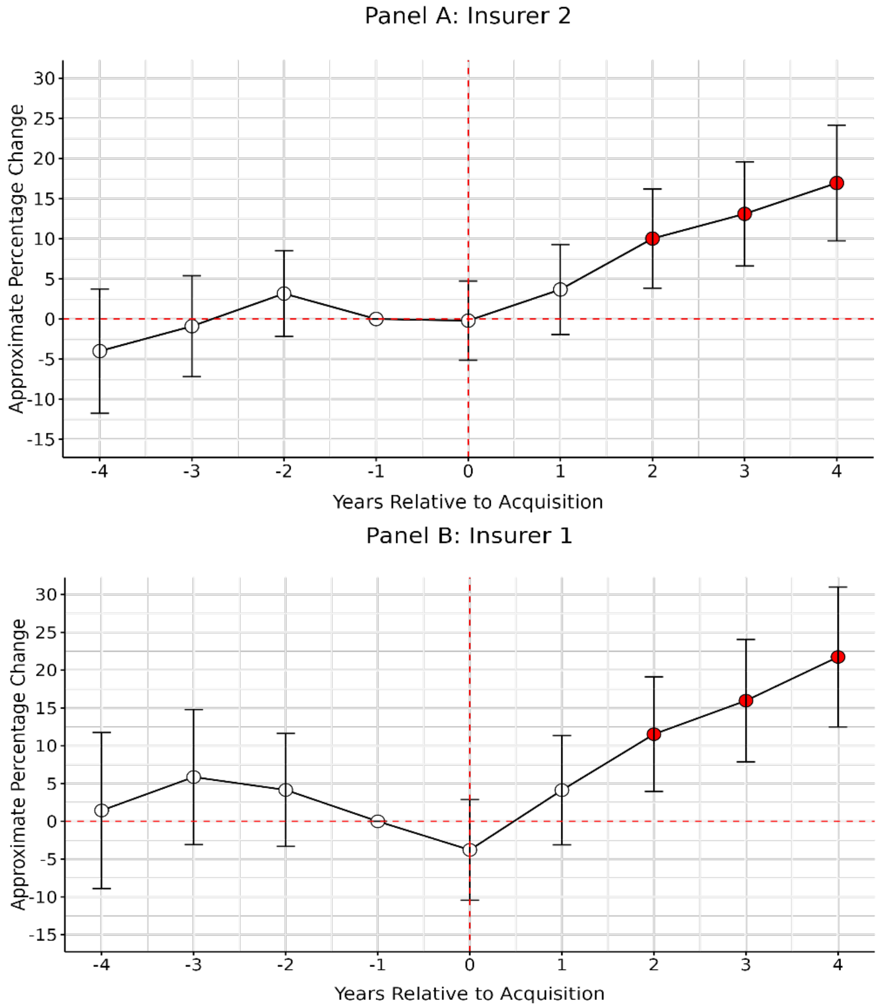
A.36 **CMA assessment.** We do not consider that trimming the top and bottom 5% of the distributions is a valid specification, for two reasons.

A.37 First, if, upon acquisition, a practice becomes more likely to issue very expensive claims, or less likely to issue very cheap claims, this should be considered as economically meaningful, and part of the hypothesis that we are seeking to test, rather than an 'anomaly' that must be corrected. Indeed, Figure A.2 below shows that, upon acquisition, a practice becomes significantly more likely to issue claims

[95] VetPartners response to the draft working paper on the impact of corporate acquisitions on treatment costs, 17 January 2025.
[96] VetPartners response to draft working paper.
[97] IVC response to draft working paper.
[98] IVC response to draft working paper.
[99] [✂] response to draft working paper.

within the top 5% of the distribution, and to charge for treatments within the top 5% of the distribution.[100]

Figure A.2: Impact of acquisition on proportion of top 5% claim values (FOPs only, all LVGs, Insurer 2 and Insurer 1 data)



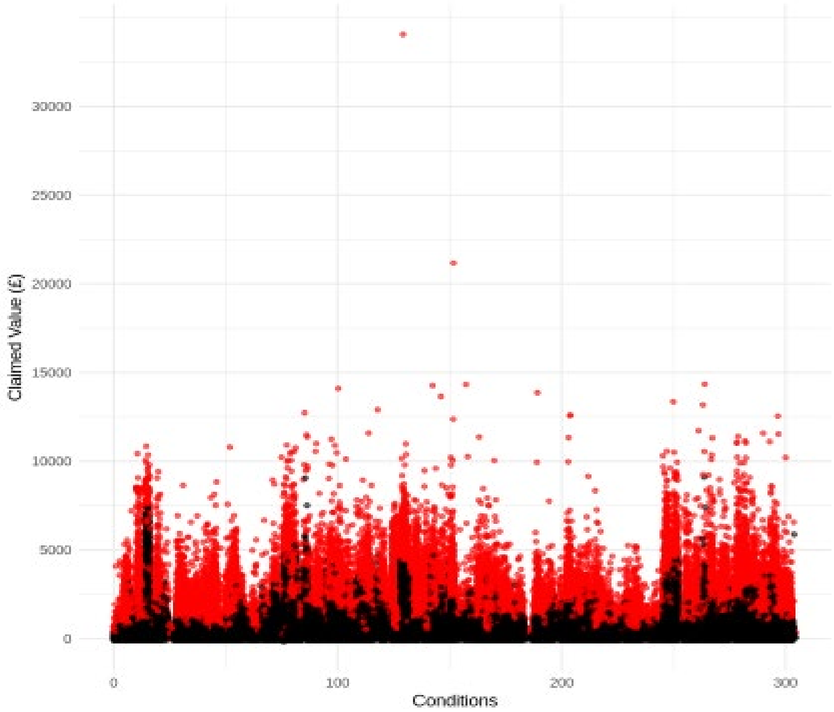
Source: CMA analysis using Insurer 2 and Insurer 1 data.
Note: The figure plots the coefficient estimates together with their 95% confidence intervals. The dots are red if coefficients are significantly different from zero. Standard errors are clustered at the practice level.

A.38 Second, while we recognise that some observations with very high or very low values might reflect data-entry mistakes, there is no reason to assume that all claims in the top and bottom 5% of the distribution reflect such mistakes. By way of illustration, Figure A.3 shows the distribution of claim values by condition in the Insurer 2 dataset, with claims that would be excluded under a 5% cutoff highlighted in red. Two extreme outliers above £20,000 stand out from the rest of the distribution, but the remaining claims are more densely concentrated between £0 and £15,000, with a substantial number of claims falling between £10,000 and

[100] In this analysis, the dependent variable is the log-transformed share of claims/treatments issued by a practice that fall within the top 5% of claims for each condition-year combination, using log(1 + x) to account for zero values. The results are broadly consistent when using either log(x) (excluding zeros) or the raw share without transformation.

£15,000. There is no prima-facie reason to consider these claims as outliers or errors.

Figure A.3: Distribution of claim values by condition (FOPs only, all LVGs, Insurer 2 data)



Source: CMA analysis using Insurer 2 data.
Note: Claims that would be excluded under a 5% cutoff are highlighted in red and represent 132,196 observations, while the black dots represent 2,511,689 observations.

A.39 We test different cutoffs, selecting claims within the following ranges: (1) £1 to £20,000, (2) £1 to £10,000, and (3) £1 to £5,000. Additionally, we apply a narrower range than the 5-95% cutoff proposed by respondents and use the 0.1% and 99.9% percentiles within each condition-year combination, which correspond to £39.6 and £2,936, respectively (averaged across all observations). As shown in Table A.2, our results remain largely unchanged when extreme observations are excluded. Specifically, when excluding claims above £5,000, the estimated coefficient remains at 0.037, just 0.001 lower than our original estimate of 0.038. Similarly, when using the 0.1% and 99.9% percentiles within each condition-year combination, the coefficient is 0.035, just 0.003 lower than our original estimate. These results indicate that excluding extreme values does not significantly alter our findings.

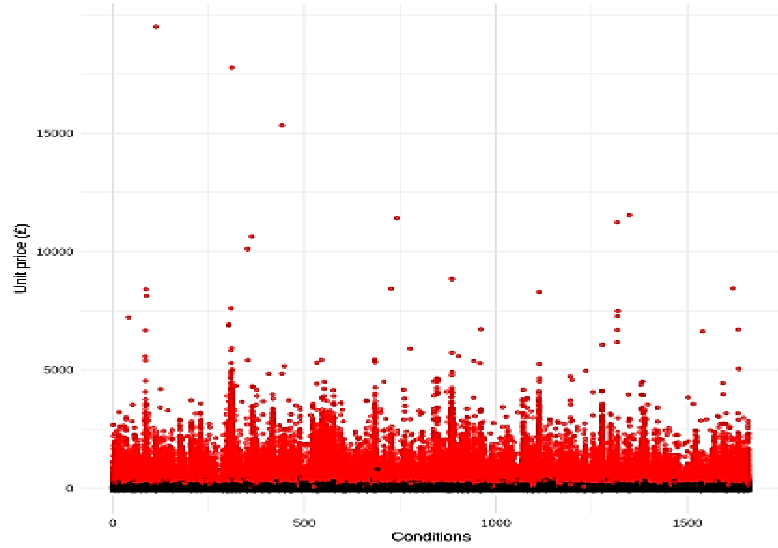
Table A.2: Effect of acquisitions on claim values after trimming outliers (FOPs only, all LVGs, Insurer 2 data)

	£1 – £20,000	£1 – £10,000	£1 – £5,000	0.1% – 99.9%
LVG	0.037*** (0.007)	0.037*** (0.007)	0.037*** (0.007)	0.035*** (0.007)
Condition-type FEs	Yes	Yes	Yes	Yes
Practice FEs	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes
Observations	2,643,308	2,643,245	2,640,730	2,634,047
Adjusted R^2	0.322	0.322	0.320	0.324

Source: CMA analysis using Insurer 2 data.
*Note: Standard errors are clustered at the practice level and are reported in parentheses. ***, **, and * correspond to statistical significance at the 1%, 5%, and 10% levels, respectively.*

A.40 Figure A.4 illustrates the distribution of treatment prices by condition in the Insurer 1 dataset, with prices that would be excluded under a strict 5% cutoff highlighted in red.[101] By removing the top 5% of prices within each condition-year, we eliminate all claims with a price above £133 on average—a cutoff that again appears too low—and exclude 1,109,627 observations in total.

Figure A.4: Distribution of treatment prices by condition (FOPs only, all LVGs, Insurer 1 data)



Source: CMA analysis using Insurer 1 data.
Note: Claims that would be excluded under a 5% cutoff are highlighted in red and represent 1,109,627 observations, while the black dots represent 21,087,694 observations.

A.41 As shown in Table A.3, our original coefficient of 0.080 decreases by 0.004 when using observation in the £1-£20,000 band (which effectively removes only one observation with a unit price of £311,151), and by 0.003 across all other specifications, including when restricting the sample to the 0.1%-99.9% range

[101] This excludes one observation with a price of £311,151, which is very likely to be an error.

within each condition-year combination. The 0.1% – 99.9% percentile range corresponds to £4.56 – £523. These minimal changes in the estimated effect further confirm that the exclusion of extreme values does not meaningfully impact our findings.

Table A.3: Effect of acquisitions on average prices after trimming outliers (FOPs only, all LVGs, Insurer 1 data)

	£1 – £20,000	£1 – £10,000	£1 – £5,000	0.1% – 99.9%
LVG	0.076*** (0.010)	0.077*** (0.010)	0.077*** (0.010)	0.077*** (0.010)
Condition-type FEs	Yes	Yes	Yes	Yes
Practice FEs	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes
Observations	1,185,512	1,185,512	1,185,508	1,186,147
Adjusted R^2	0.068	0.068	0.068	0.069

Source: CMA analysis using Insurer 1 data.
*Note: Standard errors are clustered at the practice level and are reported in parentheses. ***, **, and * correspond to statistical significance at the 1%, 5%, and 10% levels, respectively. The number of observations for the subsamples (£1 – £20,000) and (£1 – £15,000) are the same, but the coefficients differ because, in our analysis of average prices, we compute the average price by practice-treatment-year. As a result, even though more outliers are excluded in one subsample compared to the other, the total number of observations remains unchanged when taking the average.*

Effect of COVID-19 and other macro developments

A.42 Several LVG have submitted that our results might be partly or entirely driven by COVID-19 (or Brexit).

(a) CVS submitted that the effect of COVID-19 might have caused violations of the parallel trends assumption.[102] A particular mechanism put forward by CVS is that LVGs (where all vets are employed staff, rather than owners, and pricing decision are taken essentially centrally) might be able to react more swiftly to bring in additional capacity in the face of a huge uptick in demand compared to individual self-employed vets.[103] Hence the CMA's estimate of acquisition effects could reflect differential responses to COVID-19 between LVGs and independent practices. CVS pointed out that, when the sample period is restricted to 2014-2019, the acquisition effect becomes statistically insignificant in the claim value model, and reduces from 8% to 4.2% in the unit price model.[104]

(b) Vet Partners stated that the period encompassing the years FY20 to FY22 were abnormal years. During this period, compared to independents, due to

[102] CVS response to draft working paper.
[103] CVS response to draft working paper.
[104] CVS response to draft working paper.

the employment and operational benefits they offered, LVGs would have had higher (mainly fixed) costs increases, and were likely more able to effectively assess the impact of the macro-economic changes in order to make the necessary adjustments to pricing.[105]

(c) [✂] submitted that the combined effect of Brexit, COVID-19, and cost inflation led to severe staff shortages and a material increase in labour costs, and a short-term increase in demand for vet services. [✂] expects that independent practices have not increased staff pay and benefits in response to these developments to the same degree as LVGs, or they have been slower to respond. At independent practices, the most experienced vets also tend to be the owners, and these owner-vets are likely to have not fully responded to market-level veterinary pay inflation in setting their own personal remuneration. Overall, it is likely that LVGs have been more effective in responding to these shocks commercially, including in passing through cost inflation into treatment prices and establishing the market clearing price.[106]

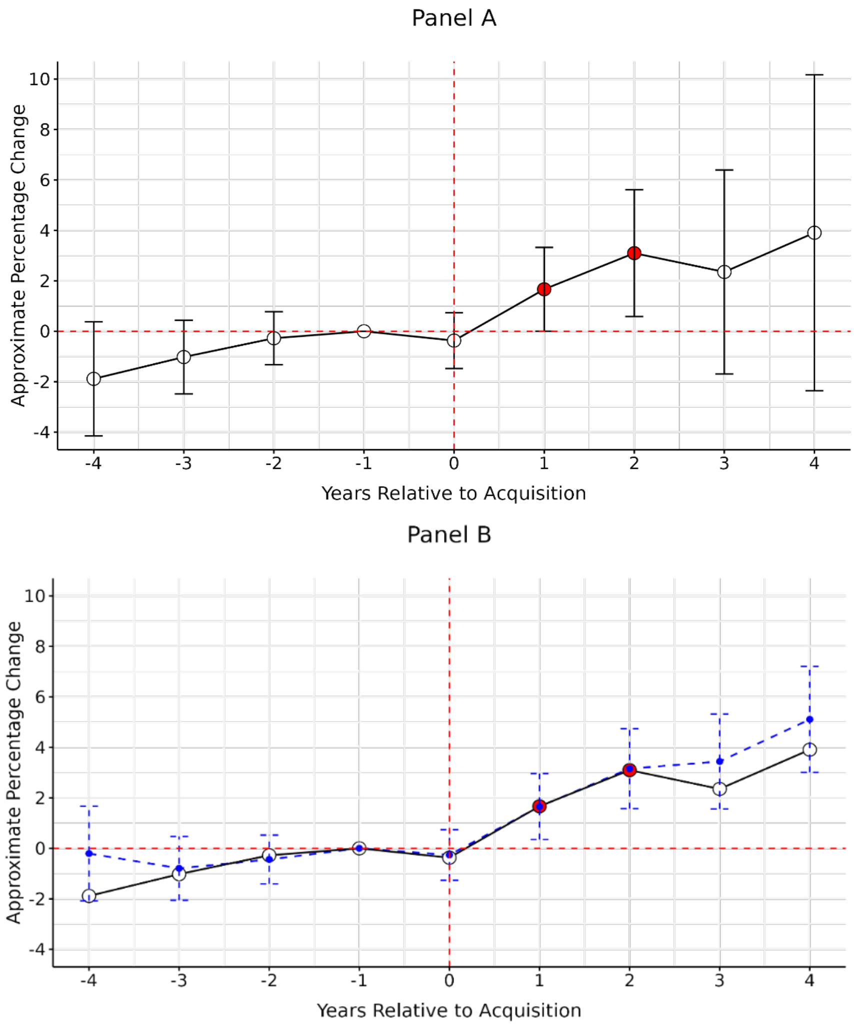
(d) Vet Partners submitted that observations above the 95% percentile are more likely to belong to LVGs after 2020, whereas there is no material difference before 2020. This suggests that there may be unobserved differences affecting sites owned by LVGs and independent sites, particularly after 2020. Vet Partners proposes to address this issue by excluding years after 2020, in which case the estimated effect of acquisitions on claim values falls to 1.5% and is statistically insignificant.[107]

A.43 **CMA assessment.** To investigate whether our results solely or primarily reflect differential responses to COVID-19 and Brexit, we re-ran our analysis after excluding observations related to the 2020-2024 period. As a matter of principle, we consider that this approach is likely to understate the total effect of corporate acquisitions on prices and treatment costs. This is because the effect of corporate acquisitions is likely to build up progressively over time, which implies that excluding the last four years of our sample period will not just 'control for the effect of COVID-19' but might essentially shut down a large part of the genuine effect of acquisitions. By removing a very large number of observations, it will also increase confidence intervals and therefore reduce the likelihood of rejecting the null hypothesis of no effect even if there is an effect. Nevertheless, it can be informative to compare the coefficients for the first few periods following an acquisition for the full sample and the truncated sample.

[105] VetPartners response to draft working paper.
[106] [✂] response to draft working paper.
[107] VetPartners response to draft working paper.

A.44 Figure A.5 shows the results of this analysis for our claim values model: Panel A shows the coefficients and confidence intervals obtained when post-2019 observations are excluded, and Panel B compares these coefficients with those obtained using the full sample (shown as the blue dashed line). Panel A shows that, even when later observations are excluded, the estimated effect for the first two years is positive and statistically significant. Panel B shows that the coefficients for the relative period -3 to +2 are nearly identical to those from the full sample. However, from period +3 onward, they diverge, which is expected given the 2019 sample cutoff. Since only practices acquired before 2017—ie in 2015 and 2016—have at least three years of post-acquisition observations, the number of available observations from +3 onward is much smaller. This reduces the precision of estimates, as seen in the larger confidence intervals at t+3 and t+4.

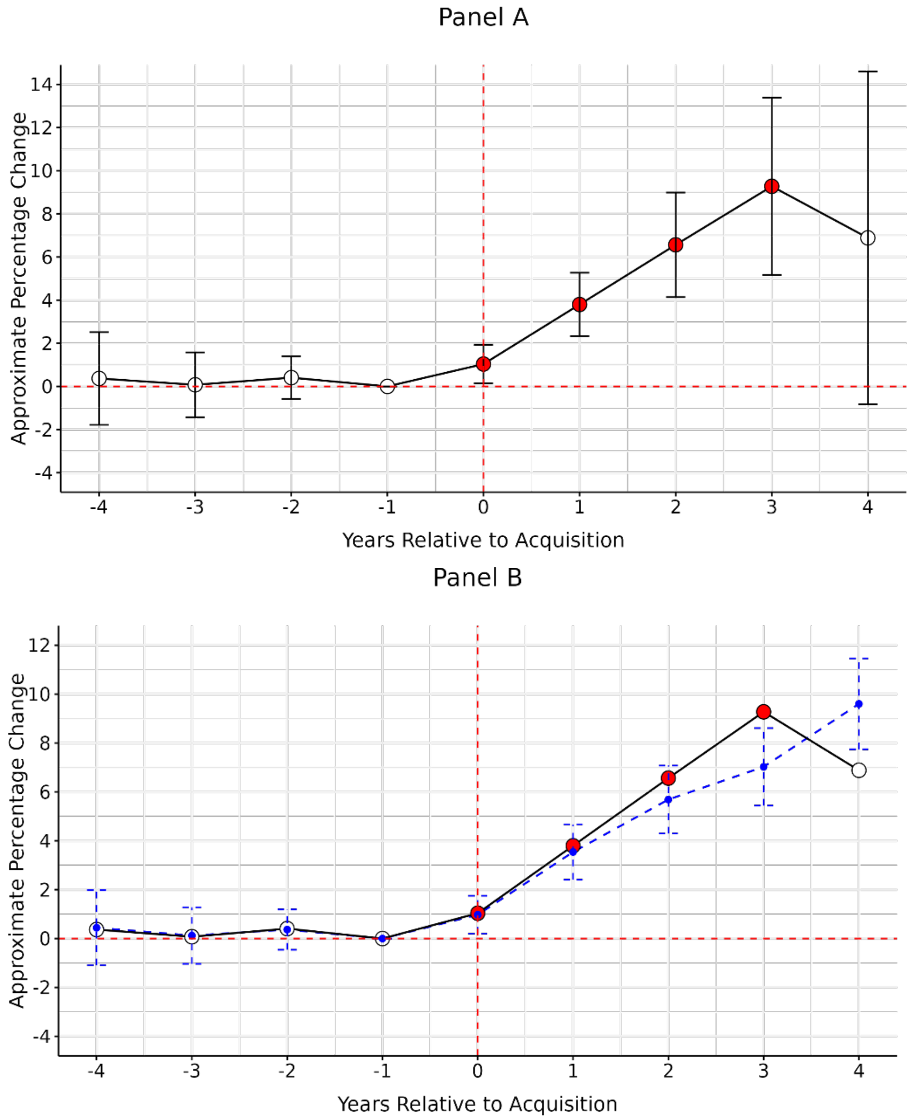
Figure A.5: Leads and lags results after excluding COVID-19 (effect on claim values, FOPs only, all LVGs, Insurer 2 data)



Source: CMA analysis using Insurer 2 data.
Note: The figure plots the coefficient estimates together with their 95% confidence intervals. The dots are red if coefficients are significantly different from zero. Standard errors are clustered at the practice level. In Panel B, the blue dashed line represents the coefficients and confidence intervals of our main model for claim values, as shown in Figure 2.1.

A.45 Figure A.6 presents a similar picture for average prices. The coefficients are largely consistent from years -4 until +2, before diverging from year +3 onward. The close alignment of the coefficients up to year +2 reinforces that the exclusion of post-2019 observations does not fundamentally alter the estimated effects in the earlier periods, but significantly reduces precision in later years due to the smaller sample size. Therefore, the fact that the estimated effects remain stable up to two years post-acquisition, even when excluding observations after 2019, provides strong evidence that our findings are not driven by COVID-19 related distortions.

Figure A.6: Leads and lags results after excluding the COVID-19 period (effect on average prices, FOPs only, all LVGs, Insurer 1 data)



Source: CMA analysis using Insurer 1 data.
Note: The figure plots the coefficient estimates together with their 95% confidence intervals. The dots are red if coefficients are significantly different from zero. Standard errors are clustered at the practice level. In Panel B, the blue dashed line represents the coefficients and confidence intervals of our main model for unit prices, as shown in Figure 2.12.

A.46 More generally, it is not clear to us that the COVID-19 period would have affected the marginal costs of LVGs to a greater extent than independents. As explained in paragraphs A.12 to A.14, LVGs have told us that they invest in (fixed) capital resources as well as staffing. LVGs did not explain why they were likely more able to adjust prices more quickly, and if they do it is not clear that it should be ignored in the analysis as a matter of principle (for example if LVGs are generally more inclined to reflect increased demand and customer willingness-to-pay in their prices).

LVG practices are very different from independent practices

A.47 Three LVGs submitted that acquisition decisions are not random, and that LVG practices are very different from independent practices.

(a) One LVG submitted that there are major differences between the average LVG practice and the average independent practice, in terms of both scale and treatment mix (as measured by the number of treatment groups claimed for). Therefore, any differences in price levels or trends between these groups cannot be ascribed to the ownership model. This also raises questions over whether the control group used for the DiD analysis would genuinely have seen the same price trends as the set of acquired independent practices (had they not been acquired).[108]

(b) Another LVG submitted that the choice of acquisition targets is by no means random, which could lead to systematic unobserved differences between the two groups. The factors considered by this in identifying potential acquisition targets include financial considerations, strategic considerations, and most importantly, ethical and cultural considerations. In this sense, the 'desirable' independent practices will be acquired, and, over time, the control group will increasingly consist of less desirable practices or at least practices that provide a different service offering and are less comparable with the acquired LVG practices.[109]

(c) A third LVG submitted that the average claim value and first-year treatment cost were consistently higher in the years preceding acquisition for clinics acquired by this, compared to clinics that remained independent, for nearly every year between 2014 and 2022. This suggests the presence of

[108] [✂] response to draft working paper.
[109] [✂] response to draft working paper.

unobserved factors that must be accounted for to ensure validity of the parallel trends assumptions.[110]

A.48 **CMA assessment.** It is uncontroversial that there are differences between practices that remain independent and practices that are acquired by LVGs at some point in the period. Our analysis of the causal effect of corporate acquisitions can accommodate differences in both observed and (time-invariant) unobserved characteristics between acquired practices and practices treated as control. The only requirement for this analysis is that, conditional on observed characteristics, the outcomes of interest for acquired and control practices change in the same way over time (the 'parallel trends assumption'). Our leads and lags tests show that this assumption holds in the periods preceding the acquisitions, and respondents have not explained why or how the differences in observable characteristics that they document might cause violations of the assumption post acquisitions, in a way that would not be captured by this test.

A.49 As a secondary point, we are not persuaded that one can interpret the observed differences between LVG-owned and independent practices in the range of treatment groups claimed for in the Insurer 1 data as a reliable proxy for differences in the clinical capability of these two groups of practices. [✂] imply that differences in the number of unique treatment groups measured in the Insurer 1 data might overstate differences in the clinical capabilities of LVG and independent practices.

A.50 Notwithstanding these points, we tested whether our findings were robust to the utilisation of a different control group consisting of not-yet-acquired practices, instead of never acquired practices. This means that, in a given year, the treated group consists of practices that have been acquired by a LVG, while the control group consists of practices that are still independent in that year but will be acquired in the future.[111] If it is correct that the independent practices that are targeted by LVGs are systematically different from those that are never acquired and might not follow parallel trends, this approach would alleviate this concern by comparing more similar practices.

A.51 We find that our results remain consistent even when using this alternative control group. We continue to observe an increase in claim values and prices immediately after acquisition, as well as parallel trends in the pre-acquisition period. This confirms that our findings are not driven by the utilisation of an inappropriate control group.

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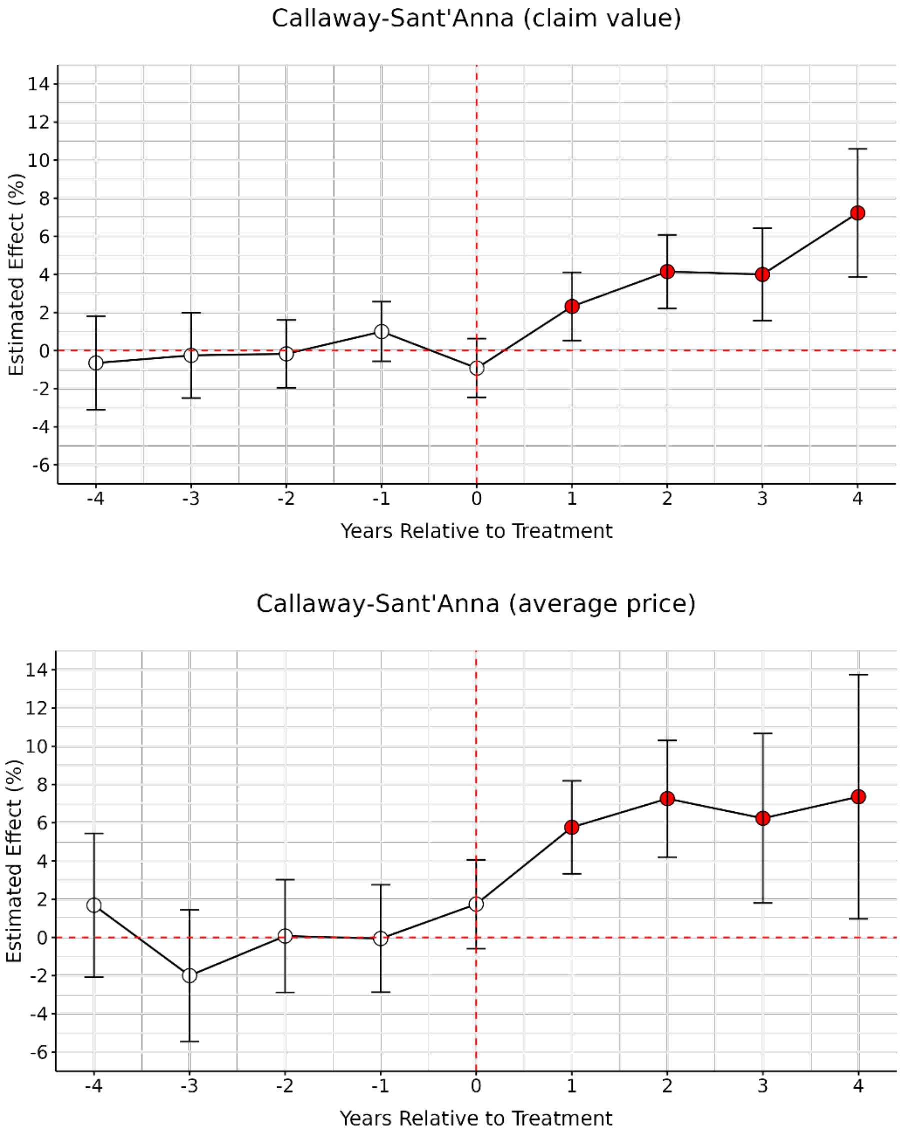
[110] [✂] response to draft working paper.
[111] All practices in the sample are eventually acquired by LVGs by the end of the sample period. As a result, the later years (from 2022 onwards) do not include any "not-yet-treated" practices that can serve as a control group. This is a common feature of staggered DiD settings where treatment eventually applies to all units. In contrast, our baseline model uses never-treated practices as the control group, which remain available throughout the entire sample period.

Table A.4: Effect of acquisitions on claim values and average prices using not-yet acquired practices as control group and the Callaway-Sant'Anna estimator (FOPs only, all LVGs, Insurer 2 and Insurer 1 data)

	Callaway-Sant'Anna estimator (2020)	
	Log (Claim value)	**Log(Average price)**
LVG	0.069*** (0.011)	0.129*** (0.017)

Source: CMA analysis using Insurer 2 and Insurer 1 data.
*Note: ***, **, and * correspond to statistical significance at the 1%, 5%, and 10% levels, respectively. The coefficients are estimated using dynamic aggregation, which averages the average treatment effects across all lengths of exposure to the treatment. Both models do not include covariates. The estimators are "doubly robust", which means they are based on first step linear regressions for the outcome variable and logit for the generalised propensity score.*

Figure A.7: Leads and lags results using not-yet acquired practices as control group and the Callaway-Sant'Anna estimator (FOPs only, all LVGs, Insurer 2 and Insurer 1 data)



Source: CMA analysis using Insurer 2 and Insurer 1 data.
Note: The figure plots the coefficient estimates together with their 95% confidence intervals. The dots are red if coefficients are significantly different from zero. Standard errors are clustered at the practice level.

Lack of geographical controls

A.52 Two LVGs said that the cross-sectional comparisons of claim values, first-year treatment costs, and unit prices were flawed because they did not control for the location of the practices.

(a) One LVG submitted that the CMA's cross-sectional regressions do not include controls for the regions in which clinics are located, meaning that a chain with a higher proportion of clinics in London and the south-east, where prices and costs to serve are higher across a wide range of industries, will be assigned by the CMA as exhibiting a higher price.[112] When including regional control variables, the estimated effect of [✂] ownership on claim values declines from [✂], as reported in Table 1.5 of the working paper, [✂]. For first-year treatment costs, the coefficient for [✂].[113]

(b) Another LVG submitted that the geographic location of practices could be important because the underlying costs of a practice may differ across areas. FOPs in urban areas likely face different costs and a different mix of pets/conditions compared to those in rural areas.[114]

A.53 **CMA assessment.** we recognise that the location of a practice can be an important determinant of its prices and claim values, and we have changed the specifications of our cross-sectional regression to include relevant controls (specifically, a series of fixed effects for local authorities and the ONS urban/rural classification of each practice).

Spillovers

A.54 As set out in paragraph 3.8, if independent practices in the control group respond to price increases in the treated group by also increasing their prices, then our analysis will underestimate the effect of corporate acquisitions (an issue known as 'spillover effects'). Two LVGs have submitted that the CMA did not present any evidence that such spillovers would be likely or material.

(a) CVS submitted that the CMA's control group consists of all independent practices and so in any case only a small portion of them would be potentially subject to spillover effects. In practice, the investments made by LVGs may also lead to an intensification of local competition, and competing practices

[112] [✂] response to draft working paper.
[113] [✂] response to draft working paper.
[114] [✂] response to the draft working paper on the impact of corporate acquisitions on treatment costs.

may instead need to reduce (or slow the increase of) their prices in order to maintain competitiveness. In this case, spillover effects would cause the effect of acquisition to be biased upwards rather than downwards.[115]

(b) IVC submitted that we have not provided any evidence that many independents have experienced an acquisition close enough that it has the potential to affect their business; that these independent could observe any price increases at the acquired practices either directly or indirectly (through a change in their demand); and that their best response would be to increase rather than decrease price (taking account of changes in quality at the acquired practices).[116]

A.55 **CMA assessment.** Almost 2000 practices were acquired by LVGs in the period considered, so it is likely that a large number of independent practices were exposed to changes in prices or quality made at these practices. It is correct in principle that, if acquired practices increase quality as well as price, this might reduce the effect of acquisitions on the residual demand faced by competing practices and on their pricing incentives. However, because the documentary evidence indicates that the initial price increases applied by at least three LVG's at acquired practices are motivated by their assessment of customer willingness to pay and comparisons with competitors rather than quality improvements (paragraph A.18), we consider that such spillover effects are more likely to imply a downward bias in our estimate of the effect of acquisitions Nevertheless, we recognise that the magnitude of spillover effects in the presence of multi-dimensional changes in product offerings is difficult to evaluate.

LVG-1

A.56 LVG-1 submitted separate arguments with respect to our analyses of claim values and unit prices, but some of the arguments are similar or identical. Where this is the case, we assess the points as part of the first sub-section on claim values, and do not repeat the discussion later.

Analysis of claim values and first-year treatment costs

Dropping one-sided acquired practice/condition combinations

A.57 LVG-1 pointed out that the sample used by the CMA includes a number of practice-condition combinations that are only observed in the data pre- or post-acquisition and proposes to drop these observations.[117] This reduces the estimated effect of corporate acquisitions from 3.8% to 3.2%. LVG-1 suggested the same adjustment

[115] CVS response to draft working paper.
[116] IVC response to draft working paper.
[117] LVG-1 response to draft working paper.

with respect to our analysis of unit prices, in which case the impact is to reduce the estimate of the acquisition effect from 8.0% to 6.4%.[118]

A.58 **CMA assessment.** We consider that this adjustment is unnecessary. In addition to fixed effects for practices and years, our model includes a separate set of fixed effects for the conditions treated (or the treatment administered, in the case of our analysis of unit prices). This means that an observation related to a particular condition in the post-acquisition period can contribute to the estimation of the acquisition effect even if there isn't an observation related to the same condition at the same practice in the pre-acquisition period.[119]

A.59 The adjustment proposed by LVG-1 will have the effect of dropping observations related to conditions or treatments that are relatively infrequent. It is possible that opportunities for increasing prices (or treatment intensity) are particularly strong for such infrequent conditions or treatments, insofar as customers may be less likely to make comparisons between competing providers. [✂].[120] For this reason, the adjustment proposed by LVG-1 might bias the estimate of the acquisition effect downward.

The CMA's control group is not suitable

A.60 LVG-1 pointed out that the CMA's control group includes independent practices that appear in the data for a limited period of time, and this casts doubt on whether the parallel trends assumption holds. To the extent that these practices have exited or entered during the period, there is a risk that their business model was either unsustainable or that their pricing did not reflect a long term equilibrium level (eg promotional pricing post-entry). LVG-1 proposed to address this issue by limiting the control group to independent practices that are present throughout the data (which reduces the estimate of the effect on claim values from 3.8% to 3.5%).[121] LVG-1 suggested the same adjustment with respect to our analysis of unit prices (which increases the estimate of the effect on unit prices from 8.0% to 8.6%).[122]

A.61 **CMA assessment**. We do not think that this observation invalidates our analysis for the following reasons.

118 LVG-1 response to draft working paper.
119 To illustrate this point with a simple worked example, suppose that, for a practice acquired in 2020, we observe a claim for condition A in 2019 for £30, and a claim for condition B in 2021 for £60. Suppose also that, based on other comparisons, we know that the average difference in claim values between condition B and condition A is £10 (conditional on other observables), and that the average difference in claim values between 2021 and 2019 is £15 (conditional on other observables). Then we would infer that the acquisition effect for this practice was £5 (=60-30-15-10). This simple example illustrates how conditioning on covariates supports meaningful comparison between claims for different conditions.
120 [✂].
121 [✂] response to draft working paper.
122 [✂] response to draft working paper.

A.62 First, practices only appear in the Insurer 2 data to the extent that they treat customers covered by Insurer 2 insurance. [✂]. Therefore, when an independent practice appears or disappears in the dataset, it should not be inferred that the practice has genuinely opened or closed. Discontinuities in the presence of practices in the dataset might reflect discontinuities in their issuance of claims to Insurer 2 rather than genuine cases of market entry and exit.

A.63 Second, the CMA's leads and lags tests show that the parallel trends assumption holds for the periods preceding acquisitions. There are cases of practice 'entry and exit' in the Insurer 2 data throughout the whole period of observation, so it is not clear why this feature of the data would specifically induce violations of the parallel trends assumption after acquisitions, which would not be detected in the periods preceding acquisitions.

Unresolved bias from the lack of control for quality

A.64 LVG-1 pointed out that the CMA's analysis does not control for changes in quality, and propose to address this issue by restricting the sample used in the analysis to include only two years following acquisition by an LVG.[123] LVG-1 also submitted that this sensitivity would capture periods over which the parallel trends assumption would hold.[124] This has the effect of reducing the estimate of the acquisition effect from 3.8% to 1%. LVG-1 suggested the same adjustment with respect to our analysis of unit prices, in which case the impact is to reduce the estimated acquisition effect from 8.0% to 2.9%.[125]

A.65 **CMA assessment.** While we recognise that our analysis does not control for quality, we do not regard the adjustment proposed by LVG-1 as a targeted or meaningful way of addressing this limitation. There is no reason to assume that any changes in pricing strategies or treatment recommendations introduced by the new owners of acquired practices are exhausted after two years. [✂].[126] If an acquired practice is deemed to charge prices under their profit-maximizing level, it might be optimal for its new owner to implement the required adjustments progressively, to make price rises less noticeable to customers, or to obtain evidence on the demand response. One internal document from another LVG [✂].[127]

A.66 It is correct that the parallel trends assumption is less likely to hold, the longer the amount of time after acquisition that is included in the analysis. However, we consider that limiting the sample to two years after each acquisition is not warranted or proportionate in this case. Our leads-and-lags results show that, for

[123] [✂] response to draft working paper.
[124] [✂] response to draft working paper.
[125] [✂] response to draft working paper.
[126] [✂] response to draft working paper.
[127] [✂].

most specifications, the parallel trend assumption holds for at least four years before each acquisition. The fact that this finding pertains to a sample with staggered treatments (that is, for pre-acquisition periods occurring at different times throughout the 2014-2024 period rather than solely for specific years in that period) makes it less likely that this property arose ‘by chance’. Moreover, our leads-and-lags results also show that our main findings – that corporate acquisitions cause significant increases in key outcomes – are not contingent on effects estimated for distant periods. On the contrary, these effects tend to materialise relatively quickly after acquisitions and then increase gradually.

A.67 Finally, some LVGs have submitted that some of the quality improvements they make upon acquisitions become effective instantly or very rapidly (for example access to specialist staff and changes in equipment – set out in paragraphs A.8 and following). If this is correct, then limiting the period of observation would not effectively address the limitation induced by the lack of good controls for quality. In our view it is better to acknowledge this limitation and consider the evidence on quality changes separately, than to try to restrict the analysis in ways that are difficult to justify.

Trimming tails of the dataset

A.68 LVG-1 pointed out that the data from 2014, early 2015, and the latter part of 2024 is incomplete, and propose to exclude observations from 2014, early 2015, and the later periods of 2024. This reduces the estimate of acquisition effect from 3.8% to 3.3%.[128] For our analysis of treatment price, this change reduces estimated acquisition effect from 8.0% to 7.3%.[129]

A.69 **CMA assessment.** We partially agree with LVG-1 and have excluded 2014 from the sample, because the number of practices in the Insurer 2 dataset is significantly lower at the start of that year, and there is a clear discontinuity between 2014 and 2015. From 2015 onward, the number of practices remains relatively stable, suggesting that the lower count observed in 2014 may reflect a specific data coverage issue affecting that year.

A.70 However, while LVG-1 also recommend dropping early 2015 observations and starting the sample in March 2015, we have retained all 2015 observations in our analysis. The data shows no significant changes or discontinuities around March 2015, providing no strong reason to exclude those months. For instance, Figure A.8 shows that both the number of practices and the average claim value per practice remain stable around March 2015. Therefore, dropping observations for

[128] [✂] response to draft working paper.
[129] [✂] response to draft working paper.

January and February 2015 appears unjustified. That said, our results remain nearly identical even if early 2015 is excluded.

A.71 We have also retained all 2024 observations, as the lower number of practices and claims in the last few months of the dataset is purely due to the timing of data collection. The dataset only includes data up to the month we requested it from insurers, meaning any missing practices or claims are simply a result of when the data was collected. This suggests that sample selection bias is unlikely to be a concern.

Figure A.8

[✂].

A.72 Although the discontinuity in terms of number of practices and average prices between 2014 and 2015 is less pronounced in the Insurer 1 dataset (Figure A.9), and we believe there is no strong reason to make the same adjustments as for Insurer 2,[130] we have excluded 2014 from the Insurer 1 analysis as well to ensure consistency across datasets.

A.73 Excluding 2014 observations also results in the exclusion of 2015 acquisitions, as our analysis requires at least one year of pre-acquisition data. Table A.5 shows that, as a result, our LVG coefficient decreases slightly—from 3.8 to 3.5 for Insurer 2 and from 8.0 to 7.6 for Insurer 1.

Figure A.9:

[✂].

[130] Specifically, [✂] the number of practices covered has consistently increased over the entire sample period, making it difficult to determine a clear cutoff point. Simply dropping all observations before March 2015, as suggested, does not provide a suitable solution.

Table A.5: Effect of corporate acquisitions on claim values and average prices after excluding observations before 2015 (FOPs only, all LVGs, Insurer 2 and Insurer 1 data)

	Log(Claim value)	Log(Average price)
LVG	0.035*** (0.008)	0.076*** (0.007)
Condition-type FEs	Yes	No
Treatment-type FEs	No	Yes
Practice FEs	Yes	Yes
Year FEs	Yes	Yes
Observations	2,473,542	1,097,376
Adjusted R^2	0.324	0.702

Source: CMA analysis using Insurer 2 and Insurer 1 data.
*Note: Standard errors are clustered at the practice level and are reported in parentheses. ***, **, and * correspond to statistical significance at the 1%, 5%, and 10% levels, respectively.*

Practice reclassification

A.74 LVG-1 submitted that the classification of practices provided in the Insurer 2 data (and used by the CMA in its analysis) does not accord with LVG-1's internal classification. Using the classification used by LVG-1 reduces the estimated impact of acquisitions from 3.8% to 3.5% for claim values and has no change on the estimated impact on unit prices.[131]

A.75 **CMA assessment.** There is no commonly agreed definition of a FOP in the industry, and in the first instance we used the classification of practices provided by Insurer 2 in its analysis, which has the merit of being consistent across providers. Notwithstanding this point, we have implemented the change suggested by LVG-1.

Dropping one-sides acquired practices

A.76 LVG-1 point out that our sample includes observations related to practices which were acquired during the sample period, but only registered claims either before or after the acquisition date. It is not possible to calculate practice-specific difference-in-difference estimates for these practices, and excluding them reduces the estimate of the acquisition effect from 3.8% to 3.7% for claim values.[132] For unit prices, the impact of the change is to reduce the estimated effect of an acquisition from 8.0% to 7.1%.

A.77 **CMA assessment.** It is correct that these observations do not directly contribute to the identification or estimation of the acquisition effect. Nevertheless, they do contribute to the identification and estimation of the parameters on control

[131] [✂] response to draft working paper.
[132] [✂] response to draft working paper.

variables, and their inclusion in the sample does not bias the estimate of the parameter of interest. As such, there is no obvious reason to exclude these observations.

Analysis of unit prices

A.78 This section reviews the arguments put forward by LVG-1 with respect to our analysis of unit prices, focusing on those arguments that are different compared to their review of claim values (and not covered in the section on common themes).

No weighting has been applied

A.79 LVG-1 pointed out that our analysis does not account for the fact that some treatment groups are more common than others and proposes to address this issue by weighting the analysis by the number of claims underlying each treatment group at each practice. This reduces the estimated effect of an acquisition from 8.0% to 6.5%.[133]

A.80 **CMA assessment.** We do not think that this is a sensible adjustment to the analysis. Weighting the analysis by the number of claims will naturally put more weight on treatments that are more frequent. If these treatments are of low value, changes in their price may not have a particularly large bearing on customer welfare. Moreover, if treatments are purchased more frequently, customers may be better informed about their prices, and vets might have a lower incentive to increase their prices (paragraph A.59).

A.81 A more appropriate sensitivity test would consist of weighting observations by value, rather than by frequency. By value, we refer to the share of total spend on each treatment. For example, the weight assigned to an observation for the treatment '*fluoroscopy*' would be calculated as the sum of invoiced amounts for fluoroscopy across all practices divided by the sum of invoiced amounts for all treatments.

A.82 The rationale for this approach is that it gives more weight to treatments that have a greater potential impact on customer welfare. Treatments that account for a larger share of customer expenditure are likely to be more economically significant and therefore changes in their prices are more relevant when assessing potential harm to consumers.

A.83 Table A.6 shows that, when applying this weighting by value, the estimated effect of acquisition increases from 8.0% to 9.5%. In addition, Figure A.10 shows an increase in prices immediately after acquisition, as well as parallel trends in the pre-acquisition period. This suggests that the impact of acquisitions is actually

[133] [✂] response to draft working paper.

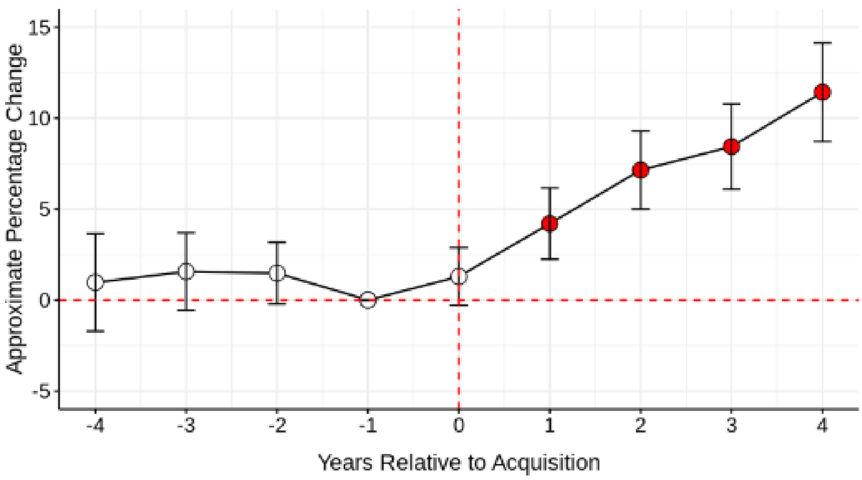
stronger for treatments that matter most in terms of customer spending and, by extension, customer welfare.

Table A.6: Effect of corporate acquisitions on average prices, with observations weighted by value (FOPs only, all LVGs, Insurer 1 data)

	Log(Average price)
LVG	0.095*** (0.013)
Treatment-type FEs	Yes
Practice FEs	Yes
Year FEs	Yes
Observations	1,189,116
Adjusted R^2	0.599

Source: CMA analysis using Insurer 2 and Insurer 1 data.
*Note: Standard errors are clustered at the practice level and are reported in parentheses. ***, **, and * correspond to statistical significance at the 1%, 5%, and 10% levels, respectively.*

Figure A.10: Leads and lags results with observations weighted by value (Insurer 1 data)



Source: CMA analysis using Insurer 1 data.
Note: The figure plots the coefficient estimates together with their 95% confidence intervals. The dots are red if coefficients are significantly different from zero. Standard errors are clustered at the practice level.

Filter treatment groups for which practices infrequently make claims

A.84 To account for outliers, LVG-1 propose to exclude treatment groups for individual practices that appear infrequently in the data (less than 12 or 52 times in a year).[134]

A.85 **CMA assessment.** We see no valid econometric argument for this adjustment. A low number of observations in certain practice-treatment combinations would be problematic if we sought to estimate differences for specific 'cells' (practice-

[134] [✂] response to draft working paper.

treatment combinations), which is not the case here. The adjustment proposed by LVG-1 has the effect of removing observations related largely to infrequent but relatively high value treatments. As customers will naturally be less informed about the prices of these treatments, the scope for increasing prices may be particularly high (paragraph A.59).

LVG-2

Increases in prices over time

A.86 LVG-2 submitted that our descriptive analysis of increases in unit prices over time is flawed in three respects: [135]

(a) it does not account for changes in the composition of treatments within the categories provided by Insurer 1;

(b) it does not account for changes in quality, and the comparison between our estimate of the increase in average prices (which does not control for changes in quality) and the (lower) increase in the ONS CPI for veterinary services (which does control for changes in quality) is consistent with the proposition that quality has improved; and

(c) the comparison with general inflation in services and wages is uninformative because these do not reflect cost increases in the veterinary sector.

A.87 **CMA assessment.** We do not think that these criticisms affect the validity of our estimates of price inflation, for the following reasons.

A.88 First, even if there are changes in the mix of treatments provided within each category (and, as a result, in the quality of the treatment provided), the resulting estimate of price inflation is still meaningful. For example, it is relevant for customers that the average price of a rhinoscopy has increased by a certain amount, even if part of the increase stems from the substitution from simple non-endoscopic techniques to video-enhanced fibreoptic rigid endoscopy (to use an example provided by LVG-2). This is particularly true if vets increasingly recommend the latter, and customers are not necessarily aware of the cheaper option.

A.89 Second, it is not clear to us that the ONS CPI for veterinary services controls for changes in quality. The ONS CPI only tracks two items, annual booster injections and dog kennel daily charges, which would seem to present limited scope for quality improvements. Given the significant difference in the scope of the treatments included in these two estimates of prince increase, it would not seem

[135] [✂] response to draft working paper.

appropriate to infer that the difference between the two can be interpreted as the impact of quality improvements.

A.90 Third, our comparison with wage inflation is meant to convey the idea that vet services are likely to account for an increasing share of household income (or more specifically, labour income), rather than that it is not justified by increases in the cost of providing the service.

More reliable controls for differences in prices between practices, conditions, and time

A.91 LVG-2 submitted that, while fixed effects are in principle a sensible way to control for differences in observable determinants of prices and claim values, it would be preferable to include a richer set of fixed effects to capture these differences. For the claim values model, LVG-2 proposed two additional specifications: [136]

(a) one that includes fixed effects for practice-condition-breed combinations and for year-condition-breed combinations (under which the estimated effect reduces from 3.8% to 2.9%); and

(b) another that includes fixed effects for breeds, practice-condition combinations, and year-condition combinations, under which the estimated effect reduces from 3.8% to 3.2%).

A.92 For the analysis of treatment prices, LVG-2 propose two additional specifications:

(a) one that includes fixed effects for practice-treatment and treatment-year combinations (under which the estimated effect reduces from 8% to 7.7%), and

(b) one that includes fixed effects for practice treatment combinations and years (under which the estimated effect reduces from 8% to 7.8%).

A.93 **CMA assessment.** While it is often preferable to use richer fixed effect structures, in this case the particular structures proposed by LVG-2 had the implication of effectively discarding a large number of observations and placing more weight on conditions or treatments that are more frequent in the data. By way of illustration, the second specification proposed by LVG-2 (which relies on fixed effects for practice-condition combinations) implies that an observation related to a particular practice and condition in the post-acquisition period only contributes to the identification and estimation of the effect of interest if there is at least one observation for the same practice and condition in the pre-acquisition period. This will naturally put more weight on conditions that are observed more frequently, but

[136] [✂] response to draft working paper.

the pricing dynamics for these conditions might be different (paragraph A.59). The first specification proposed by LVG-2 compounds this problem by considering fixed effects for practice-condition-breed combinations (by using smaller 'cells'). Thus, the difference between LVG-2's results and ours might reflect differences in the set of observations used and the implicit weighting of conditions rather than differences in the set of controls used. For this reason, in this case we do not accept that the inclusion of additional fixed effects necessarily makes the results more robust or more meaningful.

Differences between acquired and never acquired practices, and matched control group

A.94 LVG-2 submitted that there are also important differences between the types of independent practices that are typically acquired by LVGs, and those that are not, notably in terms of the number of claims and the range of treatment claimed.[137] This is consistent with LVG-2 stating that it would not normally consider smaller/less up-to-date sites for acquisition as they would not provide a platform to develop the range and quality of services that form LVG-2's offer.[138] To address this issue, LVG-2 create a 'matched control group' consisting of independent practices that are more similar on observable characteristics to acquired practices. The matching is based on measures of volumes, revenue, geographical location and range of conditions and types of pets treated. This sensitivity reduces the estimate of the acquisition effect on claim values from 3.8% to 2.4% and increases the estimate of the acquisition effect on unit prices from 8.0% to 8.4%.[139]

A.95 **CMA assessment**. We consider that the sensitivity proposed by LVG-2 is not sensible, for the following reasons.

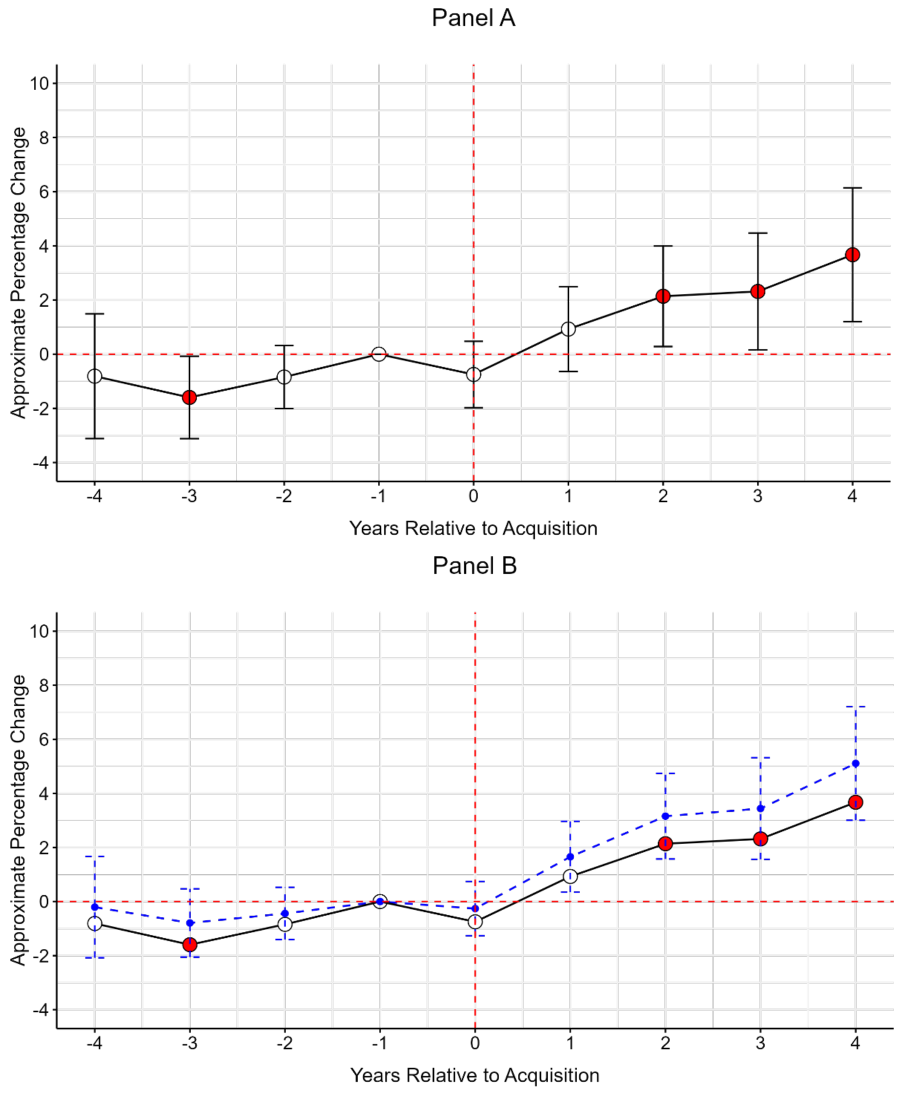
A.96 First, as explained above, our analysis of the causal effect of corporate acquisitions can accommodate differences in both observed and (time-invariant) unobserved characteristics between acquired practices and practices treated as control. This analysis only requires that acquired and control practices be similar in terms of changes in the outcome variable over time, and our leads and lags test shows that, for most specifications, this assumption holds in the pre-acquisition periods. LVG-2 has not explained why differences in the observable characteristics of acquired and control practices are likely to violate the common trends assumption in post-acquisition periods, in a way that could not be detected in pre-acquisition periods.

A.97 Second, a simple 'leads and lags' test suggests that the control group proposed by LVG-2 is not demonstrably better than the one we used. If the control group built by [✂] was better than that used by the CMA, one might expect the coefficients on

[137] [✂] response to draft working paper.
[138] [✂] response to draft working paper.
[139] [✂] response to draft working paper.

leads (the coefficient related to pre-acquisition periods) to be smaller; that is, differences in the price changes made by the treated and control groups before acquisitions should be smaller. To test whether this was the case, we replicated LVG-2's analysis on claim values using a leads and lags specification. We found that the coefficients in the pre-acquisition period in LVG-2's model are in fact larger than those in our model, with one being statistically significant (Figure A.11). This implies that the control group built by LVG-2 is not demonstratively better, and plausibly worse, than the simple control group we used.

Figure A.11: Leads and lags results with matched control group (effect on claim values, FOPs only, all LVGs, Insurer 2 data)



Source: CMA analysis using Insurer 2 data.
Note: The figure plots the coefficient estimates together with their 95% confidence intervals. The dots are red if coefficients are significantly different from zero. Standard errors are clustered at the practice level. In Panel B, the blue dashed line represents the coefficients and standard errors of our main model for claim values, as shown in Figure 2.1.

A.98 Third, we have some reservations with respect to the particular specifications used by LVG-2 to build their matched control group. LVG-2 matched control and treated practices using observations for the year 2014 only, despite acquisitions occurring at different times over the 2014-2024 period. This means that, for practices acquired toward the end of the period, the matching is not based on recent data. For example, practices acquired in 2019, were matched with independent practices based on data from five years before their acquisition.

A.99 To illustrate this point, we examined differences between matched treated and control groups in some of the covariates used for matching at two points in time: in 2014, and in the four years preceding the acquisition of each practice. We find that the gap between the treated and control groups has widened during the pre-acquisition period (-4 to -1), indicating that any balance achieved in 2014 did not persist.[140]

Table A.7: Comparing Covariate Means in 2014 vs. Pre-Acquisition (-4 to -1) for Matched Groups.

	Matched Treated group	***Matched control group***	***Difference (p-value)***
Year = 2014			
Total of claim values	7871.929	7349.378	522.55 (0.193)
Average pet age	5.902	5.888	0.014 (0.865)
Number of distinct pets	14.550	13.725	0.825 (0.134)
Number of distinct conditions	14.360	13.662	0.698 (0.125)
Pre-acquisition period (-4 to -1)			
Total of claim values	108,074.84	86,692.51	21,382.33*** (0.000)
Average pet age	7.568	7.520	0.048 (0.433)
Number of distinct pets	118.287	101.930	16.357*** (0.000)
Number of distinct conditions	70.293	62.629	7.664*** (0.000)

Source: CMA analysis using Insurer 2 data.
*Note: ***, **, and * correspond to statistical significance at the 1%, 5%, and 10% levels, respectively.*

A.100 Another important issue arising from this specification is that it limits the sample to practices that were present in the data in 2014. This results in a substantial loss of observations.

A.101 Finally, as explained in paragraphs A.50-A.51, we performed an additional sensitivity analysis using a control group consisting of not-yet-treated practices, which by construction addresses the issue raised by LVG-2. The estimated effect of corporate acquisitions is higher than in our baseline specification.

[140] We obtain similar results when focusing on individual relative years to acquisition between -4 and -1 (eg year -4 or year -1). In each case, we consistently find significant differences between the matched treated and control groups. This indicates that the balance achieved in 2014 is not reflected in any of the relative years.

Data coverage lower for LVG-2 than other LVGs

A.102 LVG-2 submits that the coverage of the insurer data received by the CMA appears to be lower for LVG-2 than for other LVGs. [✄] only around three quarters of LVG-2's practices appear in the data. The share of both insurers also appears lower for LVG-2 than for the market as a whole. Also, LVG-2 estimate that, for FY2023, the value of LVG-2 claims in the two datasets amounts to [✄]% of LVG-2's revenue, whereas the CMA estimates, for FY2024, the value of LVG claims in the two datasets amounts to [✄]% of total LVG revenues.[141]

A.103 **CMA assessment.** Differences in coverage between LVGs are expected, and may reflect differences in the penetration of different insurers (for example due to different LVGs agreeing to distribute or advertise different insurance products), differences in business models (for example due to the differences in the mix of insurable and non-insurable treatment in the turnover of different LVGs), or differences in reporting practices [✄]. LVG-2 has not explained why or how the lower coverage of the data for LVG-2 is likely to bias results.

Acquired veterinary practices will have been prepared for sale

A.104 LVG-2 has submitted that, in the run up to an acquisition, a practice can be expected to run its business in a way that aggressively manages costs to boost its short-term profitability (for example by operating with as lean a workforce as possible for a short period and not making capital intensive investments to scale-up/enhance aspects of the business etc.). This often means that newly acquired practices require substantial investment, in terms of adding staff resource, making capex spend on new equipment etc. shortly after completion.[142]

A.105 **CMA assessment.** Any anticipation effect would only be concerning if, in the run up to an acquisition, practices would reduce prices or treatment intensity. If this were the case, then any estimated increase in average unit prices or first-year treatment costs would wrongfully be attributed to the acquisition itself, whereas in fact it should be interpreted as a reversion to equilibrium. We do not think that this scenario is likely because: (i) this would be an odd decision to make from the point of view of a practice seeking to boost short-term profitability; and (ii) there is no indication of any deviation from parallel trends in the year preceding acquisition.

[141] [✄] response to draft working paper.
[142] [✄] response to the draft working paper on the impact of corporate acquisitions on treatment costs.

LVG-3

Inclusion of covariates

A.106 LVG-3 submitted that the method used by the CMA does not allow for proper statistical testing of the parallel trends assumptions when covariates are included. Specifically, LVG-3 point out that Callaway & Sant'Anna (2021) motivate differences between their method and the method used by the CMA in terms of their ability to incorporate covariates. To test for the parallel trends assumption, the CMA should exclude all covariates from its models.[143]

A.107 **CMA assessment.** It is not clear to us that the Sun & Abraham (2021) estimator cannot accommodate covariates. Sun & Abraham (2021) state that their method can accommodate covariates using the procedures of Callaway & Sant'Anna (2020), and the Stata package that they developed to implement their estimator ('eventstudyinteract') explicitly allows for covariates. The package that implements their estimator in R ('fixest', which we used for this study) also explicitly allows for covariates.

A.108 In any case, to test the robustness of our results, we have re-estimated our model using the Sun & Abraham estimator without covariates. Table A.8 shows that, when using the Sun & Abraham estimator without covariates, the acquisition effect moves from 3.8% to 3.9% for claim values, and from 8.0% to 7.5% for unit prices. Additionally, in Figure A.12, we report results from the leads and lags model. The findings show that the pre-acquisition coefficients remain statistically insignificant and are almost identical in magnitude to the coefficients estimated including covariates.

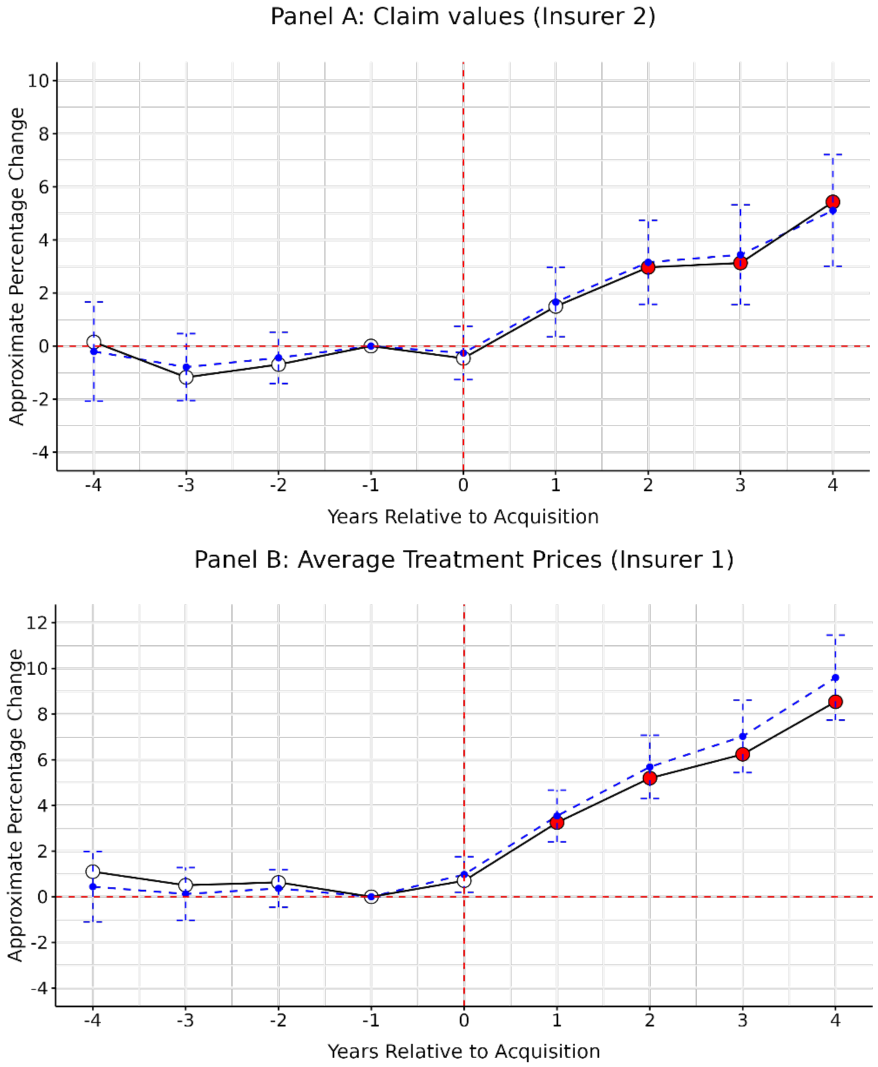
Table A.8: Effect of acquisitions on claim values and average prices without covariates (FOPs only, all LVGs, Insurer 2 and Insurer 1 data)

	Log(Claim value)	**Log(Average price)**
LVG	0.039*** (0.009)	0.075*** (0.010)
Condition-type FEs	No	No
Treatment-type FEs	No	No
Practice FEs	Yes	Yes
Year FEs	Yes	Yes
Observations	2,643,629	1,189,119
Adjusted R^2	0.058	0.068

143 [✂] response to draft working paper.

Source: CMA analysis using Insurer 2 and Insurer 1 data.
Note: Standard errors are clustered at the practice level and are reported in parentheses. ***, **, and * correspond to statistical significance at the 1%, 5%, and 10% levels, respectively.

Figure A.12: Leads and lags results without covariates (FOPs only, [✂] LVGs, Insurer 2 and Insurer 1 data)



Source: CMA analysis using Insurer 2 and Insurer 1 data.
Note: The figure plots the coefficient estimates together with their 95% confidence intervals. The dots are red if coefficients are significantly different from zero. Standard errors are clustered at the practice level. The blue dashed lines represent the coefficients and confidence intervals of our main models with covariates.

Incorrect test for parallel trends

A.109 LVG-3 submitted that our test for the parallel trends assumption is incorrect, for two reasons: first, the relevant test for parallel trends is a joint statistical test of whether differences in trends before the treatment are jointly significantly different from zero; and second, the assumption should be tested cohort-by-cohort (where a cohort refers to the practices acquired in a given year). If the assumption is violated for a particular cohort, that cohort should be excluded from the analysis.[144]

A.110 **CMA assessment.** The main identifying assumption in Sun & Abraham (2021) is indeed that the expected change in potential outcomes between any two dates should be the same between any two cohorts (including the never treated cohort).[145] In practical applications, however, researchers typically tests this assumption by considering event-study plots at an aggregate level (ie where the effect is averaged between cohorts),[146] rather than by implementing a series of F-tests for joint statistical significance cohort-by-cohort. This is primarily because parameters for individual cohorts are likely to be imprecisely estimated, and because any significant violations of the parallel trend assumption at the level of an important individual cohort are likely to be captured at the aggregate level. Moreover, the multiplication of individual tests makes it more likely that at least one test will return a 'false positive' (evidence of a breach even if there is no breach). For these reasons, we have adopted the most common approach of using aggregate event-study plots.

A.111 We also note that, even when applying the more stringent approach they proposed for testing parallel trends (and when dropping covariates), LVG-3 only found plausible evidence of a violation of the common trends assumption for two of the nine cohorts tested (one of these two cohorts being the 2024 cohort, which only had one acquisition).[147] For the seven cohorts that did not exhibit any evidence of a violation, LVG-3 found that the estimate of the effect of an acquisition was positive and statistically significant for both specifications (claim values and unit prices) in all cases bar two: the 2024 cohort (which only had one acquisition), and the 2023 cohort (which exhibited evidence of an effect for claim values but not unit prices).[148] These findings do not amount to a significant breach of the identifying assumption.

A.112 Finally, we are not persuaded that LVG-3's approach of including individual cohorts in the sample conditional on them meeting the pre-trends test is valid. Recent econometrics research has shown that conditioning the analysis on 'passing' a pre-trends test induces a significant selection bias known as pre-test bias (Roth, 2022).[149]

[144] [✂] response to draft working paper.
[145] Sun & Abraham (2021), assumption 1.
[146] See for example Braghieri, Luca, Ro'ee Levy, and Alexey Makarin. 2022. "Social Media and Mental Health." American Economic Review, 112 (11): 3660–93.
[147] [✂] response to draft working paper.
[148] [✂] response to draft working paper.
[149] Roth, Jonathan, 2022. Pre-test with Caution: Event-study Estimates After Testing for Parallel Trends. Am. Econ. Rev. Insights 4 (3), 305–322

Using the median

A.113 LVG-3 submits that, in those of the CMA's sensitivities that rely on a level of aggregation, the dependent variable should be the median rather than the mean claim value. This reduces the estimated acquisition effect on claim values from 3.8% to 2.8%.[150]

A.114 **CMA assessment**. We do not think that this adjustment is justified. If, upon acquisition, a practice becomes more likely to issue very expensive claims, or less likely to issue very cheap claims, this should be considered as economically meaningful, and part of the hypothesis that we are seeking to test, rather than an 'anomaly' that must be corrected. In any case, these models were developed as sensitivities on our main results, which in our view remain more reliable as they control for other additional determinants of claim values.

Motivation for using never treated group as control

A.115 LVG-3 submit that the CMA does not motivate a conceptual basis on which to expect that its control group (independents that are not acquired) is comparable with the treatment group. LVG acquisition decisions are not random, so it is likely that acquired sites differ to some extent from sites that are not acquired. Thus, even if a statistical parallel trends test is satisfied, there may be unobserved differences between the control and treatment groups that means that the parallel trends assumption does not in fact hold (that is, the evolution of outcomes in the two groups are not comparable after acquisition).[151]

A.116 **CMA assessment**. Our DiD model can accommodate differences in both observed and (time-invariant) unobserved characteristics between acquired practices and practices treated as control. The only requirement for this analysis is that, conditional on observed characteristics, the potential outcomes of interest for acquired and control practices change in the same way over time (the 'parallel trends assumption'). Our leads and lags tests show that this assumption holds in the periods preceding the acquisitions. Thus, for LVG-3's concern to be valid, we would need to believe that LVGs systematically targeted practices which, even if they exhibited parallel trends with the control group prior to acquisitions, were expected to increase prices to a greater extent than the control group after the acquisition (irrespective of the acquisition itself). We consider that this is a very improbable scenario, especially as our analysis relates to almost 2000 acquisitions staggered over a 10-year period, and no LVGs have provided evidence that they targeted acquisitions in this way.

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[150] [✂] response to draft working paper.
[151] [✂] response to draft working paper.

Reliability of insurance data

A.117 LVG-3 stated that it has reservations regarding the reliability of insurance data because: (i) some insurers may have sought to place restrictions on price increases charged by practices within their preferred network; (ii) some insurance policies place restrictions on the type of conditions or treatments covered, the duration of coverage, or the amount claimed; and (iii) insurance customers are more likely to demand higher quality treatment.[152]

A.118 **CMA assessment**. We are not persuaded that these features of the data invalidate our analysis. If, upon being acquired by LVG, a practice becomes part of an insurer's preferred network, and if the insurer has placed restrictions on price increases within that network, then our analysis is more likely to understate the effect of corporate acquisitions on other customers (customers who are uninsured or insured by other insurers). If some insured customers face limits on the amount they can claim or the duration of their coverage, then this reduces the difference in incentives between them and uninsured customers. Finally, we acknowledge that insured customers are more likely to accept recommendations for higher-quality treatments, but we do not consider that this invalidates the analysis as this also pertains to pre-acquisition incentives.

Definition of FOPs and referral centres

A.119 LVG-3 submitted that there is no clear industry standard on the distinction between FOPs and referral centres, and any distinction between both is arbitrary. Some LVG-3 practices provide both typical primary care services and more advanced care services. By contrast, many independent practices might only provide typical primary care services as they lack the capacity or investment required to provide the more advanced care service or even basic OOH services.

A.120 **CMA assessment.** We are aware that there is no commonly accepted definition of FOPs or referral centres. For this reason, we have used the categorization provided by Insurer 2 throughout. While this categorization may not coincide with the categorizations used internally by suppliers, it has the merit of being applied consistently across providers and across time. In any case, the results of our DiD estimation are robust to any mis-categorizations, provided these are stable over time. This is because the model includes practice-level fixed effects which control for any (time-invariant) unobserved factors. If, upon acquisition by a LVG, practices start offering more advanced treatments (including on a referral basis), this will indeed be captured in our estimate of the acquisition effect on first-year treatment

[152] [✂].

costs and, albeit to a lesser extent, our estimate of the acquisition effect on average prices.

LVG-4

No control for OOH provision

A.121 LVG-4 submitted that the Insurer 2 and Insurer 1 datasets do not identify whether a treatment is conducted in ordinary business hours or out-of-hours (OOH), which is several times more expensive and tends to take place more commonly at LVG practices. As such, groups which offer a higher proportion of OOH treatments in their treatment mix will be assigned by the CMA as exhibiting a higher price. The inclusion of a 24-hour control variable for 24-hour clinics decreases [✂].[153]

A.122 **CMA assessment.** It is correct that suppliers tend to charge higher prices for services provided OOH, and LVG-4's internal documents indicate that [✂].[154] However, our analysis of data collected from all suppliers suggests that independents are more likely to offer the OOH in-house than LVGs,[155] so we are not persuaded that the omission of this variable at the claim level analysis is likely to understate the effect. We also consider that the sensitivity put forward by LVG-4 to address this point is invalid: the variable used to control for 24-hour provision is only available for LVGs, and LVG-4 assumes that independents never provide this service, which is evidently incorrect.

Controlling for Ongoing Claims in First-Year Treatment Cost Regression

A.123 LVG-4 submitted that the first-year treatment cost regression in Table 1.5 controls for whether the customer has submitted a previous claim for the same condition, even though the intended analysis was not meant to include this variable. When this control is removed, the coefficient for [✂] first-year treatment costs decreases [✂].

A.124 **CMA assessment.** This point is correct and we have incorporated this adjustment in the specifications of our cross-sectional regressions for first-year treatment costs (by excluding the control for whether the customer has submitted a previous claim for the same condition). This adjustment reduces the coefficient for LVG-4, and increases the coefficients for [✂] other LVGs. This adjustment does not change the overall picture, as the results consistently show that [✂] LVGs have higher first-year treatment costs compared to independents.

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[153] [✂] response to draft working paper.
[154] [✂] internal presentation.
[155] CMA working paper 'Analysis of local competition', Figure 3.1

[✂]

A.125 [✂].[156]

A.126 [✂].

A.127 [✂].

A.128 [✂].

A.129 [✂].

Figure A.13:

[✂]

A.130 [✂].

RESET test

A.131 LVG-4 stated that the CMA's cross-section regressions fail a RESET test (Regression Equation Specification Error Test), and that this indicates that the CMA's regression model are mis-specified.[157]

A.132 **CMA assessment.** The RESET test adds square or cubic transformations of the independent variables to a model, and tests whether the coefficients on these additional variables are statistically different from zero. But in the CMA's cross-sectional regressions, with the exception of pet age, all the independent variables are binary 'dummies'; that is, variables that can only take the values zero or one. The RESET test is evidently not relevant or insightful for such variables (the square of zero is zero, and the square of one is one). Therefore, we do not consider that it is pertinent for these models. The failure of the RESET test for these models only indicates that claim values and unit prices rise non-linearly with pet age, which is inconsequential for the analysis.

The CMA's findings are inconsistent with actual market outcomes

A.133 LVG-4 stated that the CMA's result lead to economically implausible implications. In 2024, LVG-4 generated revenue of £[✂], with EBITDA of £[✂] and pre-exceptional EBIT of £[✂]. If the CMA's upper-end estimate of a 25% unit price differential were applied as a counterfactual, this would, all else equal, equate to an £[✂] in revenue, as there is no reason to believe that LVG-4 operates inefficiently or is incurring costs beyond what is strictly necessary for the operation of the

[156] [✂].
[157] [✂] response to draft working paper.

business. Under this scenario, LVG-4 would have recorded negative EBITDA and EBIT figures, meaning the business would have been operating at a financial loss.[158]

A.134 **CMA assessment.** We do not think that the thought experiment proposed by LVG-4 is valid or insightful. We are not claiming that the coefficients of our cross-sectional regressions should be interpreted as a measure of 'excess pricing' above fair or competitive levels. We make it clear that this simple analysis does not control for unobserved determinants of prices and claim values, which is the motivation for using a DiD method. We also note that LVGs in general, and LVG-4 in particular, tend to acquire practices that are larger and register higher claim values and unit prices pre-acquisition.

A.135 Our DiD estimate of the effect of corporate acquisitions on unit prices for LVG-4, which controls for time-invariant factors, [✂]%. Applying this reduction on LVG-4's turnover would not result in a negative EBITDA. To be clear, we are not saying that our DiD estimates should be interpreted as measures of excess pricing, insofar as they might not control for changes in costs or quality at acquired practices. But even if they were given this strong interpretation, they would not falsify the thought experiment proposed by LVG-4.

LVG-5

LVG-5 has lower 'prices' than LVGs and most independents

A.136 LVG-5 submitted that their analysis of the Insurer 2 data shows that LVG-5 has lower claim values than LVGs and most independents.[159] This claim is based on: (i) a comparison of average claim values, first-year treatment costs, and average prices between independents, LVGs excluding LVG-5, and LVG-5 (for all practice types);[160] and (ii) the same comparison with the sample split between urban and rural areas (also for all practice types).[161]

A.137 **CMA assessment**. The alternative analysis put forward by LVG-5 does not control for differences in the mix of practice types, conditions, or treatments, which are obvious determinants of the variables of interest (claim values, first-year treatment costs, and unit prices). If LVG-5 practices tend to treat less severe conditions, or apply less complex treatments than independents or other LVGs, LVG-5 might report lower average claim values (or first-year treatment cost or unit prices), even if in reality its treatment costs for specific conditions or treatments are higher. For this reason, we think that it is more appropriate to place weight on our regression

[158] [✂] response to draft working paper.
[159] [✂] response to draft working paper.
[160] [✂] response to draft working paper.
[161] [✂] response to draft working paper.

analysis, which controls for such variables, than on LVG-5 comparison of simple averages, which does not. While the treatment categories reported in the Insurer 1 data can sometimes cover different types of treatments, controlling for such variables represents an improvement on a simple comparison of unconditional averages.

A.138 With respect to the second argument, we recognise that the location of a practice can be an important determinant of its prices and claim values, and we have changed the specifications of our cross-sectional regression to include available controls.

The CMA's comparison of first-year treatment costs for LVG-owned and independent practices (Figure 1.5) is unlikely to be robust.

A.139 LVG-5 submits that the comparison of first-year treatment costs for LVG-owned and independent practices is not robust. [162] This claim is based on the following arguments: (i) [✂];[163] (ii) this analysis drops a large number of observations to form a balanced sample;[164] (iii) some conditions are only present in one of the two groups;[165] and (iv) the analysis does not control for changes in the distribution of conditions within practices.[166]

A.140 **CMA assessment.** The first point raised by LVG-5 is incorrect: [✂] The other points raised by LVG-5 are correct, and the most important one (the effect of sample restrictions) is acknowledged in the working paper. These limitations are reflected in the weight we have put on this analysis in reaching our conclusions.[167] The working paper uses this analysis as a simple visual illustration of changes in first-year treatment costs for practices that were acquired by LVGs and practices that remained independent, which motivates the use of the difference-in-difference methodology. Our conclusions regarding the effect of corporate acquisitions places more weight on the more detailed difference-in-difference analysis, which does not restrict the sample of observations in the same way, and controls for compositional effects.

The cross-sectional regression of unit prices can be simplified.

A.141 LVG-5 submits that the large number of condition and breed fixed effects used in the CMA's cross-sectional regressions of claim values and first-year treatment costs creates a risk of bias.[168] LVG-5 propose an alternative approach that groups

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[162] [✂] response to draft working paper.
[163] [✂] response to draft working paper.
[164] [✂] response to draft working paper.
[165] [✂] response to draft working paper.
[166] [✂] response to draft working paper.
[167] Paragraph 2.29.
[168] [✂] response to draft working paper.

conditions into 13 broad disciplines, under which LVG-5 does not appear to be more expensive than independents.

A.142 **CMA assessment.** This argument is incorrect from an econometric viewpoint: the addition of many covariates to a linear regression model does not create a bias (in the econometric sense that the expected value of the regression coefficient is not necessarily equal to its true value).[169] Indeed, if these covariates are correlated with the independent variable of interest and affect the dependent variable, then *not* including these variables makes the estimator of the main parameter biased. When a sample contains relatively few observations, using a large number of covariates in a linear model can reduce the precision of model estimates (in the econometric sense that it increases the variance of the estimator), but all the samples used by the CMA for the purpose of estimating these models have more than 195,000 observations, and therefore can accommodate a large number of covariates. There is therefore no valid reason not to use the granular categorization of conditions provided by the insurers.

The CMA's cross-sectional regression of unit prices is 'unstable'

A.143 LVG-5 submits that the CMA does not justify the way it 'constructed' the treatment variable used in this regression, and that controlling for the 'incident' (condition) variable instead of the treatment type produces very different results, where LVG-5 appears cheaper.[170]

A.144 **CMA assessment.** The alternative specification proposed by LVG-5 – controlling for condition instead of treatment type – is not economically meaningful. The regression put forward by the CMA seeks to answer a simple question: 'do LVG-owned practices charge higher prices for given treatment categories than independent practices'. The specification proposed by LVG-5 would not answer that question. LVG-5 argue that it might identify cases where 'a vet can successfully treat the same type of incident at a lower price than another vet',[171] but it is difficult to see how the specification it proposes could have that interpretation. The total cost of treating a condition is dependent on the number and type of treatments applied, and the unit price of these treatments. If a vet applies a larger number of cheaper treatments to treat a condition than other vets, it may come across as cheaper in the analysis proposed by LVG-5, even if the total

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[169] In support of their argument, LVG-5 cite a paper from an epidemiology review (Brennan C. Kahan, Michael O. Harhay, "Many multicenter trials had few events per center, requiring analysis via random-effects models or GEEs, Journal of Clinical Epidemiology", Volume 68, Issue 12, 2015, pp. 1504-1511). This paper simply makes the points that methods like fixed effects 'often require a large number of patients or events per centre to perform well', which is uncontroversial. The paper highlights that, in the medical studies it reviewed, the median number of observations per 'centre' was 3, which is evidently too low and significantly lower than the number of observations per covariate used in the CMA's model (which is above 200 in all specifications). The paper does not demonstrate that the inclusion of many fixed effects creates a risk a bias in the econometric sense (of the expected value of the coefficient not being equal to its true value).

[170] [✂] response to draft working paper.

[171] [✂] response to draft working paper.

cost of treatment is the same or higher. Our analysis of first-year treatment costs attempts to capture all observed drivers of the costs of a course of treatment.

A.145 The treatment variable used by the CMA in this regression is not 'constructed' - it is based on the categorisation used [✂] by Insurer 1 and reported in its dataset. [✂]. The CMA simply concatenated the [✂] variables to obtain a unique treatment variable for each observation. For example, an item may be coded as 'Diagnostics' under the first level of aggregation, 'Endoscopy' under the second level, and 'Laparoscopy' under the third level. This item will receive the treatment identifier 'Diagnostics_Endoscopy_Laparoscopy'. This is a simple, mechanical manipulation that does not add or subtract any information. It is necessary because the model can only use one variable to denote a treatment (and some treatments are only coded under the second level of aggregation, so we cannot solely use the third).

The CMA's cross-sectional regression of prices shows that LVG-5 has lower prices than independents when all observations are included

A.146 In the hearing on [✂] 2025, LVG-5 presented analysis showing that if we run a cross-sectional regression on prices using all individual observations, instead of measuring average prices for each treatment category, [✂] practices appear to be cheaper than independents. Specifically, the regression coefficient for [✂] becomes negative when using this approach.

A.147 **CMA assessment.** The sample used for our initial analysis, which LVG-5 also used to estimate their proposed sensitivity, contained all treatment categories reported in the Insurer 1 data, including those labelled as 'other'. As discussed in paragraph A.28, we have now removed these categories from the sample. When these categories are removed, the coefficient on LVG-5 remains positive whether the regressions are performed on average prices or on all observations. When using average prices, we find that prices are [✂] higher at LVG-5 practices than at independents for all practice types (and for FOPs only, respectively) (paragraph 2.42). When we use all observations, the estimated difference is slightly smaller but remains positive—[✂]% for all practice types and [✂]% for FOPs only (Table A.9 below). The effect is smaller because, by using all observations, we assign greater weight to more frequent treatments, which have a stronger influence on the results.

A.148 In the sections above, we continue to report results based on average prices, but for completeness, we also present the results below using all observations.

Table A.9: Cross-sectional regression for prices using all observations (Insurer 1 data)

	Log (Price)		Log (Price)	
	All Types	FOPs	All Types	FOPs
LVG	0.165***	0.160***		
	(0.012)	(0.012)		
[✂]			[✂]	[✂]
			[✂]	[✂]
[✂]			[✂]	[✂]
			[✂]	[✂]
[✂]			[✂]	[✂]
			[✂]	[✂]
[✂]			[✂]	[✂]
			[✂]	[✂]
[✂]			[✂]	[✂]
			[✂]	[✂]
[✂]			[✂]	[✂]
			[✂]	[✂]
Controlling for:				
Pet Characteristics (Age, Breed, Condition)	Yes	Yes	Yes	Yes
Practice Type	Yes	No	Yes	No
Ongoing Claim	Yes	Yes	Yes	Yes
Rural/urban classification	Yes	Yes	Yes	Yes
Local authority	Yes	Yes	Yes	Yes
Treatment	Yes	Yes	Yes	Yes
Observations	5,073,714	4,453,156	5,073,714	4,453,156
Adjusted R^2	0.507	0.517	0.508	0.519

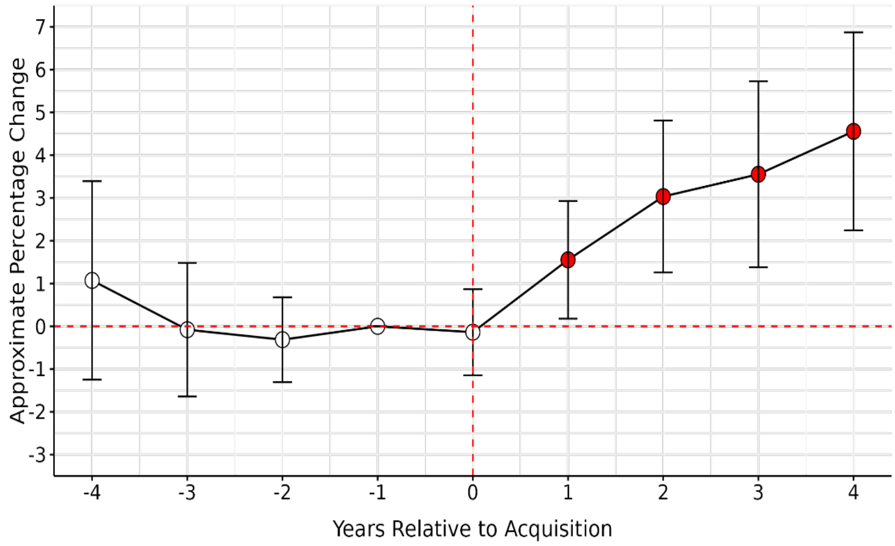
Source: CMA analysis using Insurer 1 data.
*Note: Standard errors are clustered at the practice level and are reported in parentheses. ***, **, and * correspond to statistical significance at the 1%, 5%, and 10% levels, respectively.*

Annex B: Further Robustness Checks

Inclusion of other practice types in the claims value model.

B.1 In this sensitivity, we expand the sample to include referral centres. The 'leads and lags' results for claim values for all practice types are shown in Figure B.1, while Figure B.2 provides the 'leads and lags' coefficients for individual LVGs. These results also point to an increase in claimed values after acquisitions, of a similar order of magnitude to that estimated for FOPs.

Figure B.1: Leads and lags results for claim values, all practice types, all LVGs (Insurer 2 data)



Source: CMA analysis using Insurer 2 data.
Note: The figure plots the coefficient estimates together with their 95% confidence intervals. The dots are red if coefficients are significantly different from zero. Standard errors are clustered at the practice level

Figure B.2:

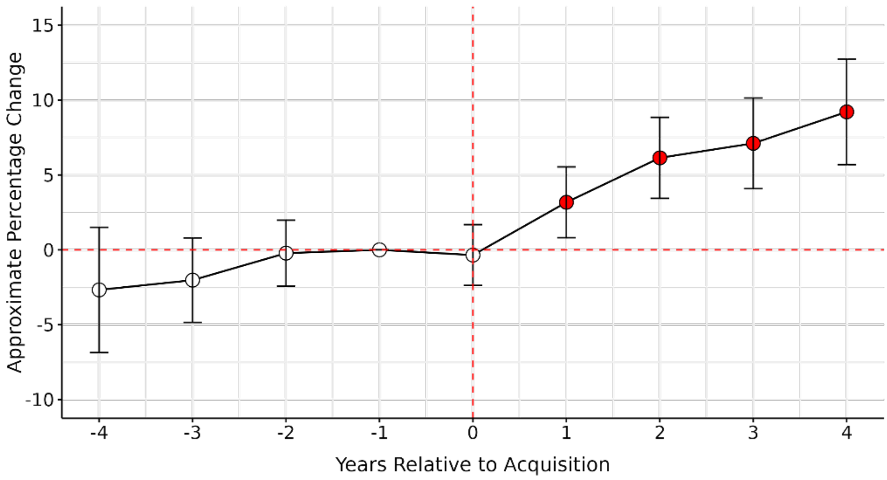
[✂]

Using a DiD model for claim values with panel data at the practice level.

B.2 In this sensitivity, we estimate a simpler DiD model where the dependent variable is the average claim value at a practice and there are no claim-level covariates. The resulting data set has an explicit 'panel structure', in that it has successive observations for each unit of observation (each practice), instead of a 'repeated cross-section'. This is the more 'traditional' way of presenting DiD models. We present these results for completeness, albeit in our case this approach is likely to be less efficient than our baseline model, insofar as it does not make full use of the information we have about the determinants of claim values.

B.3 Figure B.3 shows that this approach generates a higher estimate of the treatment effect compared to our baseline specification (10% instead of 5%, four years after an acquisition). This difference in results between our baseline approach (which controls for the condition treated) and this panel data approach (which does not) might imply that acquired practices start treating a larger share of conditions that are more expensive to treat.

Figure B.3: Effect of a corporate acquisition on claim value: analysis using simple averages (Insurer 2 data)

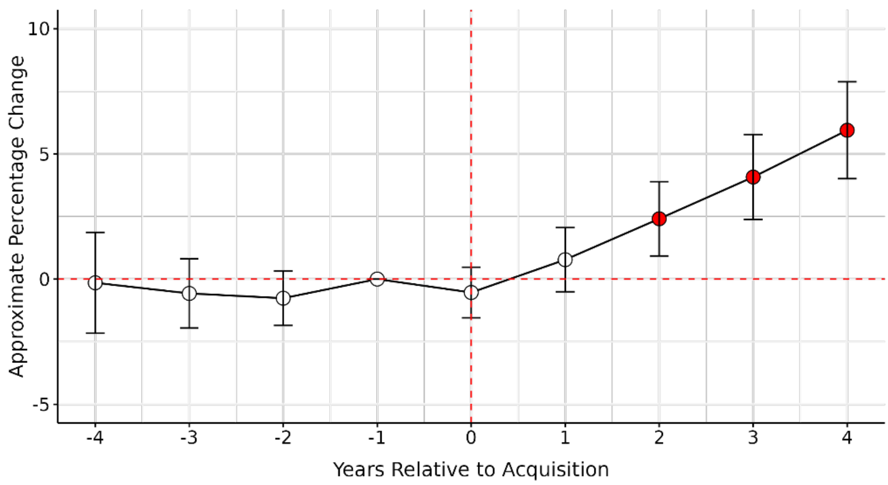


Source: CMA analysis using Insurer 2 data.
Note: The figure plots the coefficient estimates together with their 95% confidence intervals. The dots are red if coefficients are significantly different from zero. Standard errors are clustered at the practice level.

Using a DiD model for claim values with panel data at the practice-condition level.

B.4 The previous model with panel data does not account for potential shifts in the types of conditions treated. For instance, if practices begin treating more complex conditions after the acquisition, relying solely on simple averages could yield confounded results. To address this, we conduct a DiD analysis on panel data at the practice-condition-year level, thereby controlling for condition composition over time. The results, displayed in Figure B.4, closely align with those from earlier models.

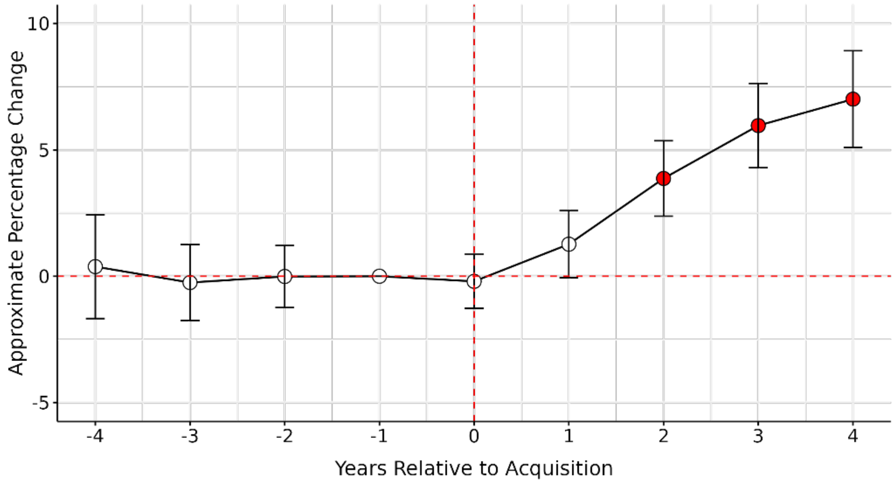
Figure B.4: Effect of a corporate acquisition on claim value: condition-specific averages by practice-year



Source: CMA analysis using Insurer 2 data.
Note: The figure plots the coefficient estimates together with their 95% confidence intervals. The dots are red if coefficients are significantly different from zero. Standard errors are clustered at the practice level.

B.5 Furthermore, in Figure B.5 we again analyse the average claimed amount by practice-condition-year, this time weighting the DiD regression by the ratio of new claims to total claims. Claims can represent either new or ongoing conditions, with ongoing claims generally being much smaller in value than new claims. Therefore, the proportion of new versus ongoing conditions should be considered when aggregating data and using a panel data approach. Figure B.5 shows that the results remain consistent with those reported previously.

Figure B.5: Effect of a corporate acquisition on claim value: condition-specific weighted averages by practice-year



Source: CMA analysis using Insurer 2 data.
Note: The figure plots the coefficient estimates together with their 95% confidence intervals. The dots are red if coefficients are significantly different from zero. Standard errors are clustered at the practice level.

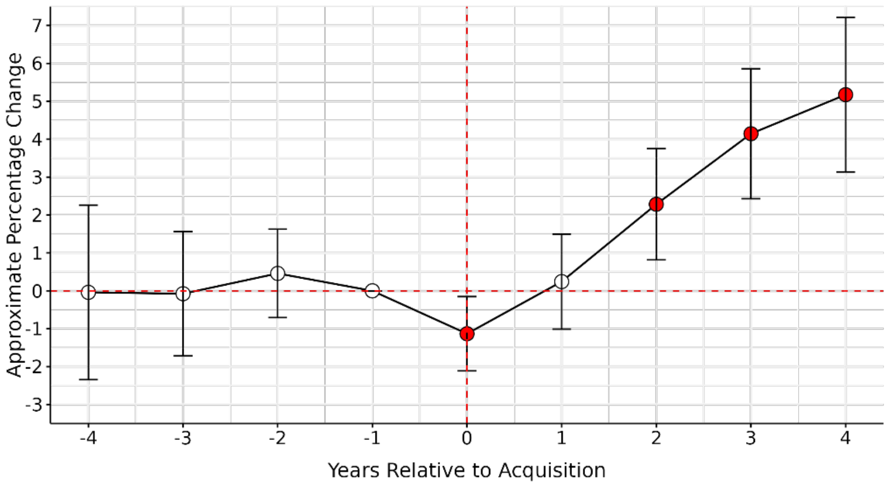
Using first-year treatment costs as the dependent variable.

B.6 We applied the model used for claim values (paragraph 3.10) using first-year treatment costs rather than claim values as the dependent variable (the rationale and methodology underpinning this metric is set out in paragraph 2.20). The effects averaged over all years of observations are presented in Table B.1 below, and the corresponding 'leads and lags' results for individual years are presented in Figure B.6 (all LVG5s) and Figure B.7 (individual LVGs). The aggregate results for all LVG5s are broadly consistent with our baseline specification, indicating an increase in treatment costs of 2.6% across all years, and 5% after four years. The results for individual LVGs are less precise and more nuanced: there is clear evidence of a positive effect for [✂] and [✂], some evidence of a negative effect for Linnaeus, and no clear evidence of an effect for [✂]. For [✂], the results are ambiguous: only one of the lags coefficients is positive and statistically different from zero, and the coefficient for the year of acquisition is negative and statistically significant.

Table B.1:

[✂]

Figure B.6: Leads and lags results for first year treatment cost, FOPs only, all LVG5s (Insurer 2 data)



Source: CMA analysis using Insurer 2 data.
Note: The figure plots the coefficient estimates together with their 95% confidence intervals. The dots are red if coefficients are significantly different from zero. Standard errors are clustered at the practice level.

Figure B.7:

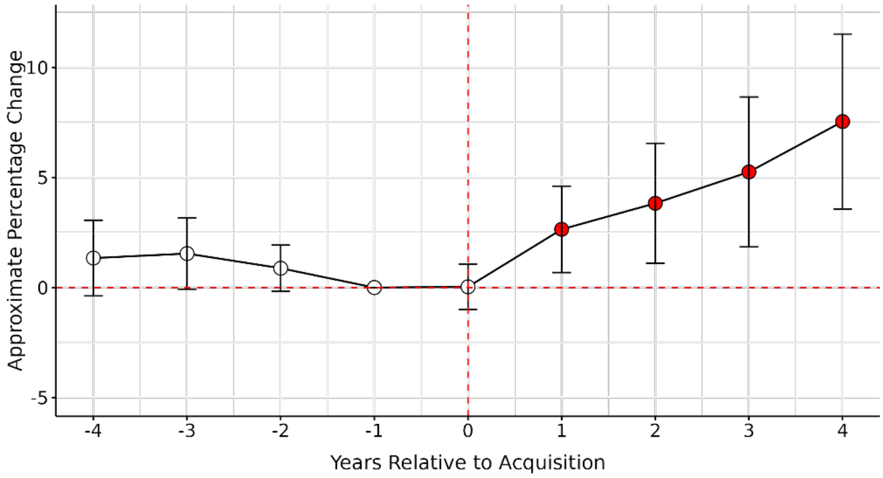
[✂]

DiD analysis of prices using all observations and covariates

B.7 In this robustness test, we re-estimate the DiD regression for prices using all observations, rather than measuring an average price per treatment category. We include the same covariates as in the claim value model: the condition treated, the breed and age of the pet, and a binary indicator for whether the customer has previously submitted a claim for the same condition. In addition, we control for the treatment type, as in the earlier analysis of average prices. Standard errors are double-clustered by practice and treatment type.[172] The figure below shows that the results remain consistent. The estimated acquisition effect on prices is 5.5% and statistically significant at the 1% level.

[172] In the current model, which includes all observations, many treatments exhibit little or no price variation across claims or pets within a given year. For example, some drugs have fixed prices that do not vary with pet characteristics or conditions. This limited variation can lead to correlation in the residuals within treatment categories, both within and across practices. To account for this potential clustering of errors, we cluster standard errors by both practice and treatment type. This ensures that standard errors are not underestimated and that statistical significance is not overstated.

Figure B.8: Acquisition effect on prices, using all observations and including covariates (Insurer 1 data)



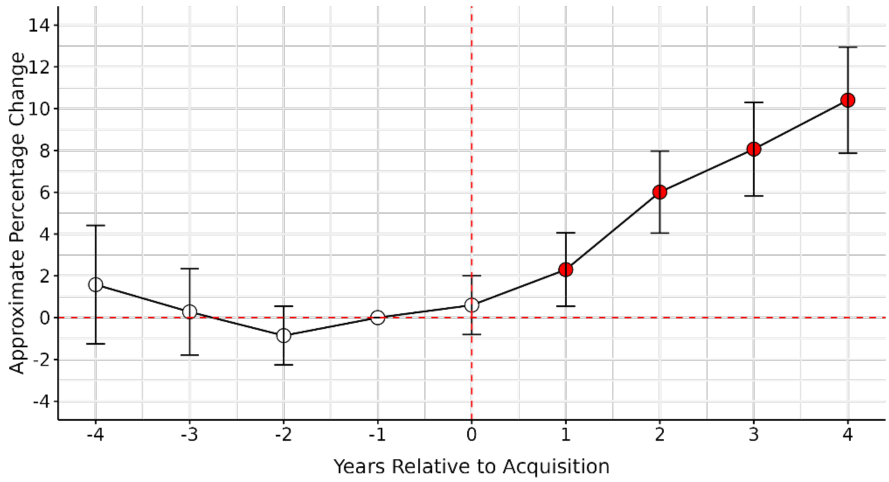
Source: CMA analysis using Insurer 1 data.
Note: The figure plots the coefficient estimates together with their 95% confidence intervals. The dots are red if coefficients are significantly different from zero. Standard errors are double-clustered by practice and treatment type.

DiD analysis of claim values after restricting the sample to treatments where the start and end dates are identical

B.8 In this robustness test, we restrict the analysis for claim values to treatments with identical start and end dates. These treatments are more likely to include only one invoice because the treatment occurred within a single day or visit, reducing the likelihood of multiple invoices being aggregated over time. This is an alternative way of mitigating potential issues arising from different practices using different policies in terms of grouping invoices into claims (in addition to including a control for the ongoing status of a claim in our main model of claim values and the estimation of an alternative model for first-year treatment costs). The estimated acquisition effect for claim values is now higher (7%) and more comparable in magnitude with the price effect estimated when using the Insurer 1 data (7.4%).

B.9 We acknowledge that this result is based on a smaller sample of claims and conditions and may not necessarily be generalisable to all claims or conditions. Specifically, the sample size decreases from 2,531,659 to 1,037,855 observations, with the number of practices included in the analysis decreasing from 4,941 to 4,659, and the number of conditions considered reducing slightly from 303 to 301. Nevertheless, this remains a substantial sample for the analysis, suggesting that our baseline model for claim values may underestimate the acquisition effect.

Figure B.9: Acquisition effect on claim values, restricting the analysis to treatments with identical start and end dates (Insurer 2 data)



Source: CMA analysis using Insurer 2 data.
Note: The figure plots the coefficient estimates together with their 95% confidence intervals. The dots are red if coefficients are significantly different from zero. Standard errors are clustered at the practice level.

DiD analysis of claim values using new data from Insurer 1

B.10 On 4 December 2024 we received a new dataset from Insurer 1 that includes all their claim submissions, [✂]. This dataset is comparable to the Insurer 2 dataset, in that it reports total claim amounts but does not include detailed information on individual items claimed. [✂].

B.11 Insurer 1 claims are submitted through [✂]. As discussed in paragraph 2.17, [✂]. Using a DiD approach, we tested whether this proportion changed differently for acquired versus independent practices. The dependent variable is the proportion of [✂], and the analysis focuses on FOPs. The results (set out in figure below) show a significant [✂].

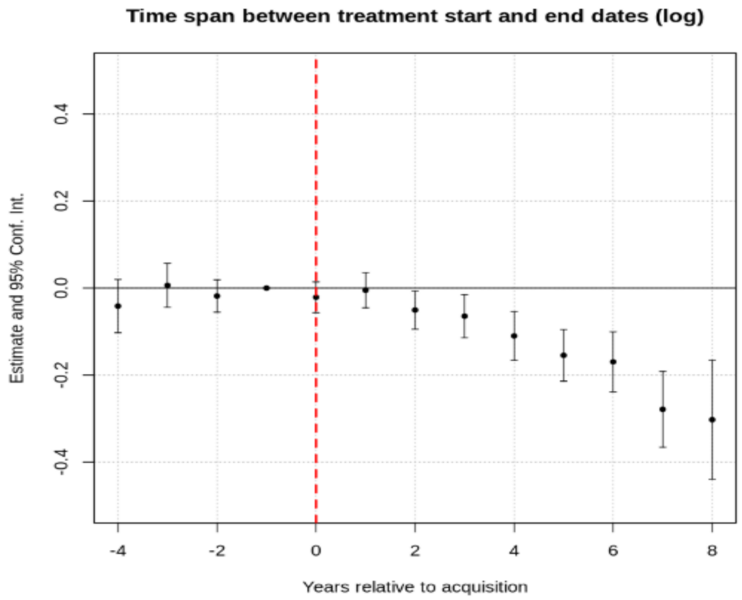
Figure B.10:

[✂]

B.12 When comparing [✂] in the new [✂] dataset with the [✂] data used in the working paper, we observed some inconsistencies in how invoices are grouped across claims. Specifically, it appears that some vets may submit multiple invoices within a single claim, while others may submit claims as expenses arise. This variation seems to be influenced by the submission method used. [✂].

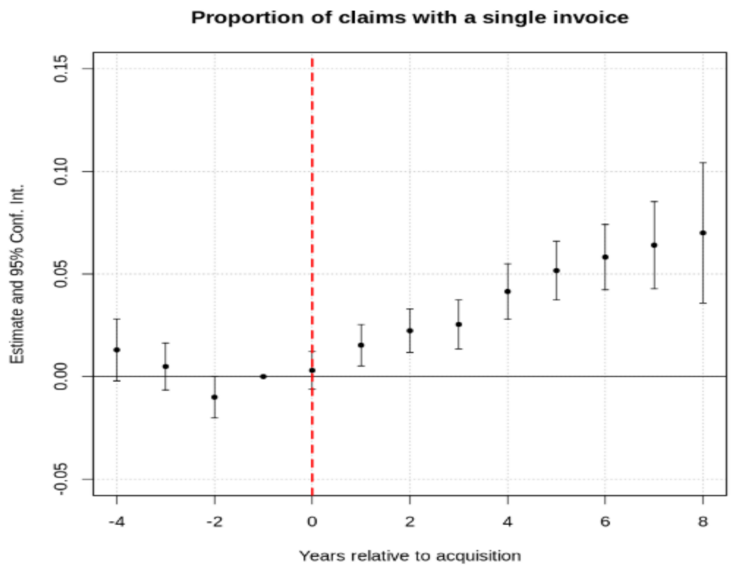
B.13 To formally test this, we analyse the time span between treatment start and end dates as a proxy for the number of invoices aggregated within a claim. The new dataset includes these dates, which correspond to the first and last invoice in each claim, but does not indicate how many invoices are bundled together. Using this time span as a proxy for invoice grouping, we run a DiD regression and find that, while the proportion of [✂]. This suggests that acquired practices [✂], aggregating fewer invoices. Consistent with this, we also observe an increase post-acquisition in the proportion of claims where the start and end dates are the same—more likely representing single-invoice submissions.

Figure B.11: Acquisition effect on the time span between treatment start and end dates (Insurer 1 data)



Source: CMA analysis using Insurer 1 data.

Figure B.12: Acquisition effect on the proportion of claims with a single invoice (Insurer 1 data)

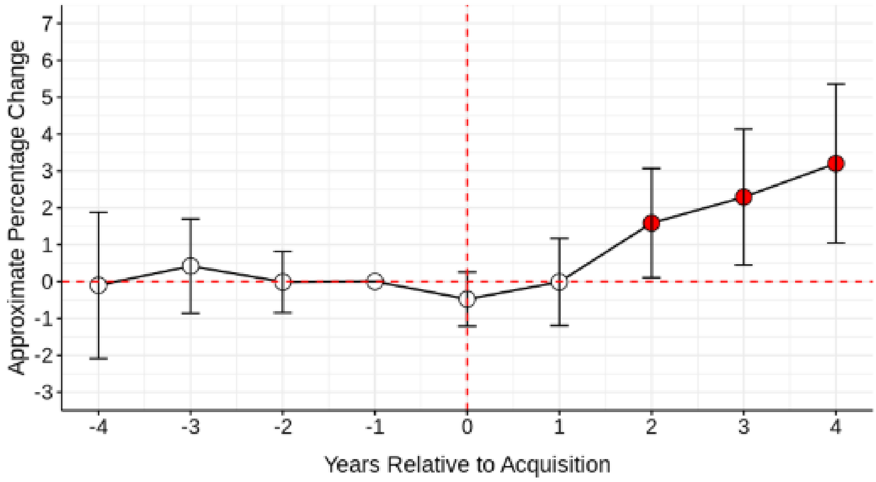


Source: CMA analysis using Insurer 1 data.

B.14 A simple model of claim values, of the type we have used for the Insurer 2 data, could lead us to observe lower claim values around the acquisition—not because of actual changes in prices, but due to differences in how invoices are grouped. This highlights the importance of controlling for both the time span between treatment start and end dates and the submission method used in the DiD regression of claim values. These controls help mitigate potential bias arising from differences in invoice grouping across practices, ensuring that observed changes in claim values reflect true price effects rather than submission practices.

B.15 We then estimate the DiD regression for claim values using this newly expanded Insurer 1 dataset. We include the same covariates as in the Insurer 2 claim value model: the condition treated, the breed and age of the pet, and a binary indicator for whether the customer has previously submitted a claim for the same condition. In addition, we control for the time span between treatment start and end dates ("treatment duration"), which is log-transformed, and include indicators for the submission method used. The figure below shows that the results are broadly consistent with our previous findings, confirming an increase in claim values following acquisition. However, the estimated effect of the acquisition is smaller than what was previously found using the Insurer 2 data.

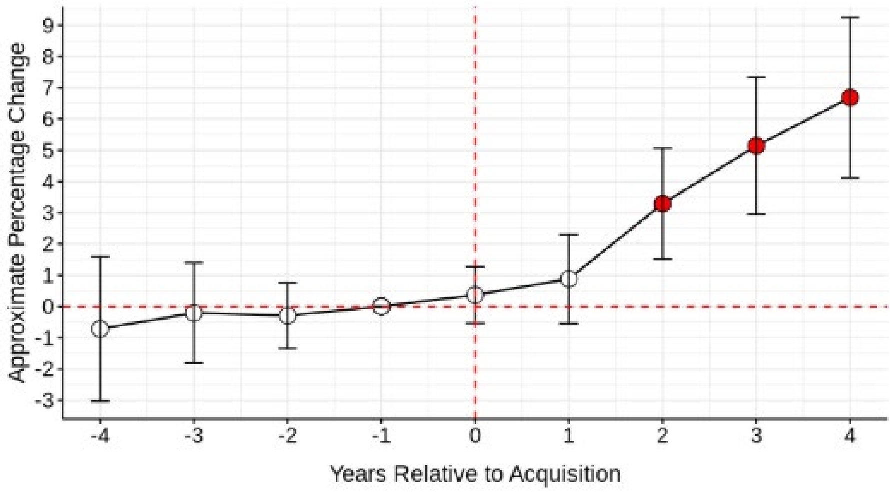
Figure B.13: Acquisition effect on claim values (Insurer 1 data)



Source: CMA analysis using Insurer 1 data.
Note: The figure plots the coefficient estimates together with their 95% confidence intervals. The dots are red if coefficients are significantly different from zero. Standard errors are clustered at the practice level.

B.16 In addition, as previously done for Insurer 2 in the robustness test described in paragraph B.8, we restrict the analysis of claim values to treatments where the start and end dates are identical. Consistent with the Insurer 2 findings, we observe a much stronger acquisition effect when applying this restriction. However, this result is based on a smaller sample of claims and conditions, and may not be fully generalisable to all claims or conditions. Specifically, the sample size decreases from [✂] observations, the number of practices included in the analysis declines from 4,225 to 4,084, and the number of conditions considered reduces slightly from 105 to 102.

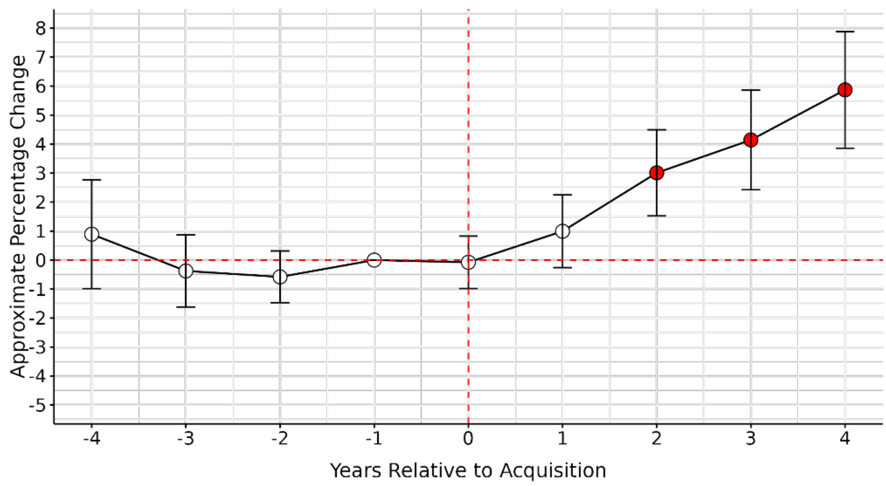
Figure B.14: Acquisition effect on claim values, restricting the analysis to treatments with identical start and end dates (Insurer 1 data)



Source: CMA analysis using Insurer 1 data.
Note: The figure plots the coefficient estimates together with their 95% confidence intervals. The dots are red if coefficients are significantly different from zero. Standard errors are clustered at the practice level.

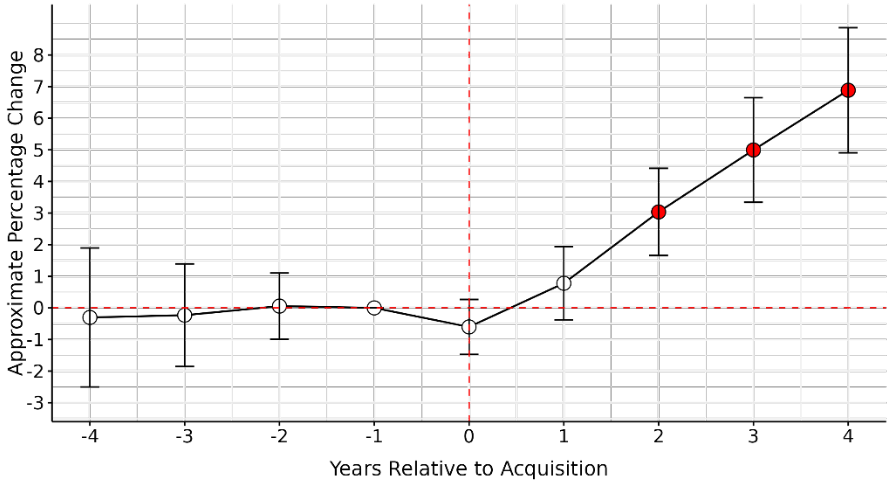
B.17 For transparency, we also re-estimate the DiD models for claim values and first-year treatment costs using the Insurer 2 data, this time controlling for the time span between treatment start and end dates. The results show that the estimated acquisition effect is stronger when we control for treatment duration. Specifically, the estimated effect on claim values increases from 3.2% in the baseline model to 3.6%, and remains statistically significant at the 1% level. Similarly, for first-year treatment costs, the acquisition effect increases from 2.6% to 3.7%, also statistically significant at the 1% level.

Figure B.15: Acquisition effect on claim values, controlling for treatment duration (Insurer 2 data)



Source: CMA analysis using Insurer 2 data.
Note: The figure plots the coefficient estimates together with their 95% confidence intervals. The dots are red if coefficients are significantly different from zero. Standard errors are clustered at the practice level.

Figure B.16: Acquisition effect on first-year treatment costs, controlling for treatment duration (Insurer 2 data)



Source: CMA analysis using Insurer 2 data.
Note: The figure plots the coefficient estimates together with their 95% confidence intervals. The dots are red if coefficients are significantly different from zero. Standard errors are clustered at the practice level.

Annex C: Assessment of second-round submissions from LVGs

C.1 We shared a second, revised version of this analysis as a working paper with the LVGs' advisers (within a confidentiality ring) on 6 May 2025. We have received responses from four of the six LVGs — namely, CVS, IVC, Medivet, and VetPartners.

C.2 An extract of the revised working paper was also shared with BVA, BSAVA, BVNA, FIVP and the RCVS on 14 July 2025.

C.3 The LVGs and their advisers have raised a number of concerns, some relating to the interpretation of our findings, others of a more technical nature. Many of the comments reiterate points previously made in response to the first draft of the working paper. These points were addressed in Annex A, and we consider our earlier responses to remain valid; accordingly, we do not repeat them.

C.4 Some of the other stakeholders' comments also overlap with issues previously raised by the LVGs or their advisers, and these are also not discussed here.

C.5 One recurring theme is the claim that the observed price effects are driven by improvements in quality, specifically, that LVGs raise clinical or service standards and make material investments at the practices they acquire. This argument was discussed in response to the previous submissions (see paragraphs A.7–A.21).

C.6 Another recurring theme is the claim that our price analysis is not conducted on a like-for-like basis, due to the treatment categories used being heterogeneous in terms of the types of items they include. This point was addressed in our previous response (set out in paragraphs A.22–A.31).

C.7 A further repeated concern is that our results may be driven by outliers. This issue was addressed in our earlier response (set out in paragraphs A.35–A.41). Another concern raised is that our sample period includes the COVID-19 pandemic, which could potentially bias our results. This point was addressed in paragraphs A.42–A.46.

C.8 The remainder of this annex focuses on the new arguments, evidence, or analysis that go materially beyond what was submitted previously. We address these points in turn below.

LVG-1

Effect of COVID-19 and other macro developments

C.9 LVG-1 argues that our sensitivity analysis presented in Annex A (paragraphs A.43 – A.46) is not robust, as it is based on the dataset used to generate results for the first draft of the working paper, which differs from the dataset used in the updated version. When LVG-1 repeats the same test using the updated dataset, excluding the 2020–2024 period, the results appear considerably weaker: the estimated effects on claim values are no longer statistically significant for either [✂] or LVG-1, and the estimated impacts on average prices are substantially smaller.

C.10 **CMA assessment**: The additional tests and arguments presented by [✂] LVG-1 do not materially add to what we have already discussed in Annex A (paragraphs A.43–A.46) for the following reasons.

C.11 Our sensitivity analysis was based on the original dataset used for the first draft of the working paper, which was appropriate given that we were responding to comments on that draft. It is not surprising that removing an additional year from the sample (ie 2014) results in larger standard errors and less precise estimates—particularly as this also leads to the exclusion of practices acquired in 2015, which are now treated as always-treated units and therefore dropped from the analysis. In total, this sensitivity test removes five years of data, and all practices acquired in 2015. The lack of statistical significance in this context simply reflects the reduced sample size and increased variability, not a change in the underlying trend.

C.12 The figures presented by LVG-1 (ie Figures 1–4) continue to show an increase in both claim values and average prices in the three years following an acquisition. For claim values, the time-to-acquisition coefficients on the first three lags are very similar to those in Figure 4.1 and Figure A.6. For average prices, the coefficients are now even larger than those in Figure 4.12 and Figure A.6 and remain statistically significant. The only difference is in the confidence intervals, which are now wider — as expected, given the smaller sample size used in LVG-1's analysis. These results are clearly consistent with an acquisition effect on both claim values and prices, and make it evident that COVID-19 does not explain our results.

C.13 As already noted in Annex A, we consider that excluding the 2020–2024 period significantly understates the impact of corporate acquisitions on claim values and prices. This is because the effects of acquisitions are likely to materialise gradually over time. Removing the final four years of data does not simply control for potential COVID-19 related effects—it also risks excluding a substantial portion of the actual acquisition effect.

C.14 Moreover, even if one assumes that COVID-19 influenced pricing trends to some degree, it is highly unlikely that it accounts for the entire observed effect in the

2020–2024 period. In this sense, the test is highly conservative—and yet, the results remain consistent with an acquisition-driven increase in both claim values and prices.

[✂]

C.15 [✂].

C.16 [✂].

C.17 [✂].

C.18 [✂]

Metacam analysis

C.19 LVG-1 argues that our analysis of Metacam prices is not reliable for three main reasons. First, the quantity variable used in the model is often missing. Second, the CMA analysis does not account for key components of the full cost to consumers, which includes not only the price of the drug itself but also additional fees such as the dispensing fee and, where applicable, an injection fee when the drug is administered in-clinic. Finally, LVG-1 claims that the CMA's results cannot be generalised to other treatments.

C.20 **CMA assessment**. We consider that the three objections put forward by LVG-1 are unfounded.

C.21 On the first point, while it is true that the quantity variable is missing in many cases, we still retain a substantial number of observations where it is available, allowing us to draw reliable inferences. Moreover, the results remain robust even when quantity is not directly controlled for, provided we include variables that serve as reasonable proxies for quantity—such as pet condition, age, and breed.[173] We discuss this issue in more depth and provide supporting robustness tests later in this Annex (set out in paragraphs C.42–C.45).

C.22 Regarding the second point, we do not believe that additional fees can fully explain the acquisition effect we document for Metacam. Several of the control variables included in our model are likely to capture part of the impact of these fees on the final price paid by the customer. For example, we control for the method of administration (oral vs injection), which should partially account for the injection fee when the drug is administered in-clinic. In addition, we include practice and year fixed effects, which help control for time-invariant practice characteristics that may

[173] The prescribed dosage of Metacam is influenced by factors such as the pet's species, age, weight, and the severity of the condition being treated (see, for example, SPC_2177020.PDF)

influence fee levels, as well as for changes in fees that occur uniformly across practices in a given year.[174]

C.23 Moreover, the claim descriptions [✂] often contain information on fees, as these likely need to be justified to the insurer. For instance, one Metacam claim included the description: [✂]. Based on this, we conduct a robustness test that explicitly controls for the presence of fees by identifying drug descriptions containing the word "fee." When we include a dummy variable equal to one if a fee is mentioned and zero otherwise, our results for Metacam remain unchanged.[175]

C.24 Finally, we have seen some evidence suggesting that LVGs are less prone to waive charges for small items and ancillary fees compared to than independents (paragraph A.26). If this is correct, then this might bias our assessment of the acquisition effect on Metacam downward. This is because, as a practice is acquired, it will start to charge for small, ancillary fees, and this will bring the average price of Metacam in claims down.

C.25 Regarding the third point, the purpose of the Metacam analysis is to provide a well-defined 'case study' that illustrates that our general findings cannot be dismissed solely on the basis that we do not observe unit prices or that treatment categories are not precisely defined. In the case of Metacam, these concerns are well addressed: the treatment category is clearly defined, pricing-relevant characteristics are accounted for, and the analysis isolates the impact of acquisition on unit prices. The fact that the acquisition effect remains significant in this context is sufficient to suggest that acquisitions can lead to material price increases— even if only for a single drug.

The Callaway and Sant'Anna estimator: undisclosed change to the CMA's analysis by LVG-1

C.26 LVG-1 argues that our implementation of the Callaway and Sant'Anna estimator in Annex A suffers from three key limitations. First, the CMA does not include covariates in the model. Second, the CMA does not use clustered standard errors. Third, the CMA has treated both the Insurer 2 and Insurer 1 datasets as repeated cross-sections, whereas [✂] considers the Insurer 1 data to be more appropriately treated as an unbalanced panel. After addressing these issues, [✂] finds that the estimated average impact of acquisitions on claim values falls to 2.3% and is no

[174] Additionally, to the extent that ancillary fees are reported separately from the drug price, this could introduce a downward bias in our estimates. There is evidence that LVGs are less likely to waive ancillary fees compared to independent practices, which would reduce the average drug price reported for acquired practices—even if the total amount paid by customers is higher.
[175] While we cannot rule out the possibility that some fees are not separately itemised in the claim description and are instead included in the total price reported, this test provides some reassurance that fees are unlikely to be a major driver of the acquisition effect we observe.

longer statistically significant, while the estimated impact on average prices is reduced to 5.5%.

C.27 **CMA assessment**: None of the three "limitations" highlighted by LVG-1 explain the weaker results it reports for average prices. Instead, the change that materially affects the results—yet was not disclosed or discussed by LVG-1—is as follows.

C.28 LVG-1 added a line of code to the R script used by the CMA to estimate the Callaway and Sant'Anna estimator, specifying "*min_e = -4*" and "*max_e = 4*". This code restricts the estimation to a narrower event-study window—only four years before and after the acquisition—significantly changing the specification and excluding important longer-term effects. It was not part of the CMA's original code and was neither disclosed nor discussed by LVG-1 in any of their submissions. Nor did LVG-1 offer any rationale or justification for imposing this restriction, despite its clear implications for the results. This adjustment is also not recommended in the econometric literature. As Sun and Abraham (2021) demonstrate, excluding post-treatment periods from event-study specifications can introduce bias into estimates of dynamic treatment effects—a concern also echoed by Baker *et al.* (2022) and others.

C.29 It is therefore entirely unsurprising that the estimated effect is smaller under this restriction: by excluding part of the sample period, the analysis excludes any long-term impacts of the acquisitions that would otherwise be captured in the full event window. As a result, the estimated effect can only be smaller. In addition, the smaller sample affects both the magnitude and precision of the estimated effects, leading to differences in both point estimates and standard errors.

C.30 If we replicate the changes [✂] claims to have made—namely, treating the Insurer 1 data as panel data and clustering standard errors by practice—but correct the key issue discussed above (ie removing the "*min_e = -4*", "*max_e = 4* restriction"), the estimated acquisition effect is 13% and statistically significant. This is identical to the 13% estimate previously reported in Table A.4, clearly demonstrating that the changes LVG-1 claims to have implemented have no material impact on our results.[176]

C.31 There are also concerns with LVG-1's DiD regression analysis of claim values. LVG-1 claims to have included the covariates from our baseline model (Table 4.1); however, in practice, they only control for pet age and whether the claim is ongoing, omitting other important covariates such as pet condition and breed. Despite this, their analysis suffers from the same methodological issue noted earlier: the results are biased due to the restricted sample resulting from the

[176] For this specification, there are no covariates that can be included. [✂] also does not control for covariates in their implementation.

addition of “*min_e = -4*” and “*max_e = 4*” to our code. When this restriction is removed, the results are once again consistent with our previous findings.

Weighting by cell value

C.32 LVG-1 argues that weighting by cell value (ie the value of a unique combination of practice, year and treatment category) rather than volume, as done by the CMA in Table A.6, is more appropriate. It shows that this reduces significantly the estimated acquisition effect for average prices from 9.5% to 2.1%.

C.33 **CMA assessment**: We do not consider LVG-1’s proposed approach—weighting by cell value, specifically the average price for each practice-year-treatment combination—to be appropriate, either econometrically or conceptually. From an econometric standpoint, this method effectively uses the dependent variable as a weight in the regression. This is not standard practice and is generally discouraged.

C.34 In particular, the weights should be independent of the dependent variable to avoid introducing bias. When the dependent variable is used to assign weights, it creates a circular logic: the variable we aim to explain also determines how much influence each observation has in the estimation —a choice that is, by definition, endogenous. As a result, the coefficient estimates may become biased and inconsistent, undermining the validity of the regression.[177]

C.35 From a conceptual perspective, our weighting approach is designed to reflect the treatments that matter most in practice—both in terms of customer expenditure and practice profitability. Weighting by frequency alone would overemphasise low-cost, routine treatments that may have limited economic relevance. Conversely, weighting by average treatment value at the practice level gives disproportionate importance to rare, high-cost procedures that make up only a small share of total spending. This approach is neither sensible nor meaningful. Our volume-based method strikes the right balance: it captures treatments that are both economically significant and commonly used. This ensures that the estimated effects reflect the treatments that matter most in the market.

[177] Weighting by share of total treatment value—as done in Table A.6—does not suffer from the same endogeneity concerns as weighting by the dependent variable itself. The weights are based on observed expenditure shares across the full sample and reflect the relative economic importance of each treatment category in the market, not the individual outcome values in the regression. This avoids circularity and ensures that more representative, economically relevant categories are given appropriate influence in the estimation.

LVG-2

Coding error in LVG-2's Metacam analysis

C.36 LVG-2 submits that the results for Metacam shown in Table A.1 and Figure A.1 of Annex A do not hold when controlling for drug concentration. It shows that the DiD coefficient changes from 10.7% to 1.6% and becomes statistically insignificant. LVG-2 also shows that including observations for [✂], reduces the overall acquisition effect from 10.7% to 7.7%. Furthermore, LVG-2 argues that selecting a sub-sample representing only 16% of total Metacam-related observations—as the CMA did in Table A.1—may introduce bias into the analysis. Finally, LVG-2 claims that the coefficient on quantity is not economically plausible, and therefore the model should be disregarded entirely.

C.37 **CMA assessment**: Due to a coding error, LVG-2's sensitivity analysis yields invalid results. After correcting the error, we confirm that the original Metacam results hold, even when controlling for drug concentration [✂] in the sample.

C.38 LVG-2 included all practices acquired from 2015 onward, along with those never acquired, in the sample used for their DiD analysis, applying the R condition "*year_acquired >= 2015 | is.na(year_acquired)*". However, because the dataset only covers the years 2015 to 2024, this approach unintentionally misclassifies practices acquired in 2015 as never acquired.

C.39 This issue arises from how the sunab() function in the fixest package estimates dynamic treatment effects: it generates event-time indicators (eg event time 0, +1, etc.) and requires at least one pre-treatment observation, from the year prior to acquisition (event time -1), to establish a valid baseline. Since the sample begins in 2015, practices acquired that same year do not have data from 2014.

C.40 As a result, the sunab() function cannot compute the required time-to-acquisition indicators for these practices. However, rather than being excluded, the 2015-treated practices are mistakenly included in the control group by default, alongside those never acquired. This misclassification biases the estimates and weakens the acquisition effect by contaminating the control—which should consist solely of never-acquired practices—with practices that are, in fact, owned by LVGs. This issue can be verified by re-running the analysis while explicitly assigning 2015-acquired practices to the control group.[178] The results will be identical to LVG-2's, confirming that these practices were included in the control group.

C.41 Tables C.1 and C.2 replicate LVG-2's Tables 3 and 4, respectively. In both cases, excluding practices acquired in 2015 (set out in columns 5 and 6 in Table C.1, and

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[178] This is achieved by setting the acquisition year to NA when it equals 2015, using the following R code: (*year_acquired <- ifelse(year_acquired == 2015, NA, year_acquired)*).

columns 4 to 6 in Table C.2) leaves the acquisition effect intact and of similar magnitude—9.1% in Table C.1 and 9.2% in Table C.2—compared to our baseline model (9.9%). The lower statistical significance is due to the fact that controlling for drug concentration reduces the sample size by 77% relative to our specification, thereby decreasing the precision of the estimates. The sample is smaller than in LVG-2's analysis because practices acquired in 2015, previously (and incorrectly) included in the control group, are now properly excluded. In Table C.1, we also report our baseline model, noting that the estimates differ slightly from those in Table A.1. This is because, in Annex A, we conducted sensitivity tests using the data from the first draft of the working paper to respond to comments from LVG's advisers, whereas LVG-2 used the data from the revised version of the working paper. The results remain consistent regardless of which dataset is used.

Table C.1: Replication of LVG-2's Table 3

	Log(Price)					
	Baseline		LVG-2's analysis		After correcting the error	
	(1)	(2)	(3)	(4)	(5)	(6)
LVG	**0.094*** **	**0.099*** **	**0.015**	**0.029**	**0.074***	**0.091*** **
	(0.019)	**(0.023)**	**(0.040)**	**(0.042)**	**(0.043)**	**(0.044)**
Bottle size (ml)	0.010***		0.009***		0.009***	
	(0.000)		(0.001)		(0.001)	
Quantity	0.017	0.008	0.278***	0.278***	0.270***	0.270***
	(0.009)	(0.008)	(0.067)	(0.067)	(0.068)	(0.069)
Cat	0.253***	0.087*	0.010		0.007	0.002
	(0.044)	(0.051)	(0.032)		(0.034)	(0.035)
Dog	0.170***	0.216***	0.133***	0.135**	0.141***	0.143***
	(0.042)	(0.055)	(0.050)	(0.053)	(0.052)	(0.055)
Oral administration	0.187***	-0.094***	-0.361***	-0.167*	-0.337***	-0.144
	(0.044)	(0.023)	(0.077)	(0.099)	(0.072)	(0.100)
Bottle size dummies	No	Yes	No	Yes	No	Yes
Concentration dummies	No	No	Yes	Yes	Yes	Yes
Condition-type FEs	Yes	Yes	Yes	Yes	Yes	Yes
Practice FEs	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Observations	156,351	156,351	39,594	39,594	36,061	36,061
Adjusted R^2	0.715	0.821	0.816	0.828	0.819	0.831

Source: CMA analysis using Insurer 1 data.
*Note: Standard errors are clustered at the practice level and are reported in parentheses. ***, **, and * correspond to statistical significance at the 1%, 5%, and 10% levels, respectively.*

Table C.2: Replication of LVG-2's Table 4

	Log(Price)					
	LVG-2's analysis			After correcting the error		
	(1)	(2)	(3)	(4)	(5)	(6)
LVG	**0.744*****	**0.009**	**0.028**	**0.092*****	**0.069***	**0.092****
	(0.018)	**(0.036)**	**(0.038)**	**(0.017)**	**(0.041)**	**(0.042)**
Bottle size (ml)	0.010***			0.010***	0.010***	
	(0.000)			(0.000)	(0.001)	
Quantity	0.016**	0.292***	0.293***	0.015**	0.284***	0.284***
	(0.007)	(0.063)	(0.064)	(0.006)	(0.065)	(0.067)
Cat	0.230***	0.007	0.001	0.233***	0.003	-0.003
	(0.040)	(0.028)	(0.029)	(0.042)	(0.030)	(0.031)
Dog	0.144***	0.119***	0.121**	0.145***	0.124***	0.127**
	(0.039)	(0.046)	(0.048)	(0.041)	(0.048)	(0.051)
Oral administration	0.146***	-0.244**	-0.124	0.159***	-0.218**	-0.099
	(0.035)	(0.099)	(0.102)	(0.037)	(0.098)	(0.103)
Bottle size dummies	No	No	Yes	No	No	Yes
Concentration dummies	No	Yes	Yes	No	Yes	Yes
Condition-type FEs	Yes	Yes	Yes	Yes	Yes	Yes
Practice FEs	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Observations	203,270	46,359	46,359	189,333	41,797	41,797
Adjusted R^2	0.722	0.797	0.809	0.722	0.801	0.814

Source: CMA analysis using Insurer 1 data.
*Note: Standard errors are clustered at the practice level and are reported in parentheses. ***, **, and * correspond to statistical significance at the 1%, 5%, and 10% levels, respectively.*

C.42 LVG-2 argues that selecting a sub-sample that represents only 16% of total Metacam-related observations may introduce bias into the analysis, implying that this could lead to an upward bias in the estimated acquisition effect.[179] We agree that such a reduction in sample size could, in principle, result in a biased estimate of the acquisition effect. However, we believe that any resulting bias would likely be downward, as shown in Table C.3.

[179] This represents 25% of the Metacam-related observations that are usable for the DiD analysis.

C.43 Approximately 75% of observations are excluded because of missing values in the quantity variable. To address this, we rely on a comprehensive set of control variables—namely pet breed, age, and medical condition—that act as proxies for treatment intensity and are likely to influence the quantity prescribed. By using these controls instead of quantity, along with drug characteristics such as bottle size and administration method, we retain a much larger sample, losing only 26% of total Metacam observations. The number of observations decreases from [✂].

C.44 It is reasonable to expect that the amount of Metacam prescribed depends on the pet’s condition, species, and other observable characteristics, all of which are accounted for in our models. According to the UK Veterinary Medicines Directorate, the prescribed dosage of Metacam is influenced by several factors, including the pet's species, age, weight, and the severity of the condition being treated (see, for example, SPC_2177020.PDF). For instance, Metacam 1.5 mg/ml oral suspension is intended for use in dogs only and is contraindicated in dogs less than six weeks of age. This underscores the importance of controlling for both species and age in the regression analysis. Dosage is typically calculated based on the pet's weight, with larger animals—proxied in part by breed and age—generally requiring higher doses. For example, for dogs, the recommended initial treatment with Metacam 1.5 mg/ml is a single dose of 0.2 mg meloxicam per kg of body weight on the first day. Additionally, the severity of the condition plays a significant role, as more severe or chronic conditions may necessitate longer or higher-dosage treatments. These variables are therefore very likely to capture much of the variation in quantity, even when quantity itself is not included in the model. As such, they allow us to control indirectly for differences in prescribing behavior that could otherwise confound the estimated acquisition effect.

C.45 Column 3 of Table C.3 shows that when using this larger sample, the estimated acquisition effect remains statistically significant and very similar in magnitude: we find a 12.6% price increase, compared to 10.2% when controlling directly for quantity. This reinforces the conclusion that sample size is not driving the results and that the observed price effect is robust to alternative model specifications. In fact, reducing the sample size by controlling for quantity slightly lowers the estimated effect, rather than inflating it. In addition, Column 6 of Table C.3 shows that even when we restrict the sample to observations with non-missing quantity, without controlling for quantity or other covariates (aside from condition), the DiD estimate is already lower than in Column 1, which uses the full sample. This suggests that simply limiting the sample to cases with non-missing quantity reduces the estimated effect, rather than increasing it, contradicting LVG-2’s suggestion that sample selection biases the results upward.

Table C.3: Acquisition effects for Metacam across different sample sizes and model specifications

	Log(Price)				
	(1)	(2)	(3)	(4)	(5)
LVG	**0.151*** **	**0.140*** **	**0.126*** **	**0.102*** **	**0.090*** **
	(0.017)	**(0.016)**	**(0.014)**	**(0.025)**	**(0.020)**
Pet age		0.021***	0.002***	0.002***	
		(0.000)	(0.000)	(0.001)	
Oral administration			-0.007	-0.165***	
			(0.016)	(0.026)	
Log(Quantity)				0.224***	
				(0.041)	
Bottle size dummies	No	No	Yes	Yes	No
Breed dummies	No	Yes	Yes	Yes	No
Condition-type FEs	Yes	Yes	Yes	Yes	Yes
Practice FEs	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes
Observations	617,162	617,035	458,118	156,258	156,363
Adjusted R^2	0.430	0.517	0.825	0.844	0.401

Source: CMA analysis using Insurer 1 data.
*Note: Standard errors are clustered at the practice level and are reported in parentheses. ***, **, and * correspond to statistical significance at the 1%, 5%, and 10% levels, respectively.*

C.46 Another criticism raised by LVG-2 is that the coefficient on quantity lacks economic plausibility, as it is too small. LVG-2 suggests this may be due to omitted variables, and on that basis argues the model should be disregarded entirely. We believe this conclusion is extreme and not well supported, for several reasons.

C.47 The purpose of including control variables was to ensure that they are not correlated with selection into treatment. For example, if LVG-owned practices began prescribing larger quantities of the same drug for the same condition and pet characteristics (such as age or breed), failing to control for these variables could bias our estimate of the acquisition effect. While we consider this scenario unlikely—since there is no clear clinical justification for prescribing more of the same drug for the same condition post-acquisition—we nonetheless included these controls to enhance the robustness of our results and to directly address the advisers' concern that, in the absence of unit price data, our estimation may be biased.

C.48 Our results clearly show that the estimated acquisition effect remains both statistically significant and very similar in magnitude after controlling for quantity and other relevant covariates, despite a substantial reduction in sample size. This aligns with our expectations: we would not anticipate significant changes in quantities prescribed or in drug characteristics (eg bottle size) for a given condition solely as a result of ownership change.

C.49 While the discussion above demonstrates that our results hold after correcting LVG-2's coding error and controlling for drug concentration, we believe it is important, as a matter of principle and completeness, to address LVG-2's proposed model and its limitations. In particular, LVG-2 argues that our model should include Metacam concentration as a separate control variable. However, doing so is problematic from both a conceptual and a statistical standpoint.

C.50 Metacam concentration can be effectively identified using a combination of pet type (dog, cat, or other), bottle size, and administration method (oral vs. injection), all of which are already included as control variables in our regression. As shown in Table C.4, in most cases, knowing these three variables is sufficient to uniquely determine the drug concentration. As a result, including drug concentration as a separate variable in the regression would introduce near-perfect multicollinearity, leading to unreliable estimates and inflated standard errors.

C.51 This is clearly illustrated in the case of Metacam for dogs, where the drug concentration is uniquely determined by the combination of administration method and bottle size. As shown in Table C.4, the oral suspension form is available in two concentrations: 0.5 mg/ml and 1.5 mg/ml. The 0.5 mg/ml formulation is sold in 15 ml and 30 ml bottles, while the 1.5 mg/ml version comes in 10 ml, 32 ml, 100 ml, and 180 ml bottles. In contrast, the injectable form for dogs is always 5 mg/ml and is sold in 10 ml and 20 ml bottles. Thus, for dogs, the concentration can be uniquely inferred from the combination of administration method and bottle size. For example, if the administration method is oral and the bottle size is 32 ml, the concentration must be 1.5 mg/ml; if the method is injection, the concentration is always 5 mg/ml. Therefore, by controlling for pet type, administration method, and bottle size in our regression—as we do—we indirectly, but effectively, control for drug concentration.

C.52 A similar pattern holds for cats receiving Metacam as an oral suspension, where the concentration is consistently 0.5 mg/ml across all bottle sizes. The only exception is in the case of injectable forms for cats, where both 2 mg/ml and 5 mg/ml formulations are available in the same bottle sizes (10 and 20 ml). We do not consider this a significant concern, as our regression also controls for the pet's medical condition. It is reasonable to assume that the choice between these two concentrations is at least partially driven by the underlying condition—an effect captured by the condition fixed effects in our model. Furthermore, this subgroup—cats receiving injectable Metacam—represents only 14% of the total sample. As such, even if there is some residual variation in concentration within this small group, it is highly unlikely to meaningfully bias the estimated acquisition effect.

C.53 Ultimately, as noted, the results remain robust whether or not drug concentration is explicitly controlled for, further reinforcing the robustness of our findings.

Table C.4: Metacam for Dogs and Cats: Concentration, Administration Method, and Bottle Size

Panel A: Metacam for dogs

Administration method	Concentration (mg/ml)	Bottle size (ml)
Oral suspension	0.5	15 and 30
Oral suspension	1.5	10, 32, 100, and 180
Injection	5	10 and 20

Panel B: Metacam for cats

Administration method	Concentration (mg/ml)	Bottle size (ml)
Oral suspension	0.5	3, 10, 15 and 30
Injection	2	10 and 20
Injection	5	10 and 20

Source: This information is sourced from the UK Veterinary Medicines Directorate's Product Information Database.

LVG-2's analysis of Apoquel and Gabapentin

C.54 LVG-2 extended the CMA's analysis to include two additional drugs—Apoquel and Gabapentin—and reported that the estimated acquisition effects for these drugs are statistically insignificant. On this basis, LVG-2 concluded that the results observed for Metacam cannot be generalised to other drugs.

C.55 **CMA assessment**: We do not consider LVG-2's analysis of Apoquel and Gabapentin to provide reliable or relevant evidence against our findings. The analysis suffers from significant methodological flaws that prevent it from providing meaningful insight.

C.56 As previously noted, the same coding error identified in LVG-2's analysis of Metacam is also present in their analysis of Apoquel and Gabapentin. In addition, the model it estimates suffers from several important limitations.

C.57 First, as LVG-2 themselves acknowledge, the quantity variable for Apoquel is inconsistently defined—sometimes referring to a single unit, other times to an entire pack. This makes the variable unreliable, and including it as a control would introduce measurement error, which could undermine both the precision and interpretability of the regression estimates. Despite acknowledging this problem, LVG-2 made no attempt to correct for it or provide any sensitivity checks to assess its potential impact on their results. Notably, LVG-2 did not identify any such issues in the case of Metacam, where the quantity variable appears to be correctly and consistently measured.

C.58 Second, LVG-2 fails to account for several critical determinants of drug price. For example, Apoquel is sold in packs of 20, 50, or 100 tablets, information that directly affects pricing but was not extracted or controlled for in their regression. Additionally, Apoquel comes in two formulations—film-coated and chewable tablets—which also differ in price. Again, this distinction was not captured in LVG-2's analysis.[180]

C.59 We recognise that this kind of information is not always consistently recorded across all drugs. This is precisely why we focused our drug-level analysis on Metacam, where price-relevant characteristics could be reliably extracted and controlled for. While we appreciate LVG-2's effort to extend the analysis to other drugs, their analysis has too many limitations to be taken into consideration.[181]

LVG-3

Acquisitions affect how claims are submitted in the Insurer 2 dataset.

C.60 LVG-3 has provided evidence that acquisitions influence how claims are submitted in the Insurer 2 data. Specifically, it shows that the proportion of submitted claims with a single invoice increases significantly after acquisition. In addition, LVG-3 argues that the CMA should have combined the Insurer 2 and Insurer 1 datasets and conducted the analysis using the merged data. It shows that using the combined dataset yields a lower estimate of the acquisition effect.

C.61 **CMA assessment**: We agree with LVG-3 that acquisitions affect claim submission patterns in both the Insurer 1 and Insurer 2 datasets. In general, acquired practices tend to submit claims with single invoices, rather than aggregating multiple invoices into one claim, following an acquisition. This introduces a downward bias in our estimates, meaning the observed effect at the claim level is likely to underestimate the actual acquisition impact. In essence, if acquired practices begin submitting smaller, more frequent claims after acquisition, we may observe only a minor change in the difference between LVGs and independent practices during the post-acquisition period. This would not indicate that independents have become more expensive or LVGs cheaper, but rather that LVGs have changed their submission practices, submitting each invoice promptly and individually.

C.62 This consideration is supported by our robustness checks in Annex B. When we focus on claims with a single invoice, we find a statistically significant acquisition effect of 7% for Insurer 2 (Figure B.2). Notably, this effect is very similar in magnitude to the price effect observed in the Insurer 1 data at the invoice item level (7.4%) in Table 4.3. Of the range of estimated acquisition effects—from 3.2% to 7.4%—we believe the actual effect is likely to be very close to the upper end.

[180] This information was obtained from the UK Veterinary Medicines Directorate's Product Information Database.
[181] [✂] themselves acknowledge that their estimates *'do not necessarily imply that there is no price increase but rather illustrate that the data lacks sufficient granularity to control for relevant characteristics*.', implicitly indicating that we should not place much weight on these results.

C.63 We do not consider that LVG-3's approach of combining datasets is appropriate, for the following reasons. Combining datasets in which the outcome variable of interest (ie claim value) is measured differently can lead to biased estimates. Insurers use different systems to upload invoices, and the specific system used directly affects how invoices are aggregated into claims. For instance, claims submitted through one system may systematically aggregate fewer invoices and reflect lower amounts than those submitted through others. This introduces considerable noise and inconsistency into the analysis, making it difficult to draw reliable conclusions from the data. A better approach is to analyze each dataset separately and assess whether both analyses point in the same direction—which is exactly what we do. When two independently collected datasets, each with its own structure, lead to similar conclusions, it strengthens the credibility of the result and provides reassurance that our findings are not driven by a particular dataset.

Rambachan and Roth's (2023) test

C.64 LVG-3 assesses the validity of the parallel trends assumption using the method developed by Rambachan and Roth (2023), which places bounds on how much post-treatment violations can differ from the largest deviation observed in the pre-treatment period. Specifically, the method requires that any post-treatment violation be no greater than a constant multiple, M, of the maximum pre-treatment violation. For example, M = 1 implies that post-treatment deviations must be no larger than the worst deviation observed before treatment for the estimated effects to remain significant. LVG-3 finds that the estimated acquisition effects on claim values and prices are only robust when M = 0.25, meaning post-treatment violations must be no more than one-quarter the size of the largest pre-treatment deviation. According to LVG-3, the CMA should instead accept an M that allows for post-treatment violations to be of the same magnitude as those observed in the pre-acquisition period. This would correspond to M = 1 instead of 0.25. Therefore, given the low M it finds, LVG-3 questions the validity of the parallel trends assumption.

C.65 **CMA assessment**: We welcome LVG-3's use of the Rambachan and Roth (2023) method, which offers a valuable framework for assessing the robustness of the parallel trends assumption. However, LVG-3's application of the method has certain limitations. We implement a number of refinements to improve its application in this context—refinements that have a substantial impact on LVG-3's conclusions. The rationale for these adjustments is explained below.

C.66 The confidence sets generated by the Rambachan and Roth (2023) method tend to widen when there is greater uncertainty about pre-treatment differences in trends between treated and control groups—that is, when the estimated pre-treatment coefficients have large standard errors. This contrasts with conventional pre-trends tests, which intuitively become less likely to reject the null when the pre-treatment coefficients are more imprecise (Roth *et al.*, 2023). In our case, this uncertainty is particularly pronounced in periods further from the treatment year, where the number of treated firms and observations is very low. For example, Table C.5 shows that in the Insurer 1 dataset, only [✂] acquired practices are observed in year -8, compared to [✂] in year -2. Only from year -6 onward, we observe more than [✂] acquired practices. A similar pattern appears in the Insurer 2 dataset: only one acquired practice is observed in year -9, compared to [✂] in year -2, and again, it is only from year -6 onward that the number of acquired practices exceeds [✂].

Table C.5: Number of treated observations and practices by relative year to acquisition

Panel A: Insurer 1

Relative year	Number of treated observations	Number of treated practices
-8	[✂]	[✂]
-7	[✂]	[✂]
-6	[✂]	[✂]
-5	[✂]	[✂]
-4	[✂]	[✂]
-3	[✂]	[✂]
-2	[✂]	[✂]

Panel B: Insurer 2

Relative year	Number of treated observations	Number of treated practices
-9	[✂]	[✂]
-8	[✂]	[✂]
-7	[✂]	[✂]
-6	[✂]	[✂]
-5	[✂]	[✂]
-4	[✂]	[✂]
-3	[✂]	[✂]
-2	[✂]	[✂]

Source: CMA analysis using Insurer 1 and Insurer 2 data.

C.67 Including early pre-treatment years with very few observations and high uncertainty can lead to excessively wide post-treatment confidence sets. In simple terms, if we include years far from the treatment with limited data and few treated units, any violation of the parallel trends assumption might appear much worse than it truly is. This is because the coefficients from those periods are highly noisy and estimated on small, unrepresentative subsamples of acquired practices.

C.68 LVG-3 acknowledges this issue by excluding the estimated coefficient for year -9 when constructing confidence sets using the Insurer 2 data. While we agree with this decision, we believe the same logic should apply to other early pre-treatment years with very low coverage. This approach is consistent with standard practice in pre-trends testing, where coefficients based on small sample sizes—typically from distant periods—are often disregarded due to high imprecision.

C.69 Choosing how many pre-treatment years to include involves a trade-off. Using more years can increase confidence in the parallel trends assumption—if trends are parallel over a longer period before treatment, the assumption becomes more credible. However, early years often contain fewer treated practices and are less representative of the broader sample, making any detected violations less reliable due to small sample sizes and greater uncertainty. Our sample spans from 2015 to 2024 and includes only practices acquired from 2016 onward, with most acquisitions occurring from 2017 and peaking in 2018. As a result, most acquired practices have at least two to three years of pre-acquisition data: practices acquired in 2017 have at least two pre-treatment years, and those acquired from 2018 onward have at least three. Coefficients for event years -2 and -3 are therefore estimated on a broadly representative subsample of acquired practices over the 2016–2024 period. The main exception is practices acquired in 2016, for which only one pre-treatment year is available. For this reason, we focus initially on event years up to -3, which provide a reliable window to assess pre-trends while avoiding the noise from earlier, sparse periods. We then run robustness checks using extended pre-treatment windows, including event years up to -6, for which we still observe at least 100 acquired practices.[182]

C.70 Using pre-treatment coefficients up to year -3, estimated using the Insurer 1 dataset, we find a break value of M = 2 for the price analysis. This means the estimated acquisition effect on prices remains statistically significant unless post-treatment violations are more than twice as large as the worst pre-treatment deviation. This high threshold increases our confidence that the parallel trends assumption holds over this window.

C.71 Extending the pre-treatment window to year -4 or -5 slightly lowers the break value M from 2 to 1.5. This means that our results for prices remain significant unless post-treatment violations are more than 1.5 times the maximum pre-treatment deviation. Despite using a broader window in this case (-5 to +5), the M value estimated is still high and therefore supports the credibility of the parallel trends assumption.

C.72 When we include coefficients up to -6, the break value M drops to 1. This implies that results remain significant unless post-treatment violations are greater than the

[182] Observations in these earlier years reflect practices acquired later in the sample period—for example, the estimate for year -6 is based on data from practices acquired in 2021 or later.

worst pre-treatment deviation. While the estimate for year -6 is based on data from only 110 acquired practices—around 17% of the treated units observed in year -2—even this conservative test yields a break value well above LVG-3's estimate of M = 0.25.[183] As LVG-3 notes in paragraph 2.27 of their report, the CMA should at least be willing to assume that post-treatment violations are no larger than the worst pre-treatment deviation. That is exactly what we find when including up to six pre-treatment coefficients.[184]

C.73 We conduct the same analysis for claim values using the Insurer 2 dataset and also find M values higher than those reported by LVG-3. Specifically, we obtain M = 1 when using pre-treatment coefficients up to -3, and M = 0.5 when using coefficients up to -4 or -6. The lower values are expected, given the noisier estimate we obtain for claim values, which are due to changes in how practices aggregate claims, as already discussed in paragraph C.70 – C.72 of this Annex.

C.74 In summary, these tests give us confidence that our conclusions are robust to reasonable departures from the parallel trends assumption.

LVG-4

C.75 On 27 May 2025, LVG-4's advisers submitted a response to our second presentation of the analysis presented in this annex.[185] We considered that LVG-4's submission largely repeated arguments made previously, without engaging with our discussion of these arguments in the second version. As a result, we did not discuss LVG-4's submission in detail.

C.76 On 5 September 2025, LVG-4's advisers submitted further comments on the third version of this analysis, where they argued that we had not properly engaged with their comments, or followed a fair process for assessing their relevance and importance (**LVG-4 Further Comments**).[186]

C.77 We continue to consider that LVG-4's submissions on the second and third versions of our analysis mostly repeat points that were already been addressed in the second version. However, for completeness we discuss the key points raised in LVG-4's Further Comments. For the purpose of this discussion, we refer to the third version of our analysis shared with LVG-4's advisers as the '**CMA August Paper'**.[187]

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[183] [✂] lower breakdown value is clearly driven by the inclusion of pre-treatment years with very small samples—for example, only 12 acquired practices are observed in year -8 for [Insurer-1].
[184] In each test, we use a balanced number of pre- and post-treatment periods (ie (-3, +3), (-4, +4), (-6, +6)), making M = 1 a natural benchmark. Results remain consistent when including all available post-treatment years.
[185] [✂].
[186] [✂].
[187] CMA, The Impact of corporate acquisitions on treatment costs, 14 August 2025.

No rationale for treating the analysis of first-year treatment costs as a sensitivity

C.78 LVG-4's advisers state that the CMA has provided no rationale for its analysis of first-year treatment costs being treated as a sensitivity.[188]

C.79 **CMA assessment**. This criticism is incorrect. Our presentation of the DiD models clearly states that, when using the Insurer 2 data, the DiD analysis focuses on claim values rather than first-year treatment costs because our identification strategy relies on the precise timing of acquisitions, and first-year treatment costs can aggregate claim values from two different years (paragraph 3.10). This argument was clearly set out in the August Paper,[189] and LVG-4's advisers have not engaged with this argument.

No holistic presentation of the evidence base

C.80 LVG-4's advisers state that the CMA's evidence has not been assessed and presented in a way which presents a holistic and consistent evidence base. They submit that the evidence presented in the August Paper is too inconsistent to support any finding of an AEC, or the necessity of or design of any remedies set out in the Remedies Working Paper.[190]

C.81 **CMA assessment**. This is not a valid criticism of our analysis. In general, we acknowledge that LVG-specific results are measured more imprecisely (paragraph 1.13).[191] Accordingly, we show detailed results for each LVG and each metric, and discuss the relative strength of the results. [✂],[192] [✂].[193] [✂].[194] More generally, the group's provisional findings on the AECs and remedies are based on the totality of the evidence collected, and not just on the econometric analysis.

C.82 It is also incorrect to describe the DiD results for LVG-4 as 'inconsistent'. It is not the case that the different pieces of analysis unambiguously point to opposing conclusions. [✂].

C.83 [✂].

C.84 In this context it is reasonable for the CMA to conclude that LVG-4's acquisitions had an effect on bill levels for its customers.

188 [✂] Further Comments.
189 CMA August Paper.
190 [✂] Further Comments
191 This is also set out in CMA August Paper.
192 This is also set out in CMA August Paper.
193 This is also set out in CMA August Paper.
194 See part A, section 6, footnote 7, and part A, section 8, footnote 26.

Relative robustness of the different analyses

C.85 LVG-4's advisers submit that the CMA should weight the different pieces of analyses differently: (i) the DiD analyses should be preferred to the cross-sectional analyses (as the DiD analyses better control for unobserved clinic level factors); (ii) analyses based on estimates of first-year treatment costs should be preferred to those of claim values and average prices (as they better reflect the total cost to the consumer); and (iii) analyses based on Insurer 2 data should be preferred to those based on Insurer 1 data (since the Insurer 1 data may be affected by changes in billing practices).[195]

C.86 **CMA assessment**. LVG-4's advisers do not present any arguments that are not already discussed in the August Paper, and in the main body of this annex. In brief, on the three points raised by LVG-4's advisers:

C.86.1 We agree that the DiD analyses should be preferred to the cross-sectional analysis for the purpose of estimating the causal effect of LVG acquisitions (set out in paragraphs 2.44 and 3.1);[196]

C.86.2 We disagree that the analysis of first-year treatment costs should be preferred to the analyses of claim values, for two reasons: first, our identification strategy relies on the precise timing of acquisitions, and first-year treatment costs can aggregate claim values from two different years (paragraph 3.10); and second, the inclusion of a control variable indicating whether an observation relates to the first claim submitted for a specific condition or an ongoing claim mitigates the impact of changes in the relative value of first and ongoing claims (paragraph 3.10). These arguments were clearly set out in the August Paper,[197]and LVG-4's advisers did not engage with them.

C.86.3 We disagree that the analysis based on the Insurer 2 data should be preferred to analyses based on the Insurer 1 data, for two reasons: first, we have limited our utilization of the Insurer 1 data to the analysis of average treatment prices, which are less likely to be affected by changes in billing practices (paragraph 2.18);[198] and second, if anything, changes in billing practices are more likely to introduce a downward bias in our estimates of the acquisition effects (paragraphs C.16-C.18).[199] These arguments were clearly set out in the August Paper,[200] and LVG-4's advisers did not engage with them.

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195 [✂] Further Comments.
196 This is also set out in the CMA August Paper.
197 CMA August Paper.
198 This is also set out in the CMA August Paper.
199 This is also set out in the CMA August Paper.
200 CMA August Paper.

C.87 LVG-4's advisers highlight a particular statement in the August Paper, which explains that a comparison between the coefficients in the cross-sectional regressions of average prices and first-year treatment costs may conflate economically meaningful factors with less relevant factors stemming from differences in the coverage of the samples or the incentives facing the customers of the two insurers.[201] But the purpose of this statement is to warn against an interpretation of that difference as indicative of differences in treatment intensity between independent and LVGs. It only applies to the cross-sectional regressions, and is irrelevant to the difference-in-difference analyses (which compare changes in an outcome variable between independents and LVGs within a given database).

[✂]

C.88 [✂].[202]

C.89 [✂].[203] [✂].[204]

C.90 [✂].

C.91 [✂], [205] [✂].[206] [✂].

C.92 [✂].[207] [✂],[208] [✂].[209] [✂].

C.93 [✂].

Insufficient consideration of previous submissions on OOH

C.94 LVG-4's advisers originally submitted that the Insurer 1 and Insurer 2 datasets do not identify whether a treatment is conducted in ordinary business hours or out-of-hours (OOH), which is several times more expensive and tends to take place more commonly at LVG practices. As such, groups which offer a higher proportion of OOH treatments in their treatment mix will be assigned by the CMA as exhibiting a

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201 [✂] Further Comments.
202 [✂] Further Comments.
203 [✂] Further Comments.
204 [✂] Further Comments.
205 This is also set out in the CMA August Paper.
206 [✂] Further Comments.
207 There are many examples of such services advertised online. For example, a provider called 'Vet Practice Support' claim 'We provide a mobile, multi-disciplinary, advanced practitioner-led clinical service to veterinary practices. You get to keep the work in-house, your clients get maximum convenience. We aim to be flexible and responsive.' https://www.vetpracticesupport.com/. Another provider called 'MSM Vets' claim 'We also offer mobile veterinary services, when the surgery is just not feasible for the team. We come to you and operate at your site. We will have any additional surgical instruments with us. We have a comprehensive, great value, fixed price surgical list. This allows you to charge your clients normally, have the theatre time reduced to allow greater throughput, retain bigger surgeries, and clients will be satisfied and bonded to your practice.' https://msmvets.co.uk/services/
208 This is also set out in the CMA August Paper.
209 [✂] Further Comments.

higher price. The inclusion of a 24-hour control variable for 24-hour clinics decreases the LVG-4 coefficient in the CMA's cross-sectional regressions.[210]

C.95 The CMA August Paper dismissed this analysis on two grounds. First, data collected from all suppliers shows that independents are in fact more likely to offer the OOH in-house than LVGs. Second, the variable used by LVG-4's advisers to control for 24-hour provision is only available for LVGs, and this analysis assumes that independents never provide this service, which is evidently incorrect.[211]

C.96 LVG-4's advisers submit that this treatment of their argument is incorrect. First, although independents nominally offer OOH care, the share of OOH services at independents is likely to be lower than that of LVGs overall (because LVGs operate dedicated 24-hour clinics).[212] Second, LVG-4's advisers maintain that there are important differences between 24-hour clinics and non-24-hour clinics, in terms of OOH activity, costs and the mix of complexity of treatments.[213]

C.97 **CMA assessment.** We do not consider that these representations change the conclusions presented in the CMA August Paper. LVG-4's advisers do not provide any evidence for their assertion that the share of OOH services at independents is likely to be lower than for LVGs. In any case, the sensitivity put forward by LVG-4's advisers (which includes a variable identifying whether a LVG-4 practice is a 24-hour clinic) only affects the cross-sectional regressions, and it is common ground between the CMA and LVG-4's advisers that these are less probative than the DiD results in testing the effect of changes in ownership (paragraph -). LVG-4's advisers do not explain how this feature of the market could or does affect DiD results. More generally, even if LVG-4's business model involves concentrating OOH provision on a small number of clinics, it is not clear that this would bias results, as both the cross-sectional and the DiD analyses average results over FOPs.

[✂]

Limitations of relying on insurance-only data

C.98 [✂], it may be inappropriate to evaluate the effect of acquisitions on treatment costs using data from insured customers alone, as they are more likely than uninsured customers to opt for complex and costly treatments, which could bias the results.

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[210] [✂] response to draft working paper.
[211] This is also set out in the CMA August Paper.
[212] [✂] Further Comments.
[213] [✂] Further Comments.

C.99 **CMA assessment**: We discuss this limitation in paragraph 1.20 of the working paper. To our knowledge, veterinary practices do not differentiate prices based on whether a customer is insured or uninsured. As such, the price effects we identify for insured customers are likely to be directly relevant for uninsured customers as well.

C.100 We acknowledge that treatment intensity tends to be higher among insured customers, particularly for complex conditions. This is because insured customers do not bear the full cost of care and may therefore be more inclined to accept recommendations for additional or higher-cost procedures. However, our analysis does not aim to assess whether price levels at LVGs are high in absolute terms. Instead, it aims to identify whether acquisitions have a causal effect on prices by examining differences in the prices paid by insured customers at independent practices compared to those at LVGs.

C.101 Therefore, as our analysis compares insured customers across both types of practices, any potential bias from focusing on this group is likely to affect both independents and LVGs similarly. Insured customers may be generally more inclined to accept recommendations for additional or higher-cost procedures, regardless of whether the treatment is administered at a LVG or independent practice. As such, this is unlikely to introduce systematic bias into the results, unless for the same condition and pet characteristics, LVGs are systematically more likely than independents to recommend the most expensive treatment options to insured customers.

Economic significance of first-year treatment cost increases

C.102 [✂] question the economic significance of our finding that first-year treatment costs increased by 53% over the period 2016-2023, noting that this equates to an annual increase of approximately 3% above inflation. They suggest that such an increase can be explained by rising input costs in the veterinary sector, including salaries, equipment, medicines, and premises. [✂] has similarly commented that many factors have influenced treatment costs over this period. In particular, it notes that there have been significant increases in treatment costs at both independent and LVG practices, alongside advancements in both medical and surgical treatment options. It also points to changes in salary and compensation structures, such as the introduction of performance-based bonuses in many practices, as well as operational changes within practices, as additional contributing factors.

C.103 **CMA assessment**: We agree that some of the observed increase in first-year treatment costs may reflect broader cost pressures in the veterinary sector, such as rising salaries, equipment costs, and other overheads. However, our provisional view is that a significant driver of the documented price increases has been a lack of effective competition between FOPs. In forming this view, we have considered a range of evidence, which we summarise here in our Overview of working papers.[214]

C.104 In addition, we consider a yearly increase of approximately 3% to be economically significant. While such an annual increase may appear modest in isolation, the compounded effect over several years is substantial and has resulted in prices at LVGs now being significantly higher than those at independent practices.

[214] Overview of our working papers